

Belief Records and the Price of Disaster Fear

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Abstract

Option-implied disaster probabilities differ from macroeconomic-record estimates (Backus et al., 2011). We argue beliefs provide a bridge between the two. Our Lucas-economy CRRA investors keep a disaster model until its since-adoption record grows too unlikely under it (Ortoleva, 2012); the option-implied intensity factors exactly into a correct-belief benchmark—the physical law priced at the common risk aversion—times a belief multiplier. It matches the thin short tail in part because the complacent price it in calm—the power mean tilts toward them, whom calm enriches—and disasters lift it at a stroke. The slope prices revisions timed by survival records: given the spot surface, record proxies forecast it—a residual belief-sufficient models set to zero.

1. Introduction

Index options price rare disasters, and the disaster distribution they imply is not the one the macroeconomic record measures. The physical disaster sizes of Barro (2006), inverted through a frictionless representative-agent kernel, overstate the tail options price on average; yet that same risk-neutral tail spikes far above any physical benchmark after a crisis (Backus et al., 2011; Bollerslev and Todorov, 2011; Andersen et al., 2015b). This paper argues that the term structure of that disaster insurance measures two different things. Its short end prices the

disasters investors fear today. Its slope increasingly prices the *revisions* those fears are due for—the corrections that arrive as the record of calm under which today’s beliefs have survived turns against them. The spot surface reads *what the market fears*; the term structure reads *how long it has feared it*. Long-dated tail protection—multi-year deep out-of-the-money index puts, the hedges institutions hold against the next crisis—is priced off the same option-implied disaster measures a current-beliefs model summarizes with one state, so if the slope prices coming revisions, such a model prices the near tail correctly and the slope wrong, in a direction the record predicts.

Call a model *belief-sufficient* if its pricing state is the current cross-section of beliefs, however that cross-section is generated. The class is the field: a moving latent intensity (Wachter, 2013; Gabaix, 2012), a Bayesian posterior (Collin-Dufresne et al., 2016), a smoothly filtered perception (Dew-Becker et al., 2025), a heterogeneous-belief cross-section (Chen et al., 2012), and the diagnostic-state distortions that generate maturity-increasing overreaction in bonds (d’Arienzo, 2020; Wang, 2021) all read the entire surface off a current state. In the binary intensity channel the near tail pins that state exactly; with richer states the identification is by the *current option state*—a vector of short-dated surface readings taken as a structural filter for it—and once that state is pinned, the slope is its functional and carries no information of its own. We argue that when disaster beliefs are revised *lazily*—held while the record accrued since adoption stays unsurprising under the held model, abandoned for the best alternative at a surprise (Ortoleva, 2012)—the slope decouples: two markets can imply the *same* near tail and carry different forward surfaces, because their beliefs sit on different records. Section 2 exhibits the decoupling in two dates and one formula; the body of the paper proves that it survives a continuum of investors, general records, disasters, and stationary turnover, and that no belief-sufficient model produces it.

The economy is a complete-market Lucas economy with rare consumption disasters and a continuum of CRRA investors, each holding a single model of the disaster rate and revising it lazily,

who from common priors and common data—heterogeneous only in their revision tolerances—come to disagree endogenously (the homogeneous-start baseline; the turnover economy of Section 6 adds newborn model dispersion). In equilibrium the risk-neutral disaster intensity factors exactly,

$$U_t = U^{\text{RA}} \kappa_t,$$

into a representative-agent benchmark U^{RA} —the intensity a correct-belief agent prices off the macro-measured physical law, fixed by that law and risk aversion—times a *belief multiplier* κ_t . The kernel does not price the average belief: it aggregates the held models by a wealth-weighted *power mean* of their disaster likelihood ratios, of order $1/\alpha$ equal to the common risk tolerance, so that risk aversion damps belief before it reaches prices and the arithmetic average is only the log-utility knife-edge (Jouini and Napp, 2007; Cvitanic and Malamud, 2011). Inverting the near tail through this kernel returns not the physical intensity λ^* but $\lambda^* \kappa_t$: what the disaster-option literature recovers is a wealth-weighted *aggregate* of beliefs—the power-mean functional the cross-section S_t generates, not the cross-section itself—and the multiplier is the wedge between it and the physical law that Baron et al. (2024) and Chen et al. (2012) document. Figure 1 summarizes the mechanism.

The decoupling is the central result (Theorem 1), and it rests on one property of the rule: a lazily held belief is not a sufficient statistic for its own revision—the since-adoption record governs the revision hazard—so the pricing state enlarges from the cross-section S_t to Ψ_t , the cross-section together with the records, ages, and revision hazards under which its models have survived (Tegua, 2026). The enlargement is the source of the result, not the result. Its pricing content is a split. *First*, the fear factor U_t , the multiplier κ_t , the short rate, and the risk-neutral size shape are functionals of S_t alone. *Second*, the price–dividend ratio, the level of the disaster premium, and the term structure of the tail load on the full Ψ_t : two dates with the same cross-section S_t —hence the same near tail—carry identical spot risk-neutral tails and short rates, yet different price–dividend ratios, premium levels, and forward surfaces. *Finally*,

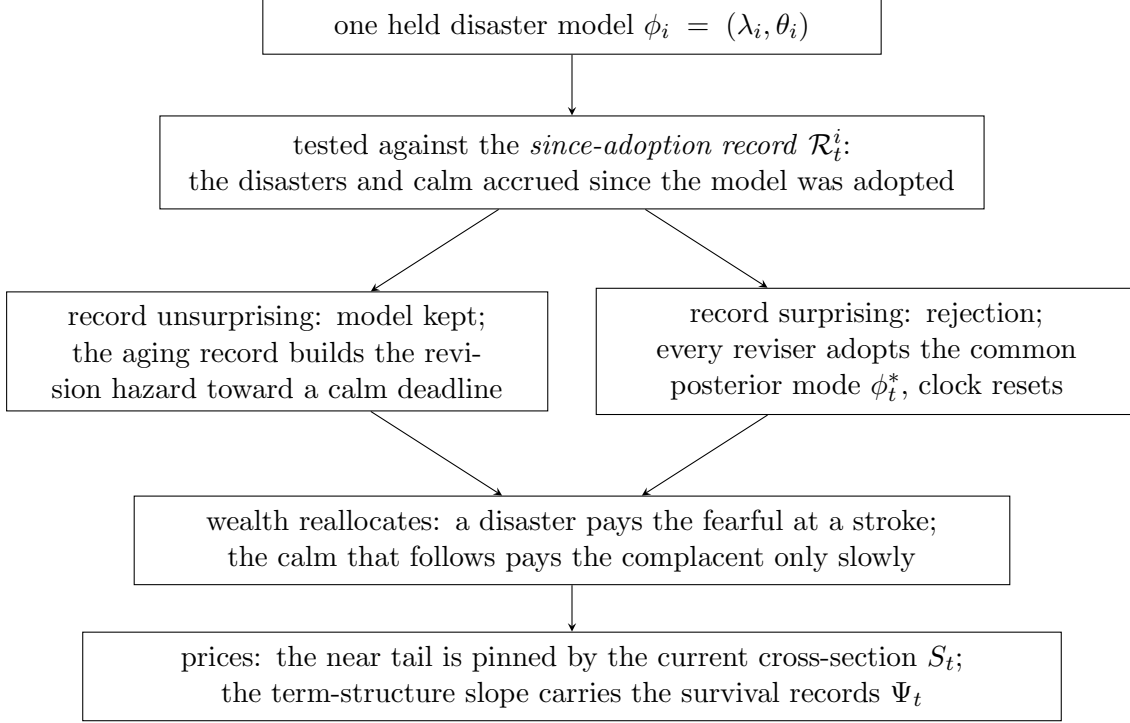


Figure 1: The mechanism. An investor holds one disaster model and tests it only on the record accrued since adoption. An unsurprising record keeps the belief and ages toward a scheduled calm deadline; a surprising one triggers revision to the common posterior mode. Wealth meanwhile reallocates toward whichever views the path vindicates. The near tail prices the current cross-section of surviving beliefs; the term-structure slope prices the records under which they survived—the coordinate no belief-sufficient model carries.

every belief-sufficient benchmark pins the entire surface to S_t , so the residual channel is present under lazy revision and identically zero across the class. What sets lazy revision apart is not a rival belief model but a property its pricing state lacks: sufficiency of the current cross-section.

The slope is the paper’s falsifiable claim, and it is open—the data have not yet tested it. The record’s imprint grows with maturity—two dates with the same near tail share the front of the curve and part only further out—so the test is a residual one: after controlling for the current option state, survival-record proxies—time since the last disaster, accumulated calm surprise, distance to a rejection boundary—should forecast the slope of the left-tail term structure, while any belief-sufficient model sets that residual to zero. The raw path-dependence documented in the surface runs the other way—strongest at the short end and *weakening* with maturity (Guyon and Lekeufack, 2023), the current-state-spanned part any such model

reproduces because S_t already encodes the recent path—so ours is the orthogonal residual that survives the control and *grows* as that raw effect fades. A nonzero residual rejects belief-sufficiency *given* the maintained spanning of the current option state (Remark 1), a class that contains the latent-intensity, Bayesian, filtering, and diagnostic accounts above but not every history-bearing alternative; attributing the residual to *testing* in particular rests on two signs no exogenous clock produces—calm deadlines that shorten in the held intensity, so the fearful revise down sooner, and a disaster-triggered *upward* re-anchoring wave arriving together with the wealth reweight. And the test can be run: unlike a latent disaster intensity—the disaster literature’s dark matter (Campbell, 2018; Chen et al., 2024), unobservable and inseparable from the belief that reads it—the coordinate that moves the forward tail is the survival record, a functional of the realized path given the maintained rule: time since the last disaster is read from the path alone, the remaining proxies are model-based reconstructions (Remark 1), and the restriction is testable where a latent-intensity account’s persistence is not.

Three things in this economy are shared with existing work, and we claim none of them. *The level.* A power mean of order below one is pulled toward the smaller likelihood ratios, so a cross-section right *on average* about how often disasters strike still prices the tail strictly below the representative-agent benchmark—dispersion alone is *complacency*, the shortfall growing in the spread of views and in risk aversion, and the inverted intensity $\lambda^* \kappa_t$ runs below the macroeconomic disaster frequency, the thinner far tail index options imply on average (Backus et al., 2011). A complete-market economy of heterogeneous *Bayesian* learners clears through the same aggregator and inherits the same complacency (Jouini and Napp, 2007; Cvitanić and Malamud, 2011); the far-tail gap, the investor-fears wedge of Bollerslev and Todorov (2011), and the unspanned tail factor of Andersen et al. (2015b) all sit at this front of the curve, the level any belief-sufficient model reproduces, and the disagreement-pricing content is the territory of Chen et al. (2012). *The wealth engine.* Selection—disasters paying the fearful at a stroke, calm paying the complacent slowly—is the market-selection mechanism of Blume and Easley (2006) and the price-impact logic of Kogan et al. (2006), and it is a functional of the current

cross-section, so it too lives on the shared side of the line. *The clock.* Models of costly attention and infrequent adjustment already place a time-since-last-action coordinate in the agent’s state (Abel et al., 2013; Bacchetta and van Wincoop, 2010), and belief revision by hypothesis test has been proposed as an expectations rule (Menzies and Zizzo, 2009); a staggered-clock economy is therefore also outside the belief-sufficient class, and the residual alone does not separate testing from it. What testing adds within the clock-bearing class are the two signs above: the deadline is not an exogenous interval but a surprise clock, shorter for more fearful models, and it is a *disaster* that fires the upward wave—content an attention cost does not supply and the data can check.

An individual belief does not drift—it stays fixed through calm and jumps only at a rejection—so the *held-model distribution* moves in episodic steps—synchronized across investors at a disaster, staggered through calm—while the wealth-weighted pricing state adds the continuous selection drift between them, and the disagreement driving it is manufactured from a common prior by that staggered timing rather than assumed. Two forces then move the multiplier—the power mean’s concavity sets how strongly each enters prices, market selection sets the direction: a disaster pays the investors who feared it, transferring wealth at a stroke toward the views that anticipated it and lifting the priced tail, while the calm that follows reverses this slowly, as an unsurprising record refutes the fearful only gradually and the mean held intensity declines at rate $(1/\alpha) \text{Var}_{S_t}(\lambda)$. The fear factor is therefore *fast in and slow out*—a sharp post-crisis spike eroding through calm. Left to itself, calm drives the cross-section toward agreement through three distinct channels—the selection drift erodes the mean held intensity at rate $\frac{1}{\alpha} \text{Var}_{S_t}(\lambda)$, the calm switch flow retires views as their deadlines arrive, and the survivors’ destinations converge as the public posterior concentrates—not because revision stops—a correct-on-average view is rejected and re-adopted forever (Lemma IA.2; Tegui 2026)—but because the survivors converge—so demographic turnover (Blanchard, 1985; Yaari, 1965; Gârleanu and Panageas, 2015) restores the dispersion and delivers a unique stationary law in which the cycle recurs. There the two opposed features of the tail become one: the thinner far tail on average is the

multiplier’s level through the long calm spells, the post-crisis spike the same level lifted at a stroke.

Two limits bound the interpretation. The calibration of Section 6 is illustrative—the belief block is pinned to a few stylized option-tail targets, for want of disagreement data on the option-relevant disaster intensity—and makes no out-of-sample claim. Throughout, Table II is the reduced-form *target* summary, the figure’s path values are *achieved* model moments, and the one reported mismatch—the half-life—is an achieved moment outside its reduced-form target, not a free parameter. And preferences are CRRA, not Epstein–Zin: the power-utility choice is what keeps the kernel and the bridge exact and confines the record to the forward surface—under recursive utility the record would move the near tail as well—at the cost of the usual smooth-rate/large-premium tension we do not resolve and a safe-rate level that is an artifact of the preference class. Where the intensity-channel sign is pinned a contrary cyclicity refutes the model; where the joint-channel condition is only flagged, that movement could be the condition instead.

Related literature. The paper adds no disaster process to rare-disaster asset pricing (Barro, 2006; Wachter, 2013): the physical law is fixed, and a heterogeneous, lazily revising cross-section moves the priced tail. From heterogeneous-belief pricing and market selection it borrows the consensus power-mean kernel and the wealth-reallocation dynamics, applies them to a disaster law learned only from rare events, and derives the disagreement endogenously. The broader theme is the one the companion single-agent paper develops (Tegui, 2026): a large literature lets rare events and slow learning generate *persistent*, belief-driven dynamics (Collin-Dufresne et al., 2016), but in every case the persistence is a property of the *belief*—the posterior is the pricing state, and two markets that share it share their forward surfaces. The closest neighbor, Dew-Becker et al. (2025), filters a latent value smoothly into a perceived intensity that is its own Markov pricing state, delivering the leverage effect and volatility dynamics we concede; ours needs Ψ_t , so the pricing state, not the shared aggregator, separates the two.

Against non-Bayesian revision (Ortoleva, 2012; Cho and Kasa, 2015) the contribution is the equilibrium asset-pricing consequences, not the rule; against price-extrapolation (Bastianello and Fontanier, 2026), in which traders mislearn *from prices*, our investors learn the disaster law from the rare-event record, with no price–belief feedback; against robust learning (Hansen and Sargent, 2010), whose countercyclical, history-dependent uncertainty premium is a single agent’s pessimistic *distortion* of a sufficient Bayesian posterior and stays Markov in it, ours is a wealth reallocation across *undistorted* heterogeneous beliefs, with the survival record and not the posterior as the state; and where Lanzani (2025) has confidence erode with unexplained evidence, our investors occupy the pole that commits to a single model and tests it. What the field has measured sits on the shared side of the line (Table I); the one directional prediction the level itself carries is that the inverted intensity $\lambda^*\kappa_t$ is countercyclical, below the macroeconomic frequency in calm and rising after disasters.¹

A growing literature makes *memory* a determinant of beliefs and prices: experienced outcomes are over-weighted and fade with the cohort (Malmendier and Nagel, 2011; Nagel and Xu, 2022), or are recalled associatively, cued by the present (Wachter and Kahana, 2019; Bordalo et al., 2020). The survival record is a different memory object. In those models memory moves prices by moving the belief about fundamentals—the recalled past lives inside the forecast—so memory and the forecast move together. Here the record moves the forward tail while forecasting no fundamental: market memory that prices without predicting, and that dissociation separates lazy revision from the memory models by the same test that separates it from filtering.

The paper proceeds as follows. Section 2 develops the mechanism in a two-date example: two updaters, two histories, one cross-section. Section 3 sets up the model—primitives, the revision rule, and the equilibrium kernel—and proves the bridge, the instantaneous prices visible in the near tail. Section 4 derives the synchronized belief dynamics, the endogenous emergence

¹A countercyclical *priced* intensity is consistent with the procyclical *physical* tail of Gormsen and Jensen (2025), who find the conditional return distribution more left-skewed in good times: their object is the physical distribution, ours the risk-neutral one, and the physical law here is fixed, so the countercyclicity is in what the market prices. Their finding is a challenge to disaster-based models we meet by reconciliation rather than a prediction we reproduce.

of disagreement, the revision hazard, and the law of motion of the pricing state, where the enlargement to Ψ_t is made precise. Section 5 prices the economy, separating the instantaneous prices that load on S_t from the forward-looking prices that load on Ψ_t , and assembles the fast-in/slow-out dynamics around a disaster. Section 6 adds demographic turnover, proves stationarity, and weighs the belief multiplier against the option-implied and macroeconomic evidence. Section 7 concludes; proofs, and a forecast-data signature of lazy revision, are in the appendices.

2. An example

Every disaster-pricing model in use shares one working assumption: today’s beliefs about disasters suffice to price tomorrow’s fear. This section builds, in two dates and one formula, a pair of economies that agree on every belief today and disagree about tomorrow—and locates the disagreement in a coordinate no belief-sufficient state can carry.

The priced object. Disasters arrive at the Poisson rate $\lambda^* = 3\%$ per year with sizes $J \sim F_J$; investors have CRRA utility with $\alpha = 3.5$ and entertain two rival models of the arrival rate, a complacent $\lambda_{lo} = 1\%$ and a fearful $\lambda_{hi} = 6\%$ —the physical and preference blocks of Table I. The simplest disaster claim pays one dollar at the first disaster inside a window; its price per unit of physical probability is the *risk-neutral* tail, which factors (Proposition 2) into a fixed risk-aversion markup $\mathbb{E}_f[e^{\alpha J}] = 2.82$ —common to every comparison below, so we drop it—times a belief multiplier. When a share s of wealth holds the fearful model, the multiplier is the intensity-channel form of Corollary 2,

$$\kappa = \left[s (\lambda_{hi}/\lambda^*)^{1/\alpha} + (1-s) (\lambda_{lo}/\lambda^*)^{1/\alpha} \right]^\alpha, \quad (1)$$

the wealth-weighted *power* mean of the held rates, not their average. With a quarter of wealth fearful, $\kappa_0 = 0.573$: the market prices the tail at 57 cents per benchmark dollar.

Two updaters. Hand the data to a Bayesian with an even prior over the two models—the section’s public prior. Each calm year multiplies her posterior odds on the fearful model by $e^{-(\lambda_{hi}-\lambda_{lo})} = e^{-0.05}$; five quiet years carry her weight on it from 0.500 to 0.438 and her predictive rate from 3.5% to 3.19%. Her belief *glides*—and, the operative property, her posterior is a sufficient statistic for her own future: tell me her belief today and I can price every claim at every horizon. Nothing else about her history matters, because nothing else is in her state.

The lazy investor of Definition 1 does not glide. She *holds* the fearful model and tracks one object, the record since she adopted it. Through calm the record is the quiet spell Δ , and her surprise is essentially the probability her own model assigns to a spell that quiet, $e^{-\lambda_{hi}\Delta}$ —a clock that drains deterministically. (The paper’s two-sided surprise adds the upper-count branch, and it is not small.) The clock crosses her tolerance ε at the *deadline*

$$\Delta^* \approx \frac{\log(1/\varepsilon)}{\lambda_{hi}},$$

the zero-count term of the exact criterion: at $\varepsilon = 0.65$ it alone would say 7.2 years, while the full two-sided surprise—adding the upper-count outcomes at least as deviant—sets the deadline at $\Delta^* = 11.9$ years. The exact date is the example’s clock; the display is its dominant term. The criterion is the model’s own (B4), which Proposition 7 proves deterministic and finite; nothing here approximates it. At the deadline she abandons the model and re-anchors at the posterior mode, which after any quiet spell is the *complacent* model (Lemma 4: the mode is non-increasing through calm).

Where the wealth goes. Beliefs are not the only thing that moves; wealth moves between believers, and the example tracks both engines. In calm the fearful investor is long insurance that never pays: her wealth bleeds to the complacent at the rate the held intensities differ, tempered by risk aversion. In the binary economy the fearful share obeys the replicator of Proposition 10,

$$d \log \frac{s_t}{1-s_t} = -\frac{\lambda_{hi} - \lambda_{lo}}{\alpha} dt = -1.43\% \text{ per year}, \quad (2)$$

so five quiet years carry s from 0.250 to 0.237 and the multiplier from 0.573 to 0.558—a smooth cent-and-a-half decline. At a disaster the insurance pays: wealth is reweighted by each model’s likelihood ratio, $s \mapsto s(\lambda_{hi}/\lambda^*)^{1/\alpha}/G$, jumping the fearful share from 0.250 to 0.357. The channel is smooth in calm and proportional at crashes—wealth never lurches on a schedule—and it is a functional of the current cross-section alone: hand me today’s s and I know today’s drift and today’s reweight exactly. A Bayesian economy has the *same* wealth dynamics; selection is the engine the benchmark shares, not the one that separates.

Up by surprise, down on schedule. At a crash the two engines fire the same direction in opposite gears: the reweight lifts the multiplier from 0.573 to 0.706 through wealth alone, while one disaster against five quiet years puts the posterior odds on the fearful model at 4.7 to 1, so the public mode flips and every investor whose updated record leaves the typical set re-anchors *upward* at once. Fear rises through the purse and through the mind together; the purse moves in proportion, the mind in lumps. In calm the engines also align—selection bleeds fearful wealth continuously, deadlines remove fearful believers in steps.

Two histories, one cross-section. Now build two tester economies identical today: in each, a quarter of wealth is fearful, so both price the spot tail at $\kappa_0 = 0.573$. They differ only in what today’s cross-section cannot see, the records. In *Aged*, the fearful cohort adopted its model eight years earlier, so it carries eight quiet years of record and its deadline arrives in year four ($\Delta^* - a \approx 3.9$). In *Fresh*, the cohort adopted at date 0 with a blank record; its deadline lies beyond the horizon. Price the five-year-forward tail claim—the tail over years five to six, conditional on calm: a calendar spread of deep out-of-the-money puts.

	spot κ_0	forward κ_5 (calm)	forward / spot
Aged	0.573	0.333	0.58
Fresh	0.573	0.558	0.97

The Aged decline decomposes along its own path: selection drifts the multiplier down 1.2 cents over the 3.9 years to the deadline, and the scheduled step carries the remaining 22.7; against

the Fresh counterfactual, the shared selection over the full five years accounts for 1.5 cents and the deadline for the 22.4-cent gap between the rows. Selection is the channel the two economies share while both run it; what separates the rows is beliefs re-anchoring, and nothing else.

Example Proposition. In the two-date binary economy, let Aged and Fresh share the cross-section s_0 and differ only in the fearful cohort’s adoption date. Then (i) the spot tails coincide, $\kappa_0 = \kappa'_0$; (ii) the calm-conditional forward tails differ, $\kappa_5 = \lambda_{lo}/\lambda^* < \kappa'_5$; and (iii) in any economy whose date-0 pricing state is the date-0 cross-section—the Bayesian of this section included—the two forwards coincide.

(Proof in Appendix A.)

The benchmark, exactly. Replace the testers by Bayesians and the comparison cannot even be posed. A Bayesian’s belief *is* her state; she carries no adoption date, no deadline, no record beyond what her posterior encodes—so “same beliefs, different histories” describes one economy, not two, and her forward is a single number pinned by the spot state. Nor is the wedge a wealth effect in disguise: the Bayesian economy runs the identical selection drift and the identical disaster reweight—the wealth channel is a functional of the current cross-section, common to every model in the benchmark class—so conditioning on the spot surface nets it out, in the table as in the data. What separates the tester from the Bayesian is not the level, which a Markov intensity model matches at will, but the state: the spot surface reads *what the market fears*; the records read *how long it has feared it*, and only the second prices the forward curve.

The example assumes what the model must deliver: that testing produces deterministic calm deadlines and a common re-selection destination; that the two-date split survives a continuum of tolerances, general records, disasters, and stationary demographic turnover; and that no finite statistic of the cross-section recovers what the records carry. The remainder of the paper shows that it does—the deadlines in Proposition 7, the common destination in Lemma 4, the split and its Markov benchmark in Theorem 1—and the residual test of Remark 1 conditions on

the spot surface to expose the residual in data. Beyond the Example Proposition’s arithmetic nothing here is proved, and nothing here claims what Theorem 1 does not; every number is the calibration of Table I.

3. The model and the disaster-fear bridge

We build the economy on one modeling choice: investors agree about everything except the law of disasters. The setting is a continuum-of-agents Lucas tree whose dividend carries Brownian risk and rare downward jumps drawn from a continuous size distribution. Investors share the diffusion—which the continuous record pins down on any interval—and dispute the disaster *law*. In the baseline they dispute its arrival rate λ , the size law held common; disagreement over the size index θ is a transparent extension, developed in closed form in the Internet Appendix. We carry the joint law $\phi = (\lambda, \theta)$ as the general object and read every result in the *intensity channel* ($f_\theta \equiv f$) unless the size extension is flagged. Every quantitative result is intensity-channel; size disagreement is what makes the pricing state infinite-dimensional (Proposition 3) and is otherwise developed in the Internet Appendix. The object to track is the cross-section of held ϕ and how it moves; every *instantaneous* price object—the kernel’s calm drift and disaster jump ratio, the short rate, the disaster-fear wedge—is a functional of it; the revision dynamics and switch flows read the record-resolved state. Why the disaster law, and not the diffusion, is the admissible thing to disagree about—the diffusion and the realized jumps are pinned down by the consumption path itself, whereas the intensity and the size law are learned only from the rare disasters as they accrue—is established in Appendix A.

A. The aggregate endowment

Fix a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ satisfying the usual conditions, with $\mathbb{P}$ the physical law. On it live a standard Brownian motion W , a Poisson process N of intensity λ^*

independent of W , and an i.i.d. sequence of disaster sizes $\{J_k\}_{k \geq 1}$ independent of (W, N) . Write $\{T_k\}_{k \geq 1}$ for the jump times of N and collect the disasters in the integer-valued random measure

$$\mu(dt, dx) = \sum_{k \geq 1} \delta_{(T_k, J_k)}(dt, dx)$$

on $[0, \infty) \times \mathcal{J}$, a marked Poisson process in the sense of Brémaud (1981).

Assumption 1 (Primitives of the endowment): *The drift $\mu_0 \in \mathbb{R}$, the diffusion coefficient $\sigma > 0$, and the disaster intensity $\lambda^* > 0$ are constants known to the modeler (the fixed physical law; μ_0 and σ are common knowledge among investors, while λ^* and F_J are known to the modeler alone—the objects investors dispute), and the size support $\mathcal{J} = [\underline{J}, \bar{J}] \subset (0, \infty)$ is a known compact interval. The disaster-size law F_J is a Borel probability measure on $\mathcal{J}$ with continuous, strictly positive Lebesgue density f . Aggregate consumption (the Lucas-tree dividend) $C_t > 0$ solves*

$$\frac{dC_t}{C_{t-}} = \mu_0 dt + \sigma dW_t - \int_{\mathcal{J}} (1 - e^{-x}) \mu(dt, dx), \quad C_0 > 0, \quad (3)$$

so that, in logs,

$$d \log C_t = \left(\mu_0 - \frac{1}{2} \sigma^2 \right) dt + \sigma dW_t - \int_{\mathcal{J}} x \mu(dt, dx) : \quad (4)$$

between disasters C is geometric Brownian, and at T_k it falls by $C_{T_k} = C_{T_k-} e^{-J_k}$, i.e. $\Delta \log C_{T_k} = -J_k$. The filtration $(\mathcal{F}_t)$ is the $\mathbb{P}$ -augmentation of the natural filtration of the path C .

Assumption 1 draws a deliberate line between what investors agree on and what they may dispute. Under $\mathbb{P}$ the disaster measure μ has compensator

$$\nu(dt, dx) = \lambda^* dt F_J(dx), \quad (5)$$

the arrival rate $\lambda^* dt$ times the size law $F_J(dx)$. Investors share the drift μ_0 , the diffusion σ , and the support $\mathcal{J}$, and agree that disasters arrive as a marked Poisson process; what they dispute is the disaster *law* itself—both how often disasters strike and how large they tend to

be. The reason the line falls here is informational: the diffusion is common knowledge because the continuous record pins it down on any interval in the high-frequency limit (Appendix A), whereas the disaster law is learned only from the rare disasters as they accrue.

B. The disputed object and the belief rule

An investor holds one dogmatic model of the disaster process: a pair $\phi = (\lambda, \theta)$ consisting of an arrival intensity λ and an index θ of the size law F_θ . Both are parameters of a distribution—the intensity of an arrival law, the index of a size law—and both are disputed. The true process has intensity λ^* and size law F_J ; neither is known to the investor, and her held pair is tested jointly—counts and sizes together—against its since-adoption record; the full public history enters only through the re-selection posterior.

The held model is a point $\phi = (\lambda, \theta)$ in the compact parameter set $\Phi := \Lambda \times \Theta$, with $\Lambda = [\underline{\lambda}, \bar{\lambda}] \subset (0, \infty)$ containing λ^* in its interior and $\Theta \subset \mathbb{R}$ a compact interval. Under ϕ the disaster process is a marked Poisson process with intensity λ and i.i.d. marks drawn from F_θ on $\mathcal{J}$, where $\{F_\theta : \theta \in \Theta\}$ has Lebesgue densities f_θ satisfying the identification and regularity conditions of Assumption 4 (Appendix A), verified there for the families used in the paper.

Two facts about the disputed object are used throughout. First, the held and physical laws are mutually absolutely continuous on every finite horizon (Lemma 2), so the likelihood ratios that drive revision are well posed. Second, the truth diverges from a held model ϕ at a per-period Kullback–Leibler rate $\mathcal{K}(\phi)$ (Definition 5) that splits additively into a timing part and a size part and is minimized at a unique pseudo-true model $\phi^* = (\lambda^*, \theta^*)$ (Lemma 3): the held intensity is asymptotically pinned to the truth λ^* , while whether the size family is correctly specified is left open and matters only in Section 4.

The belief rule is the dynamic hypothesis-testing rule of Ortoleva (2012), applied to the joint model $\phi = (\lambda, \theta)$; the dynamics it generates are derived in Section 4.

Definition 1 (Dynamic Ortoleva rule on the joint model): *All objects are adapted to the public filtration $(\mathcal{F}_t)$, the $\mathbb{P}$ -augmentation of the natural filtration of the path C (Assumption 1). Investor i has a permanent surprise threshold $\varepsilon^i \in (0, 1)$ and a prior over models: a second-order posterior ρ_t^i on the menu Φ , the Bayesian update of a prior ρ_0 over models by the likelihood of the entire marked-Poisson record on $[0, t]$,*

$$\rho_t^i(\phi) \propto \rho_0(\phi) e^{-\lambda t} \lambda^{N_t} \prod_{k: T_k \leq t} f_\theta(J_k), \quad \phi = (\lambda, \theta), \quad (6)$$

which is $\mathcal{F}_t$ -measurable and accumulates on the full history. Her held model $\phi_i(t) \in \Phi$ is piecewise constant, equal to the mode of her posterior ρ_t^i over models at her most recent adoption date and held until the next rejection. The rule runs as follows.

Step 0 (initial model—the best given the prior over models). *At $t = 0$ the investor adopts the maximum-a-posteriori model of the prior ρ_0 ,*

$$\phi_i(0) = \phi_0 \in \arg \max_{\phi \in \Phi} \rho_0(\phi), \quad a_0^i = 0,$$

the common starting model under the homogeneous start of §3.C. Write a_t^i for the most recent adoption date up to t and $\tau_t^i = t - a_t^i$ for the regime age.

Step 1 (the test—the since-adoption history against the held model). *The held model is tested only on the history accrued since it was adopted: the record on the window $(a_t^i, t]$,*

$$\mathcal{R}_t^i = (n_t^i, (J_k : a_t^i < T_k \leq t)), \quad n_t^i = N_t - N_{a_t^i}, \quad (7)$$

its count and realized sizes, both $\mathcal{F}_t$ -measurable.² Under the held model $\phi = (\lambda, \theta)$ —which makes the disaster process a marked Poisson process of intensity λ with i.i.d. marks F_θ (As-

²The record is recovered from the path by Lemma IA.1(ii), which rests only on Assumption 1 and is independent of the rule, so the forward reference is non-circular, and the since-birth count N_t enters only through n_t^i . Because the held model is time-homogeneous, the predictive law $Q_{\phi, \Delta}$ below depends on the window only through its length $\Delta = \tau_t^i$, so the test window is never empty once $\tau_t^i > 0$.

sumption 4)—a window of length Δ has predictive law $Q_{\phi,\Delta}$, the law of the window record under the held model's own measure $\mathbb{P}^\phi$, with density (against counting measure on the count and Lebesgue measure on the sizes)

$$q_{\phi,\Delta}(n, (x_1, \dots, x_n)) = \underbrace{e^{-\lambda\Delta} \frac{(\lambda\Delta)^n}{n!}}_{\text{count: Poisson}(\lambda\Delta)} \cdot \underbrace{\prod_{j=1}^n f_\theta(x_j)}_{\text{sizes: i.i.d. } F_\theta}. \quad (8)$$

Departure of the realized record from the held model is measured by the deviance—the saturated count deviance at the empirical rate n/Δ plus the size log-likelihood ratio of the record against its best in-family fit $\hat{\theta}$ (the size MLE on the realized sizes),

$$D_{\phi,\Delta}(\mathcal{R}) = \underbrace{\Delta d_P\left(\frac{n}{\Delta} \parallel \lambda\right)}_{\text{count}} + \underbrace{\sum_k \log \frac{f_{\hat{\theta}}(J_k)}{f_\theta(J_k)}}_{\text{size}}, \quad d_P(a \parallel \lambda) = a \log \frac{a}{\lambda} - a + \lambda, \quad (9)$$

the Poisson count deviance (with $d_P(0 \parallel \lambda) = \lambda$) plus the parametric size deviance, each non-negative and zero only at a record the held model fits exactly. The surprise of a record is its predictive deviance tail under $Q_{\phi,\Delta}$,

$$\bar{\varepsilon}(\mathcal{R}; \phi, \Delta) = Q_{\phi,\Delta}(\{\mathcal{R}' : D_{\phi,\Delta}(\mathcal{R}') \geq D_{\phi,\Delta}(\mathcal{R})\}). \quad (10)$$

The held model passes the test at t iff the realized record is unsurprising, $\bar{\varepsilon}(\mathcal{R}_t^i; \phi_i, \tau_t^i) > \varepsilon^i$ —equivalently, iff $\mathcal{R}_t^i$ lies in the mass- $(1 - \varepsilon^i)$ typical set, the acceptance region $A_{\tau_t^i}(\phi_i, \varepsilon^i) = \{\mathcal{R} : D_{\phi_i, \tau_t^i}(\mathcal{R}) < r_{\phi_i, \tau_t^i}(\varepsilon^i)\}$ with $r_{\phi_i, \tau_t^i}(\varepsilon)$ the $(1 - \varepsilon)$ -quantile of D_{ϕ_i, τ_t^i} under Q_{ϕ_i, τ_t^i} (at atoms of D under Q_{ϕ_i, τ_t^i} —the count atoms, the zero-count record included—the p -value rule $\bar{\varepsilon} > \varepsilon$ governs and the closed-threshold form is its shorthand; conventions preceding Definition 5).

Step 2 (keep, or update and re-select). As long as the held model passes, it is kept and only the regime age advances ($\dot{\tau}_t^i = 1$). It is rejected at the first instant the record leaves the typical set, $\tau^{\text{rev}} = \inf\{t > a^i : \bar{\varepsilon}(\mathcal{R}_t^i; \phi_i, \tau_t^i) \leq \varepsilon^i\}$, an $(\mathcal{F}_t)$ -stopping time. At a rejection the investor (i) updates the prior over models on the full history—the posterior ρ_t^i of (6) evaluated at that

instant—and (ii) re-selects the new model as the tie-broken mode of that posterior. Fix once and for all the lexicographic total order $\preceq$ on Φ : $(\lambda, \theta) \preceq (\lambda', \theta')$ iff $\lambda < \lambda'$, or $\lambda = \lambda'$ and $\theta \leq \theta'$. Let $\mathbf{m}(\rho) := \min_{\preceq} \arg \max_{\phi \in \Phi} \rho(\phi)$ denote the $\preceq$ -smallest maximiser; it exists and is unique because $\arg \max_{\phi \in \Phi} \rho$ is a non-empty compact subset of the compact Φ (the posterior is continuous on Φ), on which the lexicographic minimum is attained. The investor adopts

$$\phi'_i = \mathbf{m}(\rho_t^i) \in \arg \max_{\phi \in \Phi} \rho_t^i(\phi), \quad (11)$$

resets the adoption date to that instant (so τ^i restarts at 0), and returns to Step 1.³

The rule’s dynamic content is what the rest of the paper turns on. Because the tolerance is held *fixed*, even a correct-on-average model is rejected and re-adopted without end (Lemma IA.2): revision is recurrent, not terminal. The second-order posterior, updated on the full record and never reset, nonetheless concentrates the re-selected mode at the pseudo-true ϕ^* in the long run (Berk, 1966)—slowly, at the rare-disaster rate of Lemma IA.1—so recurrent rejection and a consolidating destination coexist, the trigger being frequentist and the re-selection Bayesian.⁴ The since-adoption window gives a held model a fresh sample on which to fail, and the surviving record—not the current belief—is what organizes the dynamics derived in Section 4.

C. The population, and a homogeneous start

A continuum of investors, indexed by $i \in [0, 1]$, watch the same path (3) and share everything in it but one thing. They share the undisputed primitives of Assumption 1—the drift μ_0 , the diffusion σ , the support $\mathcal{J}$ —and the menu of Assumption 4, but not the physical disaster law (λ^*, F_J) , which is known to the modeler and is precisely what they learn and may dispute. The

³The menu Φ permits re-selection in either direction in each coordinate, so no model is absorbing; since re-selection may return a model near the incumbent, the belief-moving event is the *switch* $\phi'_i \neq \phi_i$, not the rejection itself.

⁴Appendix A places the rule in the sequential-testing and learning literatures: the foregone-conclusion results that make fixed-tolerance rejection recurrent, the merging results on the consolidating side, and the sequential sufficiency of the since-adoption record.

one thing that distinguishes them is a *tolerance for surprise*.

Assumption 2 (One-dimensional heterogeneity; homogeneous start): *Each investor i has a surprise tolerance $\varepsilon^i \in (0, 1)$; the cross-section of tolerances is the primitive (non-random) continuum law G on $(0, 1)$, with continuous and strictly increasing G , so ε^i is investor i 's coordinate under G rather than an independent draw. This tolerance is her one permanent type. At $t = 0$ all investors share a common second-order prior $\rho_0 = \rho_0^\Lambda \otimes \rho_0^\Theta$ on $\Phi = \Lambda \times \Theta$, a product of a continuous prior on intensity and one on the size index, with full support and a continuously differentiable log-density; all hold the common model $\phi_0 = (\lambda_0, \theta_0) \in \arg \max_\phi \rho_0(\phi)$, differing only in ε^i .*

The environment and the disputable object are now fixed: a Lucas disaster economy in which investors share the diffusion and the occurrence of disasters but hold rival, identifiable laws $\phi = (\lambda, \theta)$ for the *intensity and size* of disasters, revised only when the rare-event record makes the held law surprising. We now ask what such a cross-section prices, deferring to Section 4 how otherwise-identical investors—differing only in tolerance—come to hold it and how it moves. The object to track is the *risk-neutral disaster-size law*—the distribution of disaster sizes priced into options, the empirical “investor fears” of Bollerslev and Todorov (2011). We build it in three steps: the change-of-measure density each held model induces and the equilibrium pricing kernel, a power-mean aggregate of those densities formed with the *fixed* Negishi multipliers (§3.D); and the risk-neutral disaster-size law it implies. That law, we show, is the physical law tilted by risk aversion and by a size-belief factor in which the *stochastic* consumption shares weight the held models’ likelihood ratios—a tilt that moves with the cross-section while the physical disaster process stays fixed (§3.E). Preferences are common CRRA and disaster risk is spanned, as in the heterogeneous-belief disaster economy of Chen et al. (2012); the departure is that the beliefs are lazy-Ortoleva held models, not Bayesian, and the dispersion is endogenous (Section 4).

D. The market and the pricing kernel

Each investor prices under her current held model $\phi_i(t-)$ (Definition 1); her belief is a change of measure from the truth with density M_t^i , the point-process likelihood ratio of Brémaud (1981, Ch. VI) (Definition 6).⁵

Assumption 3 (Market and preferences): *A continuum of investors $i \in [0, 1]$ have common CRRA utility $u(c) = c^{1-\alpha}/(1-\alpha)$, $\alpha > 0$ with $\alpha \neq 1$ (the log case $\alpha = 1$ is the continuous extension $u(c) = \log c$; the disaster results use $\alpha > 1$), and common discount rate $\delta > 0$. Markets are complete: investors trade a riskless bond, the Lucas tree (a claim to the aggregate endowment C of Assumption 1), and a complete set of disaster-contingent (Arrow) claims spanning the jump risk, so there is a unique common state-price density ξ_t (Cox and Huang, 1989; Karatzas et al., 1990). Budget (Negishi) multipliers $\zeta_i > 0$ satisfy $\int_0^1 \zeta_i^{-1/\alpha} di \in (0, \infty)$.*

(iv) Symmetric allocation. *We select the symmetric, equal-Pareto-weight equilibrium: $\zeta_i \equiv \zeta$ (normalized $\zeta = 1$) for every date-0 investor. The allocation is selected in consumption-share space; we do not impose equal initial financial wealth, which is inconsistent with a common ζ under heterogeneous beliefs (Remark 5). The constant ζ is fixed once, independently of the economy's realization: the same number in any two economies compared below. This selection is without loss for what the wedge delivers: its form, the sign and range of its magnitude, and its entire stationary law are invariant or robust to the Pareto weights (shown in the Internet Appendix); the weights fix only which cross-section enters the kernel at a date.*

⁵Between revisions the held model is fixed and the disaster process is marked Poisson with intensity λ_i and size law F_{θ_i} ; globally the subjective law is a marked point process with predictable compensator $\lambda_i(t-) F_{\theta_i(t-)}(dx) dt$ (Brémaud, 1981). Because the intensity is bounded and the per-disaster ratio is continuous and strictly positive on the compact $\mathcal{J} \times \Phi$, M^i is a strictly positive $(\mathcal{F}_t, \mathbb{P})$ -martingale with $\mathbb{E}[M_t^i] = 1$ (Appendix C). M^i is built from the realized held-model process—the density that makes M^i a $\mathbb{P}$ -martingale and feeds the Negishi construction of Appendix F. The valuation primitive is *sophisticated with respect to the rule*: the agent evaluates plans under the density built from her realized revision process, her own future revisions included, so the martingale method applies without qualification and her wealth at every revision is the equilibrium continuation value by definition. Lemma 1 shows that none of this sophistication is load-bearing for prices: on every inter-revision interval the realized path satisfies her naive Euler equation—the first-order condition of the currently held model treated as permanent—and those conditions, concatenated along the realized re-adoptions, support the state-price density for every claim; sophistication has bite only in the consumption level at each re-solve, and at logarithmic utility not even there.

The pricing weights here are *stochastic* consumption shares, so the share process is part of the state: the consumption-share-weighted cross-section S_t of held models, the wealth-weighted distribution whose emergence from otherwise-identical investors and law of motion are derived in Section 4.⁶

The kernel comes from individual optimization and market clearing. Each investor solves the standard complete-market problem, maximizing $\mathbb{E}_i \int_0^\infty e^{-\delta t} u(c_t^i) dt$ subject to the budget $\mathbb{E} \int_0^\infty \xi_t c_t^i dt \leq W_0^i$ —the objective taken under her subjective belief, the budget under the common state-price density ξ . Writing the subjective expectation as $\mathbb{E}_i[\cdot] = \mathbb{E}[M_t^i \cdot]$, the first-order condition $e^{-\delta t} u'(c_t^i) M_t^i = \zeta_i \xi_t$ equates discounted marginal utility under her belief to the shadow price of wealth, so with $u'(c) = c^{-\alpha}$ optimal consumption is $c_t^i \propto (M_t^i)^{1/\alpha}$: the reciprocal of risk aversion applied to her belief density. Market clearing $\int_0^1 c_t^i di = C_t$ then aggregates these individual demands across the cross-section into the single belief factor $\Xi_t = \int_0^1 \zeta_i^{-1/\alpha} (M_t^i)^{1/\alpha} di$ that the kernel carries—a *power mean* of the held densities, not their average. The next result records this equilibrium and its uniqueness.

Proposition 1 (Equilibrium kernel): *Under Assumptions 1–4, and the maintained integrability condition that lifetime utilities and present-value budgets are finite (implied by the transversality condition (T) of Section 5, $\delta > \sup_S m(S)$, where $m(S)$ is the equilibrium value-growth rate defined there), a competitive equilibrium (Definition 7) exists and is unique; its state-price density is*

$$\xi_t = e^{-\delta t} C_t^{-\alpha} \Xi_t^\alpha, \quad \Xi_t := \int_0^1 \zeta_i^{-1/\alpha} (M_t^i)^{1/\alpha} di \in (0, \infty), \quad (12)$$

and the equilibrium consumption shares are

$$s_t^i := \frac{c_t^i}{C_t} = \frac{\zeta_i^{-1/\alpha} (M_t^i)^{1/\alpha}}{\Xi_t}, \quad \int_0^1 s_t^i di = 1. \quad (13)$$

⁶This is the complete-market, heterogeneous-belief Lucas economy of Basak (2005); Jouini and Napp (2007) and, with disasters, Chen et al. (2012), with one difference: the pricing weights are stochastic consumption shares, which the fixed-weight consensus of Jouini and Napp (2007)—built on exogenous diffusion beliefs—cannot absorb.

The kernel's belief factor is the multiplier-weighted $\frac{1}{\alpha}$ -power mean Ξ_t^α of the densities, which is not the share-weighted arithmetic mean $\langle M \rangle_t := \int_0^1 s_t^i M_t^i di$ (the two coincide, up to a price-irrelevant constant, only when the cross-section is degenerate—all M_t^i equal). Under the symmetric allocation (Assumption 3(iv)) $\zeta_i \equiv \zeta$ cancels from (13) and rescales (12) by a constant, so $s_t^i = (M_t^i)^{1/\alpha} / \int_0^1 (M_t^j)^{1/\alpha} dj$.

Lemma 1 (Naive Euler support): *In the equilibrium of Proposition 1, fix investor i with revision times $\tau_0 = 0 < \tau_1 < \dots$, and on $[\tau_k, \tau_{k+1})$ write $M_{t \rightarrow s}^{(k)}$ for the likelihood ratio of the model $\phi_i(\tau_k)$ held there, treated as permanent. (i) (Naive Euler equations.) For every $\tau_k \leq t \leq s < \tau_{k+1}$ the realized path satisfies*

$$e^{-\delta(s-t)} \left(\frac{c_{i,s}}{c_{i,t}} \right)^{-\alpha} M_{t \rightarrow s}^{(k)} = \frac{\xi_s}{\xi_t}, \quad (14)$$

the first-order condition of the problem posed at τ_k under $\phi_i(\tau_k)$ held forever; and no consumption-portfolio deviation supported in $[\tau_k, \tau_{k+1})$ and self-financing at the market prices ξ raises the objective $\mathbb{E}_{\tau_k}^{(k)} \int e^{-\delta(s-\tau_k)} u(c_s) ds$. (ii) (SDF support.) Consequently ξ is determined interval-by-interval by naive first-order conditions, $\xi_s / \xi_{\tau_k} = e^{-\delta(s-\tau_k)} (c_{i,s} / c_{i,\tau_k})^{-\alpha} M_{\tau_k \rightarrow s}^{(k)}$ on $[\tau_k, \tau_{k+1})$, and their concatenation along the realized re-adoptions is the realized-density representation of Appendix C; every claim is priced by this ξ , with no investor's forecast of her own revisions entering the price system. (iii) (Levels.) Exact implementation of consumption levels by self-financing naive re-solving—each plan chosen from the market value of the previous plan's continuation—holds if and only if at every τ_k the market value of the frozen-model continuation equals that of the equilibrium continuation; at logarithmic utility ($\alpha = 1$) the equality holds identically, and for $\alpha \neq 1$ it is an additional restriction that equilibrium does not impose. In equilibrium, wealth at each revision equals the equilibrium continuation value—exactly the level at which the naive re-solve reproduces the path.

The lemma locates exactly what naivety delivers and what it does not. Between revisions the investor is behaviorally naive: her realized path satisfies the Euler equation of the model she currently holds treated as permanent, and no deviation she can finance and evaluate improves

it—so the state-price density, and with it every option and every calendar spread, is supported by naive first-order conditions concatenated along the realized re-adoptions. What naivety alone does not pin is the consumption *level* at each re-solve: a fully self-financing naive sequence prices its own frozen tail, not the equilibrium one, and the two values coincide only at the logarithmic knife-edge. In the equilibrium the gap does not open: the sophisticated investor’s wealth at each revision *is* the equilibrium continuation value, exactly the level the naive re-solve needs—the naive reading is exact in valuation and local optimality, the sophisticated one in levels—and she carries no hedge against her own revision; none is used below.

The kernel splits into an endowment term $e^{-\delta t} C_t^{-\alpha}$ —the representative-agent kernel of the true Lucas economy—times a belief factor Ξ_t^α that the lazy beliefs contribute. The level Ξ_t is a *fixed-weight* (Negishi) power mean of the densities M_t^i ; it moves not because the weights move but because the densities do, which is what makes the priced belief object time-varying even though every primitive is fixed. The state-dependent consumption shares s_t^i are the marginal pricing weights of its *local dynamics*—they govern the disaster jump ratio and the calm drift, not the level. Three statistics must be kept distinct: the head-count mean held model $\int \phi_i m_t$, the wealth-weighted mean $\int \phi_i s_t^i di$, and the priced power mean Ξ_t^α ; the kernel prices the last, and the first two are descriptive shadows. The proof (Appendix C) is the complete-market construction: the individual static optimum (martingale method, Cox and Huang, 1989), market clearing, and strict concavity for uniqueness.

E. The disaster-fear factor

Markets are complete, so there is a unique equivalent martingale measure $\mathbb{Q}$, with density process $Z_t = \xi_t \exp(\int_0^t r_s ds) / \xi_0$ (r the equilibrium short rate). Since the bank account is continuous, Z and ξ share the same jump ratio, and the risk-neutral disaster law is fixed by that jump ratio alone—the diffusive change of measure does not touch the size law.

Definition 2 (Risk-neutral disaster law, fear factor, and the investor-fears wedge): *Let, at a*

disaster of size x occurring at t ,

$$g_t(x) := \frac{\Xi_t}{\Xi_{t-}} \Big|_{J=x} = \int_0^1 s_{t-}^i r_i(x; t)^{1/\alpha} di = \int_0^1 s_{t-}^i \left(\frac{\lambda_i(t-) f_{\theta_i(t-)}(x)}{\lambda^* f(x)} \right)^{1/\alpha} di, \quad (15)$$

the share-weighted aggregator whose α -th power $g_t(x)^\alpha$ is the $(1/\alpha)$ -power mean of the held models' disaster likelihood ratios (the formulas below use g_t^α). The risk-neutral disaster compensator is $\nu^\mathbb{Q}(dt, dx) = Y_t(x) \nu(dt, dx)$ with $Y_t(x) = e^{\alpha x} g_t(x)^\alpha$, i.e.

$$\nu^\mathbb{Q}(dt, dx) = \lambda^* e^{\alpha x} g_t(x)^\alpha dt F_J(dx). \quad (16)$$

As a predictable compensator $\nu^\mathbb{Q}$ is evaluated at the left limit: g_t is built from the pre-jump shares s_{t-}^i above, so $g_t, Y_t, U_t, f_t^\mathbb{Q}$ enter jump compensators through their predictable $t-$ versions, while S_t (and g_t, U_t) without qualification denote the current post-jump state. At a disaster epoch the post-jump values are the reweighted-and-relocated ones; (17)–(18) hold at every t under this convention, compensators reading the $t-$ versions. Writing $\mathbb{E}_f[h] := \int_{\mathcal{J}} h(y) f(y) dy$ for expectation under the physical size law, its size density, intensity, and tail are

$$f_t^\mathbb{Q}(x) = \frac{e^{\alpha x} g_t(x)^\alpha f(x)}{\mathbb{E}_f[e^{\alpha J} g_t(J)^\alpha]}, \quad \lambda_t^\mathbb{Q} = \lambda^* \mathbb{E}_f[e^{\alpha J} g_t(J)^\alpha], \quad \bar{U}_t(x) := \int_x^{\bar{J}} f_t^\mathbb{Q}(y) dy. \quad (17)$$

The disaster-fear factor is the risk-neutral disaster intensity $U_t := \lambda_t^\mathbb{Q}$; the frictionless benchmark is its correct-belief value $U^{\text{RA}} := \lambda^* \mathbb{E}_f[e^{\alpha J}]$ (the value at $g_t \equiv 1$); and the investor-fears wedge is the gap from the physical rate,

$$U_t - \lambda^* = \lambda^* (\mathbb{E}_f[e^{\alpha J} g_t(J)^\alpha] - 1).$$

Finally the belief multiplier is the normalized fear factor,

$$\kappa_t := \frac{U_t}{U^{\text{RA}}} = \frac{\mathbb{E}_f[e^{\alpha J} g_t(J)^\alpha]}{\mathbb{E}_f[e^{\alpha J}]} \in (0, \infty),$$

the $e^{\alpha J}$ -weighted mean of g_t^α relative to its correct-belief value.

Proposition 2 (The disaster-fear bridge): *Under Assumptions 1–4, in the equilibrium of Proposition 1:*

- (i) (tilted law) *the risk-neutral disaster law (16)–(17) is the physical size density f tilted by the risk-aversion factor $e^{\alpha x}$ and the belief factor $g_t(x)^\alpha$, with fear factor $U_t = \lambda^* \mathbb{E}_f[e^{\alpha J} g_t(J)^\alpha]$;*
- (ii) (the bridge) *the fear factor factorizes into a preference benchmark U^{RA} and a belief multiplier κ_t ,*

$$U_t = U^{\text{RA}} \kappa_t, \quad U^{\text{RA}} = \lambda^* \mathbb{E}_f[e^{\alpha J}], \quad \kappa_t = \frac{\mathbb{E}_f[e^{\alpha J} g_t(J)^\alpha]}{\mathbb{E}_f[e^{\alpha J}]}; \quad (18)$$

- (iii) (belief-driven variation) *holding (α, λ^*, F_J) fixed, U^{RA} is constant, so*

$$d \log U_t = d \log \kappa_t, \quad (19)$$

and $f_t^\mathbb{Q}$, $\bar{U}_t$, and U_t are functionals of the belief aggregate g_t (15) alone.

The risk-neutral size law is the BT object (Bollerslev and Todorov, 2011; Breeden and Litzenberger, 1978): the physical F_J bent two ways—by risk aversion $e^{\alpha x}$, which fattens the tail because a representative agent dreads large disasters, and by the belief factor $g_t(x)^\alpha$, the only channel through which beliefs enter the price. That factor is a wealth-weighted *power* mean of the *joint* (intensity-and-size) disaster likelihood ratios—not their head-count or arithmetic mean, and not a mean of size beliefs alone, since intensity beliefs enter even when the size law is common. With (λ^*, F_J) fixed, $d \log U_t = d \log \kappa_t$: any movement in the multiplier maps one-for-one into the fear factor, while a self-re-adoption or offsetting cross-sectional change leaves it unmoved—price moves, physical quantity fixed.

U_t loads on the belief state g_t and on no diffusive quantity—volatility does not enter (17)—the structural counterpart of the unspanned left-tail factor of Andersen et al. (2015b), which

predicts equity and variance risk premia but not volatility; the wedge $U_t - \lambda^*$ is the Bollerslev and Todorov (2011) Investor Fears index (anchors structural, not identical). The smirk prices the pair $(U_t, f_t^{\mathbb{Q}})$, whose normalized shape $\bar{U}_t$ carries the size-dispute information a binary-intensity reading lacks. Only the *variation* of U_t is belief-driven—its level retains the fixed $e^{\alpha x}$ tilt—and these objects sit at the $\tau \rightarrow 0$ *front* of a surface.

Index options trade across *maturities*, and a disaster-tail claim of maturity τ prices the expected *path* of the risk-neutral disaster law over $[t, t + \tau]$, not the instant alone—the *forward fear factor*

$$F_t(\tau) := \mathbb{E}[U_{t+\tau} \mid \mathcal{F}_t] = U^{\text{RA}} \mathbb{E}[\kappa_{t+\tau} \mid \mathcal{F}_t], \quad \tau \geq 0, \quad (20)$$

the expected risk-neutral disaster intensity at horizon τ , whose origin is the bridge $F_t(0) = U_t = U^{\text{RA}} \kappa_t$. (A maturity- τ option prices the *risk-neutral* forward intensity $\mathbb{E}^{\mathbb{Q}}[U_{t+\tau} \mid \mathcal{F}_t]$, the option-implied forward tail risk of the affine surface literature (Andersen et al., 2015b); the physical $F_t(\tau)$ equals it at the origin and departs for $\tau > 0$ only by the forward-intensity risk premium, and the survival records enter both, so the term-structure slope’s dependence on Ψ_t below is invariant to the measure.) Away from the origin, with U^{RA} fixed, the curve moves with the expected forward multiplier $\mathbb{E}[\kappa_{t+\tau} \mid \mathcal{F}_t]$ —the path the cross-section is expected to trace, the belief *dynamics*—so the surface relates to beliefs at every maturity, not only the shortest. Its slope carries the survival *records* Ψ_t of the standing models, which S_t does not encode; the precise statement, and the proof that this record-dependence is absent for every Markov benchmark, is Theorem 1, once the dynamics of Section 4 are in place.

We work throughout in the *intensity channel*: the size law is common ($f_{\theta_i} = f$), so $r_i = \lambda_i / \lambda^*$ is x -free, the aggregate is the scalar $G_t = \mathbb{E}_{S_t}[(\lambda / \lambda^*)^{1/\alpha}]$ (Corollary 2), and the multiplier is $U_t / U^{\text{RA}} = G_t^\alpha$. This is the object the disaster-option literature inverts, and the setting of every quantitative result. Disagreement over the size index θ is a transparent extension—the aggregator g_t then carries the full joint object and the whole shape $f_t^{\mathbb{Q}}$ moves with the cross-section—which we develop in closed form in the Internet Appendix; the few statements that

invoke it are flagged.

The bridge carries the fixed physical law (λ^*, F_J) the macro-disaster literature measures (Barro, 2006) into the moving risk-neutral law $(U_t, f_t^{\mathbb{Q}})$ the option literature measures (Bollerslev and Todorov, 2011; Andersen et al., 2015b), with the belief aggregate g_t the only object on it; κ_t is the *ratio* of the two measurements, equivalently of the inverted intensity $\lambda^* \kappa_t$ to the physical λ^* . Two scope limits hold throughout: the deep disaster tail, not the frequent small jumps of the near-the-money surface; and a multi-year contraction idealized, as in Wachter (2013); Gabaix (2012), by an instantaneous jump.

The factorization’s invariance to the Pareto weights, and the robustness of the wedge’s sign, range, and stationary magnitude to that allocation, are collected in the Internet Appendix; Remark IA.2 delimits which belief processes move the factor and which make it flat.

The factorization $U_t = U^{\text{RA}} \kappa_t$ prices *any* cross-section of held beliefs; it does not say where that cross-section comes from or how it moves. How otherwise-identical investors come to hold dispersed models—synchronized revisions at disasters, staggered relief in calm—and the law of motion of its wealth-weighted distribution S_t , which turns this static tilt into the fast-in/slow-out fear *dynamics*, are the subject of Section 4; and the prices it implies are Section 5.

4. Belief dynamics and the fear-wave law of motion

We turn the revision rule into a law of motion for the wealth-weighted cross-section of held models. The section states three objects—the model each investor holds and tests, the hazard at which she abandons it, and the law of motion they induce on the cross-section—and then develops their implications. We introduce only what those statements require; the single-agent foundation of the hazard is the companion paper’s result (Tegui, 2026), and the deadline structure, the common-destination lemma, and the proofs are in Appendix B.

A. The test object, the revision hazard, and the law of motion

A *model of the world* is a single candidate disaster law $\phi = (\lambda, \theta) \in \Phi = \Lambda \times \Theta$ —an arrival rate and a size index—that an investor holds and tests against her *since-adoption record* $\mathcal{R}_t^i$ (7): the count n_t^i of disasters and their realized sizes accumulated since she last revised, read at regime age τ_t^i . She keeps ϕ while that record stays unsurprising under it and abandons it for the menu’s best alternative at a surprise; the re-selection target is the mode (11) of the public second-order posterior ρ_t (6), common across investors. Two facts about this object organize what follows, stated here and established in Appendix B. First, the disaster count is informative about the held *intensity*, so the test fires not only at a disaster but also in a long calm, when too few disasters have arrived to be consistent with the held λ_i . Second, the destination factorizes: because both the likelihood and the common prior separate across λ and θ , the joint re-selection mode splits into an intensity mode and a size mode taken separately (Lemma IA.1(iii)).

The kernel (12) and the bridge (18) aggregate held models by consumption share, so the object the law of motion governs is their wealth-weighted distribution. Three objects now appear together and are worth keeping distinct: the pricing state S_t , the wealth-weighted cross-section any belief model would carry; the public posterior ρ_t , the common destination a reviser re-selects to; and the record-resolved state Ψ_t , the cross-section together with the survival records—the coordinate that governs the forward surface and that S_t does not encode.

Definition 3 (The pricing state): *The pricing state S_t is the wealth-weighted law of held models: the probability measure on $\Phi = \Lambda \times \Theta$ with $\int_{\Phi} h dS_t = \int_0^1 h(\phi_i(t)) s_t^i di$ for every bounded h . Write $\bar{\lambda}_t^S := \int_{\Phi} \lambda dS_t = \int_0^1 s_t^i \lambda_i(t) di$ for the wealth-weighted mean held intensity.*

Definition 4 (The record-resolved state and the post-disaster map): *The record-resolved state Ψ_t is the wealth-weighted law of the full per-investor state: the probability measure on $\Phi \times \mathcal{R} \times [0, \infty) \times (0, 1)$ with*

$$\int h d\Psi_t = \int_0^1 h(\phi_i(t), \mathcal{R}_t^i, \tau_t^i, \varepsilon^i) s_t^i di$$

for every bounded h , where $(\mathcal{R}_t^i, \tau_t^i)$ are the since-adoption record and regime age of Definition 1, completed by the public posterior ρ_t —a deterministic functional of the full public record, common across investors—whose mode is the re-selection destination $\phi_t^* = \mathbf{m}(\rho_t)$. Its held-model marginal is the pricing state,

$$S_t = \pi_\phi(\Psi_t), \quad \pi_\phi : (\phi, \mathcal{R}, \tau, \varepsilon) \mapsto \phi.$$

At a disaster of size x , the post-disaster state Ψ_t^x is obtained from Ψ_{t-} by, in order, (a) the share reweight $s_{t-}^i \mapsto s_{t-}^i r_i(x; t)^{1/\alpha} / g_t(x)$ (D2); (b) the record update $\mathcal{R}^i \mapsto \mathcal{R}^i \oplus x$; (c) the test/rejection evaluation (D21); and (d) the relocation of every triggered switcher to the common destination ϕ_t^* (11).

Formally, Ψ_t is a finite Borel measure on the per-investor state space $\Sigma := \Phi \times \mathbf{R} \times \mathbb{R}_+ \times (0, 1)$ of tuples $(\phi, \mathcal{R}, \tau, \varepsilon)$, with $\mathbf{R} := \bigsqcup_{n \geq 0} \mathcal{J}^n$ the space of finite size records—a standard Borel space as a countable disjoint union of compacts; Σ carries the product σ -algebra, $S_t = \pi_\phi \Psi_t$ is the pushforward under the first coordinate, and every integral against Ψ_t runs over Σ .

The public posterior is carried alongside Ψ_t because it accumulates on the entire public history and is therefore *not* recovered from the since-adoption records inside Ψ_t . The instantaneous risk-neutral objects $\bar{\lambda}_t^S$, $g_t(\cdot)$, U_t , and $f_t^\mathbb{Q}$ read only the marginal S_t ; the forward-looking valuation of Section 5 loads on the full pair, and we write $v(\Psi_t)$ and $\mathcal{A}_\Psi$ for $v(\Psi_t, \rho_t)$ and $\mathcal{A}_{\Psi, \rho}$, displaying ρ_t only where it does analytic work—the calm boundary map of Lemma 6 and the post-disaster relocation (d) above.

The rate at which an investor abandons her model is the *revision hazard*: the first passage of the deviance $D_{\phi, \tau}(\mathcal{R})$ (9) across its acceptance threshold (Propositions 7 and 11). Between disasters, with the record fixed, the deviance drifts deterministically and meets the threshold at a scheduled *calm deadline* $\Delta^*(\phi, \varepsilon)$ (B4)—a regime age at which a disaster-free record is rejected for a process expected to deliver disasters; a disaster appends a mark and triggers

rejection when the updated record leaves the typical set. The single-agent foundation—that the surviving record, not the current belief, is the sufficient state for this hazard—is the companion paper’s result (Tegua, 2026); the record-resolved passage law, which does not close in general, is in Appendix D. In the baseline ($f_\theta \equiv f$, Corollary 2) the deviance depends on the record through the count alone, and the hazard closes in the low-dimensional count-and-age state.

Between revisions the cross-section drifts by *selection* (Proposition 10): in calm the share of a below-average-intensity investor grows and that of an above-average one shrinks, while a disaster rewards whoever held a model that fits it—the same selection a Barro (2006)-style truth would impose, here on *held* models. This is the wealth channel that turns the static tilt of Proposition 2 into a moving one. The revisions themselves enter through two flows the hazard defines: a staggered *calm switch flow*, as tolerance-indexed deadlines arrive and each reviser relocates to the common destination $\phi_t^* = \mathbf{m}(\rho_t)$ (Lemma 4), and a synchronized *disaster jump*, at which every triggered investor relocates to that same destination at once (Proposition 8). The law of motion collects the three.

Proposition 3 (Law of motion of the pricing state): *In the equilibrium of Proposition 1, for every bounded h on Φ the pricing state evolves by:*

(i) (selection drift — continuous, calm)

$$\left. \frac{d}{dt} \langle h, S_t \rangle \right|_{\text{sel}} = \frac{1}{\alpha} \int_{\Phi} (\bar{\lambda}_t^S - \lambda) h(\phi) S_t(d\phi); \quad (21)$$

(ii) (calm switch flow — calm) *the staggered calm-deadline revisers leave their held models and arrive at the common destination $\phi_t^* = \mathbf{m}(\rho_t)$, the same for every investor revising at t (Lemma 4); aggregated, they contribute*

$$\left. \frac{d}{dt} \langle h, S_t \rangle \right|_{\text{sw}} = \int_{\Phi} [h(\phi_t^*) - h(\phi)] \Phi_t^{\text{sw}}(d\phi), \quad (22)$$

with Φ_t^{sw} the calm switch flow (D30) of Lemma 5—finite-variation, and continuous where

the calm-deadline law is atomless; on a flat deadline level set of positive G -mass it carries a predictable finite-variation jump, which by Lemma 6 leaves every traded price continuous; disaster-triggered switches are not part of this term and enter only at (iii);

(iii) (disaster jump — at disaster epochs) at a disaster of size x the state jumps in two steps—the marginal of the post-disaster map Ψ_t^x of Definition 4. First the shares reweight (Proposition 10(ii)),

$$\tilde{S}_t^x(d\phi) = \frac{r(\phi, x)^{1/\alpha}}{g_t(x)} S_{t-}(d\phi), \quad r(\phi, x) = \frac{\lambda f_\theta(x)}{\lambda^* f(x)}; \quad (23)$$

then every triggered switcher relocates to the common destination ϕ_t^* with record and age reset, $(\mathcal{R}, \tau) \rightarrow (\emptyset, 0)$; the final S_t is the held-model marginal of the relocated record-resolved state Ψ_t^x —the relocation acts on Ψ , not on Φ . The triggered set is read on the record-resolved state Ψ_{t-} , not on Φ so S_t is not a partition of S_{t-} over Φ , and (23) equals S_t only when the triggered set is empty. Both steps are a single jump at the disaster epoch, not a continuous rate; the continuous calm drift is the sum of (i) and (ii).

Under size disagreement the disaster tilt loads on S_t through the aggregator

$$x \mapsto g_t(x) = \int_{\Phi} \left(\frac{\lambda}{\lambda^*} \right)^{1/\alpha} \left(\frac{f_\theta(x)}{f(x)} \right)^{1/\alpha} S_t(d\phi), \quad (24)$$

which by the spanning condition (Assumption 4(v)) no finite collection of linear moments of S_t determines; the instantaneous risk-neutral law $(U_t, f_t^{\mathbb{Q}})$ is therefore infinite-dimensional in S_t .⁷

The triggered set is genuinely record-level: investors sharing a held model ϕ may carry different $(\mathcal{R}, \tau, \varepsilon)$ and so need not trigger together.

The derivation—the deadline structure, the finiteness of the switch flow, and the price-continuity of its predictable jumps—is in Appendix B.

⁷This rules out any finite collection of *linear* moments; the reduction is in fact tighter—no *continuous* finite-dimensional statistic of S_t is sufficient for $(U_t, f_t^{\mathbb{Q}})$ under size disagreement (Proposition IA.1, Appendix D, by Brouwer’s invariance of domain). The continuity qualifier is essential: in the weak topology $\mathcal{P}(\Phi)$ is Polish, so a measurable bijection furnishes a discontinuous one-dimensional sufficient statistic.

B. What the law of motion implies

The three forces give the wedge its *fast-in, slow-out* shape. Only the disaster jump is synchronized—all triggered switchers land on the single common mode ϕ_t^* (Lemma 4), so the repricing is a coordinated wave rather than idiosyncratic noise—and it moves $\bar{\lambda}_t^S$ up at one instant through the reweight (E42)—signed in the intensity channel, and whenever λ is positively associated with $r(\cdot, x)^{1/\alpha}$; the relocation adds under the complacent-switcher condition stated in Proposition 6. Selection and the calm switch flow are its staggered relaxation, both pushing $\bar{\lambda}_t^S$ down in calm (the switch flow under monotone-reselection)—a finite-history regime: the condition holds once the public mode has followed the calm evidence down, while in the locked stationary limit with a fearful projection the switch flow turns non-negative and the relaxation is selection and turnover—, only as fast as the surviving disagreement that selection consumes, spread across the population by the deadline dispersion. These signs are the wave priced in Section 5.C; the equations fix only directions, while the wedge’s magnitude waits on the stationary cross-section of Section 6.

The law of motion is not autonomous in the cross-section. The instantaneous tail reduces to the scalar G_t in the baseline (Corollary 2), but the switch flow and its destination read the record-resolved Ψ_t , so the *forward* surface depends on Ψ_t even where the instantaneous tail does not—the enlargement the rest of the paper prices, and the coordinate the term-structure test isolates.

A last implication is that the dispersion the cross-section carries is manufactured, not assumed: investors identical in priors, data, preferences, and initial model—heterogeneous only in their surprise tolerances—watching one path *become* a non-degenerate cross-section.

Proposition 4 (Endogenous emergence of disagreement): *Start from the homogeneous configuration of Assumption 2: at $t = 0$ every investor holds the common model ϕ_0 , so $S_0 = \delta_{\phi_0}$ and $\text{Var}_{S_0}(\lambda) = 0$. Disagreement is then generated by the rule, not assumed: there is a finite $t > 0$ with $\text{Var}_{S_t}(\lambda) > 0$ in the consumption-share measure S_t of Definition 3 (a positive- G -mass split*

transfers to positive S_t -mass by the absolute continuity established in the proof) whenever either

- (a) (disaster split) *the first disaster T_k precedes the earliest calm deadline—so the population is still homogeneous at T_k^- ($S_{T_k^-} = \delta_{\phi_0}$) and the post-disaster surprise of the updated one-count record is the operative trigger—the rejecting cohort $\{\varepsilon^i \geq \bar{\varepsilon}(\mathcal{R}_{T_k}; \phi_0, T_k)\}$ —the post-disaster surprise of the one-count record over $(0, T_k]$ — is a strict subset of positive G -mass of the population, and the common destination differs from the incumbent in intensity, $\lambda(\phi^{(k)}) \neq \lambda(\phi_0)$ (a size-only move, λ unchanged, leaves $\text{Var}_S(\lambda) = 0$); or*
- (b) (staggered calm) *the calm-deadline map $\varepsilon \mapsto \Delta^*(\phi_0, \varepsilon)$ is non-constant on a positive- G -mass set, the calm intensity mode $\tau \mapsto \lambda_\tau^m$ is strictly decreasing (Remark 4), and the realized path stays disaster-free long enough for two positive- G -mass threshold groups to reach their distinct deadlines (so the staggered revisions fire before a disaster resets the count).*

The disaster channel (a) needs only a partial rejection with a moved posterior mode; the calm channel (b) needs the regularity of Remark 4. The point is sharp against the merging benchmark: common priors updated on common data converge—predictive laws merge (Blackwell and Dubins, 1962) and rational learners come to agree (Kalai and Lehrer, 1993)—so persistent disagreement ordinarily needs heterogeneous priors or private signals. Here it needs neither: identical priors, identical data, and a single non-Bayesian rule whose staggered rejection timing holds the cross-section apart at any finite date. Long-run persistence is a separate matter: without demographic refresh the survivors eventually agree (Proposition 17(ii)), and the non-degenerate stationary cross-section is supplied by the turnover economy (Propositions 16–17). With common data the population is the exact deterministic transport of the tolerance distribution along the path—the cross-section the image of G , not a continuum average. That is the finite-horizon statement: dispersion emerges at a date from a homogeneous start. Left to run, the survivors’ convergence re-concentrates it; the *stationary* dispersion the quantitative sections use is the turnover object of Proposition 17.

5. Prices and the dynamics of the wedge

The kernel of Proposition 1 prices every claim and the fear factor of Proposition 2 identifies its disaster-tail component, so we can read the wedge off observable prices. We proceed in two steps: first the economy at a point in time—short rate, price–dividend ratio, disaster premium—then how those prices move as beliefs revise, the fast-in, slow-out *time series of the wedge*. The static results are state-contingent, valid for any cross-section and free of demographic assumptions; only unconditional averages await the stationary primitive of Remark 6. One distinction organizes the section: the prices split into a part any Markov pricing state already delivers and a part that requires the survival history behind it, which we record once as a decomposition every pricing result inherits.

A. The benchmark decomposition

Write S_t for the current pricing cross-section (Definition 3) and Ψ_t for the record-resolved state that also carries the since-adoption records, ages, and revision hazards (Definition 4), with $S_t = \pi_\phi \Psi_t$ its held-model marginal.

Proposition 5 (The benchmark decomposition: current cross-section versus survival history):
In the intensity channel, every equilibrium price object F_t admits an orthogonal decomposition into a part carried by the current cross-section and a part carried by the survival history,

$$F_t = \underbrace{B_t(S_t)}_{\text{current cross-section}} + \underbrace{\mathcal{H}_t(\Psi_t)}_{\text{survival-history residual}}, \quad B_t := \mathbb{E}[F_t \mid S_t], \quad \mathbb{E}[\mathcal{H}_t \mid S_t] = 0, \quad (25)$$

with B_t a functional of S_t and the residual $\mathcal{H}_t = F_t - \mathbb{E}[F_t \mid S_t]$ a functional of Ψ_t orthogonal to S_t .

- (i) (Instantaneous prices carry no residual.) *The fear factor U_t and multiplier κ_t , the risk-neutral disaster law, and the short rate r_t are functionals of S_t alone (Proposition 15); for*

each, $B_t = F_t$ and $\mathcal{H}_t \equiv 0$.

- (ii) (Forward-looking prices carry a residual that grows with horizon.) *The price–dividend ratio $v(\Psi_t)$, the level of the disaster premium, and the forward fear factor $F_t(s) = \mathbb{E}[U_{t+s} \mid \mathcal{F}_t]$ at horizons $s > 0$ have $\mathcal{H}_t \not\equiv 0$: the price–dividend ratio and the premium level carry it at date t itself, while the forward fear factor’s residual is zero at horizon zero and first order in s , its slope loading on the survival-history coordinates of Ψ_t —the calm deadline-proximity flow Φ_t^{sw} and the disaster-trigger propensity π —and it accumulates with horizon (Propositions 13 and 1).*
- (iii) (The residual is the lazy-revision signature.) *For any Markov benchmark—a model whose pricing state is Markov in the current cross-section, the belief-sufficient class of the Introduction—every forward object is a functional of that state, so $\mathcal{H}_t \equiv 0$. The residual is therefore present for any disaster law under lazy revision and absent across the belief-sufficient Markov class.*

A heterogeneous Bayesian economy is one such benchmark, because each investor’s posterior is a sufficient statistic for her own future belief and the wealth-weighted cross-section is therefore itself Markov; an affine latent-state specification and a representative filtering economy are others (the single-agent Bayesian case, where the multiplier itself merges to a constant, is Internet Appendix IA.9). It is exactly what the term-structure test isolates (Remark 1): conditioning on the current option state removes $B_t(S_t)$ and leaves $\mathcal{H}_t(\Psi_t)$.

The instantaneous price objects—the short rate, the price–dividend ratio, and the disaster premium—are placed in this decomposition and derived in Appendix E: the short rate and the risk-neutral tail carry no residual, while the price–dividend ratio and the premium level inherit the survival-history residual through $v(\Psi_t)$.

B. The term structure of the tail, and what the measured objects identify

Proposition 15 says the instantaneous tail is pinned by S_t and the forward-looking objects are not. The next result makes that gap operational. It locates exactly where the record-resolved state enters the *term structure* of the risk-neutral tail, gives the channels their signs, and shows the dependence is a property of the revision rule rather than of the disaster law—absent for any Markov benchmark.

Theorem 1 (Forward tail: level on S_t , term-structure slope on Ψ_t): *Suppose the age structure of Ψ_t admits a density, so the calm flow of held models reaching their deadline $\Delta^*(\phi, \varepsilon)$ (B4) at t is a finite measure Φ_t^{sw} (Proposition 6). The forward fear factor $F_t(s)$ of (20) decomposes as follows. In the intensity channel:*

- (i) (Level on S_t .) $F_t(0) = U_t = U^{\text{RA}} \kappa_t$ is a functional of the pricing state S_t alone (Proposition 2, Proposition 15).
- (ii) (Slope splits, into two history channels.) *The right-hand slope at the origin decomposes into a part measurable in S_t and a part measurable in Ψ_t but not in S_t ,*

$$\partial_s^+ F_t(0) = \underbrace{U_t [-b_{\text{sel}}(S_t)] + \lambda^* \Theta_{\text{rw}}(S_t)}_{\text{selection drift and disaster reweight: } b_S(S_t)} + \underbrace{U_t c_{\text{sw}}(\Phi_t^{\text{sw}})}_{\text{calm re-leveling}} + \underbrace{\lambda^* \Theta_{\text{sw}}(\Psi_t)}_{\text{disaster-triggered re-leveling}},$$

where $b_{\text{sel}}(S_t) \geq 0$ is the calm selection drift of (28) and $\Theta_{\text{rw}}(S_t) := \int_{\mathcal{J}} (U(\tilde{S}_t^x) - U_t) f(x) dx$ the compensator of the reweight-only fear jump both functionals of S_t through g_t . The two Ψ_t -channels are the staggered revisions: the calm flow

$$c_{\text{sw}}(\Phi_t^{\text{sw}}) = \frac{\alpha \mathbb{E}_f[e^{\alpha J} g_t(J)^{\alpha-1} \int_{\Phi} (r(\phi_t^*, J)^{1/\alpha} - r(\phi, J)^{1/\alpha}) \Phi_t^{\text{sw}}(d\phi)]}{\mathbb{E}_f[e^{\alpha J} g_t(J)^{\alpha}]},$$

non-positive under monotone-reselection; and $\lambda^* \Theta_{\text{sw}}(\Psi_t)$, where Θ_{sw} is the per-disaster, mark-averaged relocation increment of U —so $\lambda^* \Theta_{\text{sw}}$ is the compensator of the relocations

fired at the record-dependent rate $\lambda^* \pi(\phi, \tau, \mathcal{R}; \varepsilon)$ (D21), whose destination intensity is non-decreasing in the disaster count (Proposition 6(ii)). Under monotone re-selection $c_{\text{sw}} \leq 0$, and under the complacent-switcher condition of Proposition 6 (for $\alpha \geq 1$, or its likelihood-ratio form) also $\Theta_{\text{sw}} \geq 0$; the theorem leaves both unsigned. Both Φ_t^{sw} and the trigger propensity π are functionals of the record-resolved state Ψ_t and not of its marginal S_t .

(iii) (Two histories, one cross-section.) For record-resolved states Ψ_t, Ψ'_t with a common held-model marginal $\pi_\phi \Psi_t = \pi_\phi \Psi'_t = S_t$, the forward curves share their level and their $b_S(S_t)$ -slope, $F_t(0) = F'_t(0) = U_t$; they differ only through the two staggered-revision channels. Conditional on no disaster over the horizon—write $F_t^0(s) := \mathbb{E}[U_{t+s} \mid \mathcal{F}_t, N_{(t,t+s]} = 0]$ for the calm-conditional curve—only the calm channel operates and the state whose calm channel is larger in the weighted sense $c_{\text{sw}}(\Phi_t^{\text{sw}}) \leq c_{\text{sw}}(\Phi'_t^{\text{sw}})$ —for instance a dominating deadline flow with the common destination—carries, under monotone re-selection, the weakly steeper forward-tail decline, $\partial_s^+ F_t^{0'}(0) \leq \partial_s^+ F_t^0(0)$; under the theorem's two sign conditions the channels are oppositely signed, and the net is a cross-sectional condition on the balance of calm-deadline proximity against disaster-trigger propensity. The wedge $F_t(s) - F'_t(s)$ is zero at $s = 0$ and first order in s ; its full-horizon law is Markov in Ψ_t , not autonomous in S_t (Proposition 13).

(iv) (No Markov-state counterpart.) Replace lazy revision by any Markov benchmark—a model whose pricing state is Markov in the current cross-section. The slope's dependence on survival history is a property of the revision rule, present for any disaster law and absent across the belief-sufficient Markov class—models whose pricing state is Markov in a current belief state;

Three remarks on the statement. The forward fear factor is the expected future risk-neutral disaster intensity at horizon s , the term structure that the deep out-of-the-money tail traces across maturities. The compensator Θ_{rw} is the change in U produced by the size- x reweight (23) of the aggregator; the sign of c_{sw} under monotone re-selection is scheduled relief (Proposition 7:

the calm destination intensity is non-increasing), the condition asking that the calm destination sit weakly below every calm switcher’s held intensity. The theorem holds in the sophisticated equilibrium and rests on none of its sophistication: the price system is supported interval-by-interval by naive first-order conditions (Lemma 1)—anticipation of one’s own revisions enters consumption levels, never prices—and the record reaches the forward curve through the realized-density representation, not through any investor’s foresight. In part (iv), the worked instance is heterogeneous Bayesian learning of the disaster law: each posterior is a sufficient statistic for its own future, so the wealth-weighted cross-section is itself Markov; there is no deadline and no trigger threshold, hence neither Φ_t^{sw} nor Θ_{sw} , and $\partial_s^+ F_t(0)$ is measurable in S_t . The forward curve is then pinned by the current cross-section at every horizon (Internet Appendix IA.9); an affine latent-state model and a representative filtering economy give the same conclusion through their own Markov states. And a benchmark that enlarges its state with ages, clocks, or records is history-bearing by construction and outside the class the comparison targets.

Remark 1 (The observable signature, and what it is not). Theorem 1 yields an over-identifying restriction on the left-tail surface, and fixes the maturity at which it bites. The level of the surface—the short-dated risk-neutral tail—is pinned by the current cross-section S_t , so once one conditions on it the age structure is irrelevant at the short end; the dependence on survival history enters the *slope* and accumulates with maturity. The test is therefore a residual one: regress a forward-tail slope (a long-minus-short tail measure) on variables proxying the two staggered-revision channels—time since the last disaster, accumulated calm surprise, or an estimate of the distance-to-rejection $\Delta^* - \tau$ —after controlling for the current option state (the short-dated tail block: level, volatility, skew), realized volatility, and current skew. The rule predicts a non-zero residual loading, strengthening with maturity, that any Markov benchmark sets to zero. The record-resolved coordinate is a distribution over records, ages, and tolerances, and the three proxies span its two named channels—calm-deadline proximity and disaster-trigger propensity—not the state; a nonzero residual loading is evidence for the mechanism, while a null rejects the proxied channels rather than the class distinction itself. The

class the residual rejects is belief-sufficiency, not every history-bearing alternative: an economy of staggered adjusters on exogenous clocks (Abel et al., 2013; Bacchetta and van Wincoop, 2010) also escapes the class and would also carry a slope residual. Attributing the residual to *testing* rests on two signs no exogenous clock produces: calm deadlines that *shorten in the held intensity*, so the fearful revise down sooner, and a disaster-fired *upward* re-anchoring wave arriving together with the wealth reweight. What is over-identifying is the restriction that the residual exists and strengthens with maturity, not a claim that three regressors exhaust it.

This residual is a finer object than the raw path-dependence of the implied-volatility surface (Guyon and Lekeufack, 2023; Andr s et al., 2024), and the two must not be conflated. That literature regresses implied volatility on the past return path *without* conditioning on the current pricing state, and finds the explanatory power strongest at short maturities and weakening as maturity grows. In the present model that raw pattern is reproduced through S_t , which itself encodes the recent path—the short-dated tail is the tightest function of S_t , while longer maturities average over future selection and turnover that revert toward the stationary cross-section—and it is *generic*: any Markov benchmark also encodes the past. The rule’s signature is the orthogonal residual, history that still moves the forward-tail slope after the current state is removed, which Theorem 1 shows is present under lazy revision and absent across the Markov class. The conditioning set carries an identification assumption: in the binary-menu intensity channel the spot tail level identifies S_t exactly, so short-dated tail controls span the S_t -block; with richer menus those controls are structural filters for S_t rather than exact replacements, and the maintained assumption is that they span the S_t -measurable part of the slope—the reading under which residual loading on record proxies falsifies belief-sufficiency rather than omitted current-state components.

Frequency. The record clock here is macroeconomic: at $\lambda^* \approx 3\%$ a disaster arrives once a generation, while option samples span decades. The bridge is that the rule is not tied to one clock. An investor testing a crash model of the equity tail runs the same deadline construction

with the crash arrival rate in place of λ^* , and records, deadlines, triggers, and destinations carry over to the higher-frequency event set; at those frequencies the proxies turn over within available option histories. The macro calibration disciplines the mechanism's magnitudes; an international disaster panel supplies the cross-sectional power the single-country clock lacks.

Remark 2 (The option-implied forward object). The surface prices the risk-neutral forward tail, $F_t^{\mathbb{Q}}(s)$. Under the kernel of Proposition 1 the $\mathbb{Q}$ -compensator of disasters is $\lambda^* e^{\alpha x} g_{t-}(x)^\alpha f(x) dx dt$ —arrival intensity $\lambda_t^{\mathbb{Q}} = \lambda^* \mathbb{E}_f[e^{\alpha J} g_{t-}(J)^\alpha] = U_{t-}$, size density the normalized (17)—an S_{t-} -measurable tilt of arrival and size. Applying the tilt to the generator split behind (E58) therefore changes only the S_t -measurable loadings: the calm flow Φ_t^{sw} is untouched and the trigger set is unchanged pathwise; the compensator of triggered relocations integrates that set against the tilted mark law, so the numerical propensity changes while its record dependence does not—the slope's two staggered-revision channels remain functionals of Ψ_t beyond S_t under $\mathbb{Q}$ exactly as under $\mathbb{P}$. The residual test of Remark 1 thus applies to the option-implied forward surface itself, not only to its physical forecast.

Corollary 1 (The priced calendar spread): *Fix $0 < s_1 < s_2$ and let $P_t(s_1, s_2) := \mathbb{E}_t^{\mathbb{Q}}[e^{-\int_t^{t+s_2} r_u du} N_{(t+s_1, t+s_2)}]$ be the discounted risk-neutral forward disaster count—the deep-out-of-the-money calendar tranche. Let $P_t^0(s_1, s_2)$ be its calm-conditional version, the same expectation conditional on no disaster through $t + s_1$. For record-resolved states Ψ_t, Ψ'_t with common held-model marginal S_t and common public posterior: $P_t^0 - P_t'^0$ is first order in $s_2 - s_1$, equal to a fear-path term driven by the calm channel of Theorem 1(iii) plus the short-rate-path term that channel induces through (E1); both are functionals of Ψ_t and not of S_t , and both vanish for any benchmark whose pricing state is Markov in the current belief cross-section. The unconditional difference $P_t - P'_t$ adds the disaster-triggered channel through the state at $t + s_1$.*

The theorem's prediction thus attaches to the priced object itself, not only to its physical forecast; the proof, in Appendix E, couples the two economies as in part (iii) of the theorem.

The three option-tail objects the literature measures are each a functional of the *current* cross-

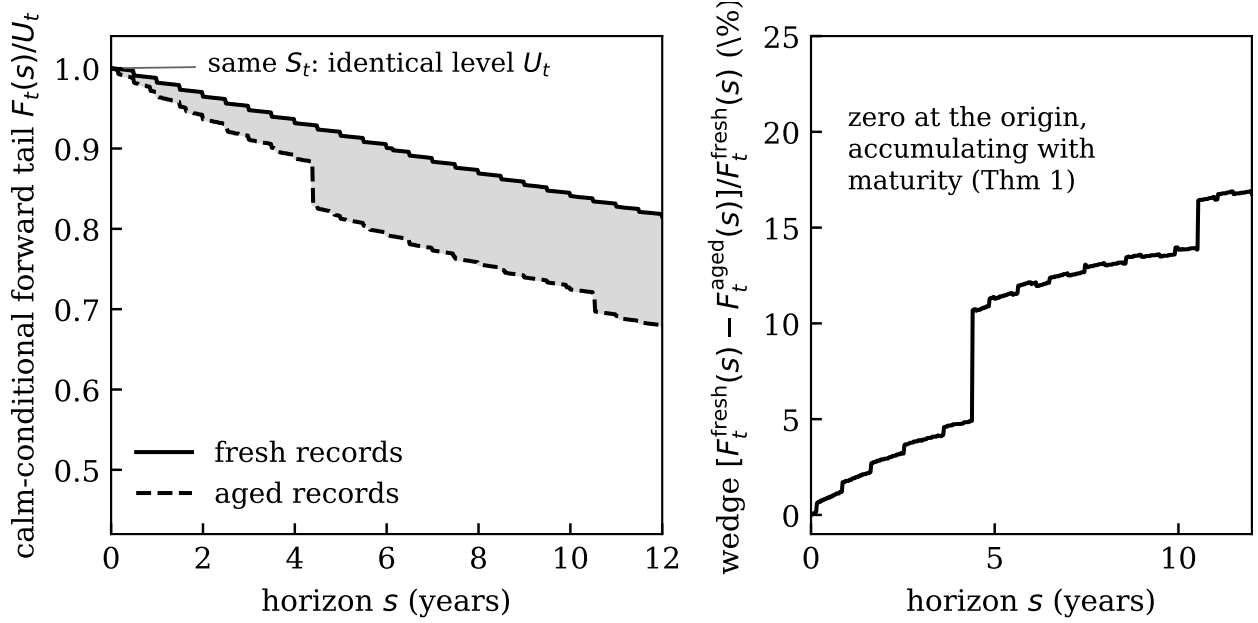


Figure 2: **Two histories, one cross-section (Theorem 1(iii))**. At $t^* = 44$ along a realized disaster record—fourteen years into the calm that follows a disaster at $t = 30$ —two record-resolved states share the same held-model cross-section S_{t^*} , hence the same near tail U_{t^*} , and differ only in their records: *aged* is the state the realized history delivers, its records carried since the post-disaster re-anchoring; *fresh* is the same cross-section with every since-adoption record reset. Left: the calm-conditional forward tail $F_t^0(s) := \mathbb{E}[U_{t+s} \mid \mathcal{F}_t, \text{no disaster in } (t, t+s]]$, normalized by the common level. Conditional on no disaster the future path is a deterministic functional of the record-resolved state—the records by Proposition 9, the weights by the selection flow—so both curves are exact, not simulated. Right: the wedge between the curves—zero at the origin and accumulating with maturity as the aged state’s calm deadlines arrive, reaching about 14% of the level at a twelve-year horizon under the calibrated belief block of Section 6.B. This is the survival-history residual of Proposition 5, the object any belief-sufficient benchmark sets to zero. Parameters: Table I, with the belief block of Section 6.B.

section S_t (Proposition 15): the far-tail level wedge is the level of κ_t (Proposition 2), the investor-fears index is the front wedge $U_t - \lambda^*$ (Bollerslev and Todorov, 2011), and the unspanned tail factor is U_t itself (Andersen et al., 2015b). Any belief-sufficient benchmark reproduces all three, so matching them certifies a sensible time-varying intensity but does not separate lazy revision from the Markov class; the object that does is the forward fear factor’s *slope* (Theorem 1), S_t -measurable in level but carrying a Ψ_t -part in slope.

The mapping also places one of their findings structurally. Andersen et al. (2015b) find the compensation for tail risk reacts far more persistently after a crisis than the underlying risk,

and fit it with a persistent latent factor; here that persistence is not a free parameter but the survival history, the fear decaying on the staggered revision schedule—the record channel, the S_t -measurable selection component being shared by any Markov benchmark—rather than at the speed of the underlying. The discriminating content is once more the slope—a function of the current factor under any Markov persistence, a function of how long the calm has run under lazy revision.

C. The time series of the wedge

A disaster is an impulse; this subsection traces the response of the priced objects assembled above. The engine is market selection on the realized path (Sandroni, 2000; Blume and Easley, 2006; Yan, 2008; Kogan et al., 2006, 2017): a disaster vindicates the fearful at a stroke, the ensuing calm vindicates the complacent only gradually, and the result is a fast jump and a slow, staggered relaxation (Andersen et al., 2015b). Throughout we keep the descriptive mean belief $\bar{\lambda}_t^S = \int_{\Phi} \lambda S_t(d\phi)$ distinct from the priced fear factor U_t : the kernel prices the power mean, not the mean (Section 3).

Proposition 6 (Disaster jump and calm erosion): *In the equilibrium of Proposition 1, with Var_{S_t} and Cov_{S_t} the variance and covariance under S_t :*

(i) (Calm erosion — exact, both calm channels.) *Between disasters the wealth-weighted mean belief moves continuously away from the predictable deadline atoms (the flat-level-set case of Lemma 5); the displayed law is the atomless-flow component, by selection and the calm switch flow,*

$$\frac{d\bar{\lambda}_t^S}{dt} = \underbrace{-\frac{1}{\alpha} \text{Var}_{S_t}(\lambda)}_{\text{selection} \leq 0} + \underbrace{\int_{\Phi} (\lambda(\phi_t^*) - \lambda) \Phi_t^{\text{sw}}(d\phi)}_{\text{calm switch flow (signed below)}}, \quad (26)$$

and, for any bounded h , $\frac{d}{dt} \langle h, S_t \rangle = -\frac{1}{\alpha} \text{Cov}_{S_t}(\lambda, h) + \int_{\Phi} [h(\phi_t^) - h(\phi)] \Phi_t^{\text{sw}}(d\phi)$. On the disaster-free-since-adoption part of the switch set the destination intensity is non-increasing (scheduled relief, Proposition 7; the calm MAP of Lemma 4 falls within a calm*

spell), so $\lambda(\phi_t^*) \leq \lambda$ and the switch-flow term is non-positive there. Call $\lambda(\phi_t^*) \leq \lambda$ on the full switch set the monotone-reselection condition.

- (ii) (Disaster jump — intensity channel.) At a disaster of size x the cross-section reweights by (23); the reweight alone raises the mean belief by

$$\bar{\lambda}_t^{\tilde{S}} - \bar{\lambda}_{t-}^S = \frac{\text{Cov}_{S_{t-}}(\lambda, r(\cdot, x)^{1/\alpha})}{\mathbb{E}_{S_{t-}}[r(\cdot, x)^{1/\alpha}]} \geq 0, \quad (27)$$

≥ 0 in the intensity channel ($f_\theta \equiv f$, $r(\phi, x) = \lambda/\lambda^*$) and strict if $\text{Var}_{S_{t-}}(\lambda) > 0$; the triggered switchers (Lemma 5) relocate to the common post-disaster mode ϕ_t^* , whose intensity coordinate is non-decreasing in the disaster count (proved by revealed preference in the appendix). Their contribution to the jump is computed under the post-reweight switcher measure (the disaster updates in the order reweight-then-relocate of Definition 4), $\Delta \bar{\lambda}_t^S|_{\text{switch}} = \widehat{s}_t^{\text{sw}}(\lambda(\phi_t^*) - \widehat{\lambda}_t^{\text{sw}})$, where $\widehat{s}_t^{\text{sw}}$ and $\widehat{\lambda}_t^{\text{sw}}$ are the switchers' mass and mean intensity after the $r(\cdot, x)^{1/\alpha}$ reweight; call $\lambda(\phi_t^*) \geq \widehat{\lambda}_t^{\text{sw}}$ the complacent-switcher condition.

- (iii) (The priced fear factor inherits the pattern — intensity channel.) Through the bridge relation (19), in calm U_t drifts down,

$$\frac{d \log U_t}{dt} = - \frac{\mathbb{E}_f[e^{\alpha J} g_t(J)^{\alpha-1} \text{Cov}_{S_t}(\lambda, r(\cdot, J)^{1/\alpha})]}{\mathbb{E}_f[e^{\alpha J} g_t(J)^\alpha]} \leq 0, \quad (28)$$

the selection part of the calm drift, strict if $\text{Var}_{S_t}(\lambda) > 0$. On a switch-active instant the calm switch flow of (i) adds a further term proportional to $\mathbb{E}_f[e^{\alpha J} g_t(J)^{\alpha-1} \int_{\Phi} (r(\phi_t^*, J)^{1/\alpha} - r(\phi, J)^{1/\alpha}) \Phi_t^{\text{sw}}(d\phi)]$, non-positive under monotone re-selection in the likelihood-ratio sense — $r(\phi_t^*, \cdot) \leq r(\phi, \cdot)$ on the switch set, which in the intensity channel is the condition of (i); U_t jumps up at a disaster with the reweighted aggregator—the reweight-only component of (ii); the triggered relocation adds under the complacent-switcher condition there, for $\alpha \geq 1$ (in its likelihood-ratio form for all α). The signs hold in the intensity channel and, generally, whenever λ and $r(\cdot, x)^{1/\alpha}$ are positively associated across the cross-section.

(iv) (Staggered recovery.) *The calm decay runs until the next disaster or the calm deadlines $\Delta^*(\phi, \varepsilon)$ (B4), at which revisers re-select a held model that Lemma 4 makes non-increasing on the disaster-free-since-adoption part of the switch set (or under the monotone-reselection condition)—a predictable step, weakly downward there. By Proposition 8 the deadlines are weakly ordered in ε and differ wherever the deadline map is non-constant, so the steps are staggered across that part of the population over a horizon set by Δ^* ,*

In words, item by item. (i) selection erodes the mean belief at a rate equal to its cross-sectional variance, scaled by the risk tolerance $1/\alpha$, while the calm switch flow relocates the staggered revisers to the calm destination ϕ_t^* . For switchers whose since-adoption records already carry non-rejecting disasters ($n \geq 1$), a disaster can have raised the public mode, so $\lambda(\phi_t^*) \leq \lambda$ is not automatic; that is the condition's content. Its regime is the finite-history mode: along calm the public mode follows the calm evidence to the complacent point—on the computed record, about seven years after a disaster—where the condition holds; in the locked stationary limit of Proposition 17 the destination is the menu's Kullback–Leibler projection, the fearful point at the binary menu, and calm relaxation there falls to selection and turnover. In the intensity channel, and under that condition generally, both calm channels push the mean down. (ii) The relocation contribution *adds* to the jump when $\lambda(\phi_t^*)$ exceeds that post-reweight mean, a cross-sectional condition we state rather than derive. The unconditional fast-in is the reweight (27) alone—a covariance signed without reference to the re-selection direction, so the fear jump does not require the disaster to raise the held mode—while the relocation is a conditional amplifier. (iii) The calm decline of U_t carries down with it the U_t -loading component of the disaster premium, the jump ratio ψ_t held fixed—the event-study reading of Section 5.C. (iv) The horizon is long when ε or the held intensity is small (the dependence on λ^* is indirect—through the held intensity coordinate and the chance a calm spell is interrupted by the next disaster—not a direct property of (B4)). The response is fast in (the instantaneous jump of (ii)) and slow out (the drift of (i),(iii) plus staggered deadline steps). In the decomposition of Proposition 5 the fast-in reweight (27) and the selection drift (28) are the cross-section part $B_t(S_t)$, while the calm

switch flow Φ_t^{sw} , the disaster-triggered relocation, and the deadline-timed steps $\Delta^*(\phi, \varepsilon)$ load on the survival-history residual $\mathcal{H}_t(\Psi_t)$ —the *centered* channels of Proposition 5, the raw flows net of their S_t -conditional mean. The staggered, record-timed part of the slow-out—switch flow, triggered relocations, deadline steps, net of their S_t -conditional mean—is the residual, while the smooth selection drift is cross-section part, which a Markov-in-cross-section benchmark also carries: such a benchmark reweights and drifts but has no deadlines and no triggers, so it carries the fast-in and the smooth drift, not the staggered slow-out (Internet Appendix IA.9).

Part (i) is market selection in its sharpest local form—the replicator drift in which the mean held belief falls at a rate equal to its own cross-sectional variance scaled by risk tolerance, the transient face of the long-run selection results (Sandroni, 2000; Blume and Easley, 2006; Yan, 2008) specialized to an economy where “accurate” is path-contingent: the complacent are vindicated in calm, the fearful at disasters. The asymmetry is the content—a disaster reweights toward fear and lifts the posterior mode at one instant (fast in), while calm erodes the wave only as fast as the surviving disagreement that selection consumes, staggered across the population by the deadline spread (slow out)—the model counterpart of the persistent post-crisis tail factor of Andersen et al. (2015b), an asymmetry visible in the beliefs themselves as well: surveyed firms’ subjective disaster probabilities spike on a shock and decay only slowly through the calm that follows (Menkhoff, 2025). The signs are intensity-channel statements—in the joint channel they need λ and $r(\cdot, x)^{1/\alpha}$ positively associated, which we flag rather than assume—and the *magnitudes* (the jump size, the decay rate $\frac{1}{\alpha} \text{Var}_S(\lambda)$, the horizon Δ^*) wait on the stationary cross-section of Section 6.

The three price functionals move together at a disaster, each driven by the same jump in the fear factor: the disaster premium’s U_t -component rises and drifts down through calm at the rate (28) (Proposition 14); the short rate moves by $dr_t = d\bar{\lambda}_t^S - dU_t$, falling *iff* the priced-tilt response dominates the mean-intensity response (Proposition 12); and the excess-variance term, generated by movement of the cross-section (Internet Appendix), bursts when the repricing ratio

ψ_t amplifies the jump. Under $dU_t > d\bar{\lambda}_t^S$ and in the intensity channel one belief object thus delivers a premium jump up, a safe-rate jump down, and a variance burst, each relaxing slowly through the staggered calm recovery. The cycle’s *shape* is pinned by the primitives; its *scale*, and the demographic refresh that makes it stationary, is the work of Section 6. A forecast-data signature—lazy revision read as state-dependent information rigidity, and its place relative to the under- and over-reaction evidence—is developed in the Internet Appendix.

D. What history prices, and what it does not

The state that prices disasters is not what the market believes but a pair: what it believes, and how long it has believed it without rechecking. The first coordinate is the cross-section S_t . The second is the cross-section of *fragility*: how each holding’s since-adoption record stands against the model it holds (Tegua, 2026), hence how near it is to revision. Two markets can share their beliefs and their near-dated tail and still differ here—one has just re-anchored, records fresh, deadlines far; the other has held the same fear through a long calm, records aged, deadlines near. They price the forward tail apart. Belief fragility is priced, and current belief does not recover it (Proposition 3).

This is economics, not bookkeeping, because of what fragility cannot stand for. The disaster law (λ^*, F_J) is fixed—known to the modeler, disputed by investors—so all the record does here is govern revision. It prices the tail—it moves the forward fear factor and the valuation $v(\Psi_t)$ —yet forecasts nothing about how often or how hard disasters strike. That is the content of the non-Markov bridge: a market-memory variable orthogonal to every fundamental forecast, yet priced. No model whose priced state is a belief about fundamentals—filtering, Bayesian learning, or an affine latent-state filter among them—can do this: there, history moves prices only by moving that belief, so the two move together; here they part.

This is also what lets the mechanism meet data where a latent disaster process cannot. A time-varying physical intensity is unobservable—the “dark matter” of the disaster literature

(Campbell, 2018; Chen et al., 2024), whose realization can never be prised apart from the belief that reads it, so its persistence is identified only jointly with the belief that reads it. Here the physical law (λ^*, F_J) is fixed and known, and the coordinate that moves the forward tail is not a latent state but the *survival record*: time since the last disaster is read off the path alone, and the remaining record coordinates are deterministic functionals of the path *jointly with* the maintained rule and the held type. Time since the last disaster is read off the path alone; accumulated calm surprise and distance to a rejection boundary are model-based reconstructions—functions of the path jointly with the held model, adoption age, and tolerance under the maintained rule—proxies an econometrician builds, not raw readings. The dark matter is in this sense partly visible, and it is that visibility which makes the over-identifying restriction of Theorem 1—record proxies forecasting the residual slope once the current surface is controlled for—a test a Markov-in-cross-section benchmark fails; latent-intensity accounts remain testable through their own price restrictions, but they do not supply these path-based residual proxies—the record channel does.

Three things follow. Through a continuing calm the same near-dated tail relaxes faster going forward where the fear has aged, nearer the deadline that revises it down (Theorem 1). A disaster need not reprice two markets of identical beliefs alike: the reprice runs through the holdings it tips into revision at once, read from the records and not from S_t (Proposition 6), so whether more revision deepens or cushions the fall is a cross-sectional condition, not a sign the model fixes. And the effect reaches the level, not the slope alone: $v(\Psi_t)$ and the premium read the whole record.

A renewal process is Markov once one tracks age, so one might ask whether “(beliefs, age)” is a Markov state in disguise. It is not. The record resets at every rejection, and both the revision hazard and its destination depend on the full record and the public posterior, not a scalar clock. These are claims of different strength, and the paper rests on the weaker one. That the forward surface reads the record Ψ_t and not the cross-section S_t alone holds in *either* channel: it follows from the non-autonomy of the law of motion, whose switch flow and trigger

read the within-cohort record distribution rather than the held-model marginal (Proposition 3). The stronger statement—that no *finite* statistic of the state is sufficient—is a size-channel result (Proposition IA.1) the intensity-channel calibration never invokes: there the instantaneous tail is the single scalar G_t (Corollary 2), and only the *forward* objects carry the record. We claim the dimensionality only where the mathematics forces it and the economics where it bites—a few measures of fragility: time since the last disaster, accumulated calm surprise, distance to a deadline. The dissociation, not the dimension, is the result.

One boundary. Fragility is priced; it does not feed back on the rule that makes it. Whether a long quiet breeds a more violent revision—fragility itself building in good times, not merely its price—is a separate question, left to other work. This is the pricing face of one fact about the rule: because rejection never stops—the calm deadline is finite even at the truth, so a correct-on-average holding is revised forever (Lemma IA.2)—fragility keeps accumulating and never collapses into the belief.

6. The wedge and the option-tail evidence

The belief cross-section alone swings the priced disaster intensity several-fold—*above and below* the representative-agent benchmark—with no physical parameter moving. This section establishes the stationary economy in which that swing lives and measures it against the option-tail evidence. The event study of Section 5.C and any unconditional moment first presuppose a *stationary* cross-section, which the present primitives do not supply: posterior consistency drives an infinitely-lived investor’s held model to the Kullback–Leibler point ϕ^* and the fear dispersion dies out (Remark 6). We add the one primitive that sustains it—demographic turnover—prove the cross-section is then stationary, and read the quantitative content as comparative statics: physical and preference parameters from the macro-disaster literature, the belief block varied and then, in the illustrative calibration of Section 6.B, pinned to a few stylized option-tail targets.

We restore a nondegenerate stationary cross-section with the device the rare-disaster literature uses for the purpose: a standard *demographic turnover*, perpetual-youth overlapping generations with fair annuities (Yaari, 1965; Blanchard, 1985; Gârleanu and Panageas, 2015). Each investor faces a constant death hazard $\Omega > 0$; newborns arrive at the same rate, drawn from a fixed type law Π over (ε, ϕ_0) with ϕ_0 -marginal Π_ϕ , and enter holding their drawn model with a reset test clock while inheriting the common public posterior. Because young cohorts hold ϕ_0 and have not yet revised, turnover continually replenishes the disagreement that selection consumes. We state the primitive as Assumption 5 and specialize to a finite menu $\Phi = \{\phi_1, \dots, \phi_K\}$ for the stationary and quantitative exercise; the assumption, the equilibrium it defines, and all proofs are in Appendix F.

Two results carry the economy of Sections 3–5 into this stationary world, and the quantitative reading needs only their conclusions. First, *the kernel keeps its form* (Proposition 16): with fair annuities the death hazard cancels from every survivor’s Euler equation and enters only through the composition of the living, so $\xi_t = e^{-\delta t} C_t^{-\alpha} \Xi_t^\alpha$ holds over the living cross-section, the short rate (E1) is unchanged—level and all—and turnover acts only by adding a mean-reversion of the cross-section toward the newborn law Π_ϕ at rate Ω . Every *instantaneous* pricing relation of Section 5 therefore transfers verbatim, evaluated at the turnover-stationary S_t , while the forward-looking valuation keeps its form with the cross-section dynamics augmented by this mean-reversion. Second, *that cross-section is stationary* (Proposition 17): in the limiting regime where the common posterior has concentrated ($\text{MAP}(\rho_t) = \phi^*$, so the re-selection destination is calendar-invariant), a finite menu and a constant inflow of fresh types make S_t a time-homogeneous functional of a stationary, ergodic disaster clock, with a unique nondegenerate stationary law whose unconditional moments equal long-run time averages.

The quantitative reading turns on two summary properties of the stationary cross-section. The stationary *dispersion*—and through Proposition 6 the amplitude of the fear wave—is governed by the newborn joint law Π —birth models *and* tolerances, through the tolerance-dependent

revision hazards—and the turnover rate Ω : more dispersed births leave more disagreement, and Ω sets how fast fresh cohorts replace the selected cross-section. Whether the stationary cross-section is *on average complacent*—the level reading we take to the option evidence below—is likewise a parameter condition on Π_ϕ , not a consequence of the primitives.

Sensitivity. Three dials govern the fear cycle, each through a displayed formula. The newborn dispersion sets the crisis amplitude through the jump covariance (E42): at a given size the jump is $\text{Cov}_{S_t}(\lambda, (\lambda/\lambda^*)^{1/\alpha})$ scaled by the mean transformed intensity, so dispersion moves it one-for-one holding that mean fixed. The turnover rate sets the demographic component of calm relaxation one-for-one: the pull in (F2) has half-life $\ln 2/\Omega$ —thirty-five years at $\Omega = 0.02$ —so halving Ω doubles it. And the menu coarseness sets the revision step: a dense menu replaces the two-point relocations by many small grid steps, so the crisis spike is milder and the post-spike plateau smooths into near-continuous decay, all other blocks unchanged.

A. Parameters and quantitative reading

We discipline the physical and preference parameters with the macro-disaster evidence and hold them fixed throughout; the belief block (Π, Φ) is not separately observed, since disaggregated disagreement data on the disaster intensity priced into options do not exist—though firm-level surveys of subjective *macroeconomic* disaster beliefs document exactly the wide dispersion and repeated revision the mechanism produces (Menkhoff, 2025). We read it two complementary ways. The comparative statics below *vary* the belief dispersion to trace the mechanism, using the option numbers only as a reference, not as inputs. The calibration of Section 6.B instead *pins* the belief block to two stylized option-tail targets (Section 6.B); this is an illustrative consistency check—whether one mechanism with near-orthogonal parameters can span the option-tail evidence—not a fit to the option surface and not an out-of-sample prediction. It establishes plausibility, not identification: the mechanism’s falsifiable content is not the fit but the residual restriction of Theorem 1, which no Markov-in-cross-section benchmark satisfies

and which the calibration neither presupposes nor exhausts. No survey data enter.

Block	Parameter	Value	Source
Physical	disaster intensity λ^*	0.03/yr	Barro and Ursúa (2008), consumption disasters (10%)
	mean disaster size $\mathbb{E}_f[b]$	0.21	Barro and Ursúa (2008)
	size law f	power law ($\gamma \approx 7.2$, $b_{\min} = 0.10$)	Barro and Jin (2011)
	trend, diffusion (μ_0, σ)	0.025/yr, 0.02/ $\sqrt{\text{yr}}$	Barro (2006); Barro and Ursúa (2008)
Preferences	risk aversion α	3.5	Barro (2006); Barro and Ursúa (2008) (equity premium)
Demography	discount δ	0.03/yr	standard
	death hazard Ω	0.02/yr (50-yr horizon)	Blanchard (1985); Gârleanu and Panageas (2015)
Beliefs (<i>free</i>)	menu $\{\lambda_{lo}, \lambda_{hi}\}$	varied (e.g. $\{0.01, 0.06\}$)	illustrative; not matched
	newborn law Π	varied	illustrative; not matched

Table I: Parameters. The physical and preference blocks are disciplined by the macro-disaster literature; the belief block is illustrative and varied. Disaster sizes are consumption contractions for the unlevered tree; for comparison Barro (2006) report $\lambda^* \approx 0.017/\text{yr}$ at a 15% GDP threshold with mean size 0.29. At the calibrated physical size law and risk aversion—menu-free objects—the kernel inputs evaluate to $\mathbb{E}_f[e^{\alpha J}] = 2.82$ and the representative-agent fear factor to $U^{\text{RA}} = 0.085/\text{yr}$.

Before beliefs enter, risk aversion alone inflates the disaster-state pricing: a single contraction of size b carries the kernel weight $e^{\alpha J}$ with $J = -\log(1 - b)$. At $\alpha = 3.5$ this multiplier is 1.4 at $b = 0.10$, 2.3 at the Barro–Ursúa mean $b = 0.21$, 3.3 at $b = 0.29$, and 6.0 at $b = 0.40$; the power-law tail (Barro and Jin, 2011) pushes the f -average $\mathbb{E}_f[e^{\alpha J}]$ above the value at the mean size.⁸ The representative-agent fear factor is $U^{\text{RA}} = \lambda^* \mathbb{E}_f[e^{\alpha J}]$: physical disasters of frequency 3% are priced as if several times more frequent. This effect is the standard rare-disaster channel;

⁸The power-law parameters describe the Barro and Jin (2011) law of the gross contraction $z = 1/(1 - b) = e^J$ ($\gamma \approx 7.2$, $z_{\min} = 1/(1 - 0.10) = 1.11$), not a Pareto on b . Under it $\mathbb{E}_f[b] = 1 - \gamma/[(\gamma + 1)z_{\min}] = 0.21$ (the Barro–Ursúa mean) and $\mathbb{E}_f[e^{\alpha J}] = \mathbb{E}[z^\alpha] = \gamma z_{\min}^\alpha/(\gamma - \alpha) \approx 2.82$ are jointly delivered; a Pareto on b would imply a smaller mean (about 0.12) and is not what is used.

our contribution is what the belief cross-section does to it.

The belief cross-section moves the risk-neutral tail several-fold. Under the binary menu, the fear factor is $U_t = G_t^\alpha U^{\text{RA}}$ with $G_t = S_t^{hi}(\lambda_{hi}/\lambda^*)^{1/\alpha} + (1 - S_t^{hi})(\lambda_{lo}/\lambda^*)^{1/\alpha}$ (D17). With the illustrative $\{\lambda_{lo}, \lambda_{hi}\} = \{0.01, 0.06\}$ around $\lambda^* = 0.03$,⁹ the ratio U_t/U^{RA} runs from 0.33 when the cross-section is fully complacent ($S^{hi} = 0$) to 2.0 when fully fearful ($S^{hi} = 1$), passing through 0.91 at $S^{hi} = \frac{1}{2}$: the belief cross-section alone swings the priced disaster intensity roughly six-fold, *above and below* the representative-agent benchmark, with no change in any physical parameter. The amplitude of this swing is governed by the menu spread and the stationary law of S^{hi} , i.e. by the free belief block through Proposition 17.

The same multiplier, written for a general cross-section in the intensity channel (Corollary 2), is $U_t/U^{\text{RA}} = G_t^\alpha$ with $G_t = \mathbb{E}_{S_t}[(\lambda/\lambda^*)^{1/\alpha}]$ (D7). Replacing the two-point menu by a continuum of held intensities filling $[\lambda_{lo}, \lambda_{hi}] = [0.01, 0.06]$ and varying the *shape* of the cross-section gives, at $\alpha = 3.5$, $\lambda^* = 0.03$, the illustrative values:

cross-section of held λ	mean belief $\bar{\lambda}^S$	multiplier U_t/U^{RA}
Beta(2, 5), complacent-skewed	0.024	0.78
Uniform[0.01, 0.06]	0.035	1.09
Beta(2, 2), symmetric	0.035	1.12
Beta(5, 2), fearful-skewed	0.046	1.51
Beta(8, 2), very fearful	0.050	1.66

The continuum fills the interval $[\lambda_{lo}/\lambda^*, \lambda_{hi}/\lambda^*] = [0.33, 2.0]$ between the binary extremes; in the intensity channel it does not raise that ceiling (the ceiling-lift of Remark IA.1 is a size-channel statement), but it makes the multiplier a smooth, dispersion-sensitive functional of the cross-section. These Beta cross-sections are illustrative instantaneous evaluations of the

⁹The two-point menu is deliberately misspecified: $\lambda^* = 0.03$ lies outside $\{0.01, 0.06\}$. No held point is correct; the consistency results, which read the destination as the menu's Kullback–Leibler projection, apply through that projection: $T(0.01) = 0.0130 > T(0.06) = 0.0092$, so the projection is λ_{hi} and the locked destination is the fearful point. The multiplier passage in the text is the pricing identity read pointwise in the fearful share s ; under the stationary law of s it inherits that law, with no learning input beyond it.

general intensity-channel multiplier (Corollary 2); they are not stationary laws of the turnover economy, whose stationarity theorem (Proposition 17) is stated under the finite menu. That sensitivity is exact: since $\alpha > 1$ and $y \mapsto y^{1/\alpha}$ is strictly concave, Jensen’s inequality applied to λ/λ^* under S_t gives

$$\frac{U_t}{U^{\text{RA}}} = G_t^\alpha = (\mathbb{E}_{S_t}[(\lambda/\lambda^*)^{1/\alpha}])^\alpha \leq \frac{\bar{\lambda}_t^S}{\lambda^*}, \quad (29)$$

with equality iff the held intensity is S_t -a.s. constant. Two consequences. First, a cross-section whose wealth-weighted mean belief equals the truth, $\bar{\lambda}_t^S = \lambda^*$, still prices the tail *below* the representative-agent benchmark—dispersion alone is complacency: at an equal-mass two-point spread $\{\lambda^* - s, \lambda^* + s\}$ the multiplier is 0.96 at $s = 1\%/yr$ and 0.82 at $s = 2\%/yr$. Second, fear ($U_t > U^{\text{RA}}$) requires $\bar{\lambda}_t^S$ to exceed λ^* by enough to overcome this Jensen discount, which a sufficiently severe disaster delivers by reweighting wealth toward and concentrating the cross-section on the fearful models (Proposition 3). Equation (29) is the analytic core of the chronic-complacency/disaster-spike asymmetry: the kernel prices the power mean of beliefs, which sits below the wealth-weighted arithmetic mean.

Selection is slow relative to disasters. The selection part of the calm erosion of Proposition 6(i), $d\bar{\lambda}^S/dt|_{\text{sel}} = -\frac{1}{\alpha} \text{Var}_S(\lambda)$, reallocates wealth at the rate $\frac{1}{\alpha}(\lambda_{hi} - \lambda_{lo}) \approx 0.014/yr$ —the decay of the high-to-low wealth *odds*, an *e*-folding of about 70 years (a heuristic scale; the full relaxation also carries the calm switch flow and demographic turnover). Disasters recur every $1/\lambda^* \approx 33$ years and cohorts turn over every $1/\Omega \approx 50$ years. Because selection is *slower* than disaster recurrence, the fear share does not fully relax between typical disasters: the stationary cross-section carries a persistently positive, fluctuating fear share, and with it a non-degenerate, time-varying disaster premium—the model counterpart of the chronic equity premium and the slow post-crisis reversion of the tail factor (Andersen et al., 2015b).

These magnitudes reconcile two option facts that look opposed. The same multiplier $U_t/U^{\text{RA}} = G_t^\alpha$ —below one in calm, above one after disasters—carries the economy between the modest option-implied disasters Backus et al. (2011) document and the large, spiking tail premium

of Bollerslev and Todorov (2011) and Andersen et al. (2015b): the literatures measure it at different points of the fear cycle, against different anchors, and are not in conflict. The countercyclical, below-macro prediction is about the inverted intensity $\lambda^*\kappa_t$, not the raw U_t , which carries the risk-aversion premium $\mathbb{E}_f[e^{\alpha J}]$ on top.

B. Confronting the option-tail evidence

The closed-form objects above line up, one for one, with three model-free option-tail literatures, so the model can be taken to that evidence without estimating the belief process. We hold the physical block at its macro-disaster calibration (Table I), which delivers the representative-agent benchmark $U^{\text{RA}} = \lambda^*\mathbb{E}_f[e^{\alpha J}] = 0.085/\text{yr}$ in closed form, and pin the belief block—newborn dispersion and disaster-trigger mass—to two stylized targets drawn from that evidence—the crisis jump, matched inside the documented band, and the calm moderation $\kappa_t < 1$, matched in deep calm though not at the literature’s point level (first row of the table); the post-crisis half-life is generated by the deadline and turnover blocks and reported as an *achieved* moment. The mapping is nearly orthogonal: dispersion sets the calm level of κ_t and the trigger sets the disaster jump. The pinned block, in the intensity channel: the binary menu $\{0.01, 0.06\}$, a newborn law 90% complacent, and tolerances on a two-block grid on $(0.01, 0.90)$ with a third of the mass below 0.12—the disaster-trigger dial. Along realized disaster records the resulting exact transport reaches crisis spikes of $U_t/\lambda^* \approx 4.5\text{--}4.8$, inside the 3.2–4.8 crisis range, with κ_t traversing the benchmark over the cycle and dipping to 0.53–0.63 in deep calm. The intensity rows of Table II are closed-form implications of the fear factor $U_t = U^{\text{RA}}\kappa_t$ (18) and the finite-menu reduction of Corollary 2; the premium rows are *structural* consequences—the dominance of the disaster over the diffusive premium and the loading of the premium on U_t rather than σ —that additionally involve the price–dividend jump ratio ψ_t , a Feynman–Kac functional of the full record-resolved state rather than a closed form (Proposition 13), and follow from its qualitative behavior rather than a solved level, as the row notes below record.

The three anchors map directly (Table II): Backus et al. (2011)’s modest option-implied disaster law is $\kappa_t < 1$ in deep calm, the option-implied intensity falling below U^{RA} as the quiet record accumulates; Bollerslev and Todorov (2011)’s time-varying fear premium is $U_t > \lambda^*$ on average, with the disaster premium nearly the whole equity premium since the diffusive $\alpha\sigma^2$ is negligible; and Andersen et al. (2015b,a)’s unspanned, persistent, premium-forecasting tail factor is κ_t , which jumps at disasters, erodes slowly in calm (28), and enters the premium through U_t while σ is held fixed.

Table II: Confronting the option-tail evidence. The physical block is fixed at the macro-disaster calibration (Table I), giving $U^{\text{RA}} = 0.085/\text{yr}$; the belief block is pinned to the crisis jump and the calm moderation; the persistence is achieved, not targeted. The model entries are closed-form in the fear factor $U_t = U^{\text{RA}}\kappa_t$, the premium rows also carrying the jump ratio ψ_t . The first block is matched—two identities and two calibrated targets; the half-life is *achieved*, generated by the deadline and turnover blocks rather than targeted.

Evidence	Documented finding	Model (closed form)
<i>Matched — identities and calibrated targets</i>		
Backus et al. (2011)	options imply a more modest disaster law than the macro record	deep-calm $\kappa_t \approx 0.53\text{--}0.63 < 1$: U_t falls below the benchmark U^{RA} late in calm; the point level $U_t \approx \lambda^*$ ($\bar{\kappa} = 0.35$) lies below the two-point menu’s re-anchoring floor and is not attained on this record
Bollerslev and Todorov (2011)	the risk-neutral jump tail exceeds the statistical one (a fear premium), time-varying	mean $U_t/\lambda^* \approx 3$ over the computed record (1.5–1.6 entering disasters); spikes 4.5–4.8 $\times$, inside the documented 3.2–4.8 crisis band
Bollerslev and Todorov (2011)	rare-event compensation is a large fraction of the equity premium	$\alpha\sigma^2 = 0.14\%/ \text{yr}$ diffusive $\Rightarrow \text{DRP}_t$ (E27) is essentially the whole premium
Andersen et al. (2015b)	the tail factor predicts the equity premium; volatility does not	exact here: DRP_t loads on U_t , σ constant $\Rightarrow$ zero volatility loading
<i>Achieved — generated by the deadline and turnover blocks, not targeted</i>		
Andersen et al. (2015a)	the tail factor is far more persistent than volatility	post-crisis wave half-life of $U_t \approx 12$ years on the computed record $\gg$ volatility’s weeks-to-months

Two of the five rows rest on a belief-free *identity*. The diffusive premium is mechanically small and fixed: $\alpha\sigma^2 = 0.14\%/yr$ at the Table I diffusion, so by (E26) the disaster premium DRP_t accounts for essentially all of the equity premium wherever it is sizable relative to that part, as at the calibration; the *share* itself is not belief-free, since the level of DRP_t moves with ψ_t and the full pricing state. The predict-the-premium-not-volatility pattern is an identity in this model: DRP_t loads on U_t (E27) while σ is constant, so the fear factor forecasts the premium with a nonzero loading and volatility forecasts it with exactly zero—the Andersen et al. (2015b) separation, an exact zero loading in the model (σ constant) rather than a fitted coefficient. Two of the remaining rows—the crisis jump and the calm moderation—are the calibrated targets: the jump matched inside the documented band, the moderation matched in deep calm though not at the literature’s point level; the half-life is an *achieved* moment, generated by the deadline and turnover blocks and reported for comparison. The row is measured on the computed record, peak to halfway back to the pre-disaster level. Its channel decomposition at this calibration: the selection drift (28) contributes about 0.5% per year (on its own a half-life above a century—the reason no free relaxation-rate parameter exists to tune), the demographic pull roughly 0.7–1.6%, and the staggered deadline flow the remainder—operative once the public mode has returned to the complacent point, about seven years after a disaster on this record: immediately after a disaster the finite-history mode sits at the fearful point, so calm deadlines re-anchor upward or re-adopt, and the monotone-reselection condition holds only once the mode has followed the calm evidence down. Persistence is generated by the revision architecture, not by a free relaxation-rate parameter. All are stylized: published estimates vary with sample and method, so we match orders of magnitude, not point values.

This is a disciplined confrontation, not a structural estimation: we show that the closed-form fear factor is consistent with the published model-free moments, not that the belief process is identified by them. The one feature the belief block cannot repair is the safe rate. Under power utility the safe rate sits at the high CRRA *level* (Section 6.A)—its realized dynamics are modest, but the level is the standard risk-free-rate artifact at $\alpha = 3.5$; that is a preference

limitation, removed only by an Epstein–Zin extension, and not something the belief calibration can absorb.

Remark 3 (What power utility buys, and what is specific to it). Power utility does two things at once. It makes the factorization $U_t = U^{\text{RA}} \kappa_t$ (Proposition 2) *exact*, and it makes the level/slope dissociation *clean*: the kernel’s disaster jump ratio is $e^{\alpha x} g_t(x)^\alpha$, which depends only on the current cross-section, so the instantaneous tail is a functional of S_t alone and the record Ψ_t enters only the forward surface (Proposition 15, Theorem 1). Both are specific to the class. An Epstein–Zin kernel carries the return on the wealth portfolio, whose disaster jump multiplies the gross return by $e^{-x} \psi_t(x)$ —the record-dependent component being the valuation-ratio jump $\psi_t = v(\Psi_t^x)/v(\Psi_{t-})$, a functional of the *full* record-resolved state Ψ_t (Proposition 13); under it the record would move the instantaneous tail as well, and the level would no longer pin S_t . The two implications cut opposite ways. The record’s *economic* role is preference-robust—indeed stronger under Epstein–Zin, where it moves the level as well as the slope—so the priced market-memory coordinate of Section 5.D is not an artifact of power utility. But the clean *separation* that makes the restriction sharp—level on S_t , residual slope on Ψ_t , isolated by conditioning on the current tail—is CRRA-specific, and a recursive-utility version would demand a different empirical design, since conditioning on the current tail would then partial out part of the record itself. Power utility is thus the tractable window through which the record’s imprint is cleanly visible, at the cost of a safe-rate *level* at the high- α artifact of the class, removable only by that same extension.

One model-free finding runs the other way. Gormsen and Jensen (2025) estimate the *physical* return distribution from option prices and find it more left-skewed and fat-tailed in good times—a *procyclical* conditional tail risk they read as hard to square with disaster models in which the physical law is elevated in bad times. This economy is not such a model: the physical law (λ^*, F_J) is fixed, and what is countercyclical is the *priced* multiplier κ_t , not the world’s disaster risk. The link is tighter than coexistence—the complacency that holds $\kappa_t < 1$ through calm *is* an under-pricing of the tail in precisely the spells where, on their evidence, the conditional

distribution is fattest—so the gap between the thin priced tail and the fat physical tail is the belief wedge, bracketed from opposite ends: its priced side the modest option-implied disasters of Backus et al. (2011) the calibration matches, its physical side the procyclical return tail Gormsen and Jensen measure. The mechanism turns this into a discriminating prediction—the physical-minus-priced gap should widen through calm and collapse at disasters, developed in the Internet Appendix—so Gormsen and Jensen’s fact is a second empirical handle on the price of disaster fear, not an embarrassment to it.

7. Conclusion

This paper has built a complete-market disaster economy in which a continuum of CRRA investors disagree about the disaster law and revise it lazily, and has shown that the risk-neutral disaster tail factors exactly into a frictionless representative-agent benchmark and a belief multiplier, $U_t = U^{\text{RA}} \kappa_t$ (Proposition 2). The mechanism is one line: the kernel prices the *power* mean of beliefs, which sits below the wealth-weighted *arithmetic* mean, so a cross-section correct on average still prices the option tail below the benchmark in calm—dispersion alone is complacency—and reprices it upward in coordinated jumps when a disaster proves the fearful right. Lazy revision makes this a fast-in/slow-out fear factor (Proposition 6)—synchronized disaster jumps, slow staggered calm relaxation as selection reallocates wealth—with a unique stationary cycle once demographic turnover refreshes the disagreement (Propositions 16, 17).

The option surface measures beliefs, not the physical disaster law: a moving priced tail over a fixed quantity of disaster risk. It reconciles the modest option-implied disasters of Backus et al. (2011) with the spiking risk-neutral tails of Bollerslev and Todorov (2011) and Andersen et al. (2015b) as one belief cycle, predicting that the representative-agent-inverted intensity $\lambda^* \kappa_t$ is countercyclical—below the macroeconomic frequency in calm, rising after disasters. And it leaves an over-identifying mark: because a held belief is not a sufficient statistic for its own revision, the forward-looking surface is priced off the records under which current beliefs have

survived, not the beliefs alone, and forecasts that surface even after the contemporaneous option state is controlled for—a residual that any Markov benchmark—any belief-sufficient model—sets to zero (Proposition 5), the cleanest line separating lazy revision from that class; among history-bearing alternatives, the deadline and crash signs of Remark 1 do the separating.

The limitations are explicit: CRRA preferences fix the safe-rate *level* at the high- α artifact of that class (removable, we conjecture but do not prove, by an Epstein–Zin extension), and the same choice is what confines the record to the forward surface, so the clean level/slope separation behind the residual test is itself specific to power utility; the disaster-dynamics signs are intensity-channel results under a positive-association condition we flag rather than assume; and the belief block is calibrated against the option-tail moments rather than estimated, since disagreement data on the disaster intensity *priced into options* do not exist, though surveyed disaster beliefs display the dispersion and revision the mechanism produces (Menkhoff, 2025). A fully structural calibration awaits such a measurement, and the nearest handle is the gap between the priced tail and the *physical* return tail, whose procyclical higher-moment risk (Gormsen and Jensen, 2025) is the far side of the same wedge—the surface under-pricing the tail in the very calm spells where the conditional distribution is fattest. Generating that procyclical tail from the primitives, rather than only measuring it against them, is the natural next step this fixed-law economy poses but does not settle. What the wedge between the option surface and the macroeconomic record measures, throughout, is not a mismeasurement of either, but the price of disaster fear.

Notation

Primitives (the fixed physical law)

C_t	aggregate consumption, the Lucas-tree dividend
μ_0, σ	drift and diffusion of log consumption
λ^*	physical disaster intensity
F_J, f	physical disaster-size law and density on $\mathcal{J} = [\underline{J}, \overline{J}]$
T_k, J_k	time and size of the k th disaster

Beliefs and the revision rule

$\phi = (\lambda, \theta)$	a held model: disaster intensity λ , size index θ
$\Phi = \Lambda \times \Theta$	model space; F_θ the belief size law
θ^*	pseudo-true size index (minimizes $\text{KL}(F_J \ F_\theta)$)
ε^i, G	investor i 's surprise tolerance; its cross-section law
ρ_t, ϕ_t^*	common public posterior; re-selection destination $\phi_t^* = \mathbf{m}(\rho_t)$
$\mathcal{R}_t^i, \tau_t^i$	investor i 's since-adoption record and regime age

Preferences and the pricing kernel

α	coefficient of relative risk aversion
s_t^i	investor i 's wealth (consumption) share
Ξ_t	belief aggregator (enters the kernel $\xi_t = e^{-\delta t} C_t^{-\alpha} \Xi_t^\alpha$)
$g_t(x)$	share-weighted aggregator; g_t^α is the $(1/\alpha)$ -power mean of the disaster likelihood ratios

The fear factor and the belief multiplier

$\lambda_t^\mathbb{Q}, f_t^\mathbb{Q}$	risk-neutral disaster intensity and size density
U_t	disaster-fear factor, $U_t = \lambda_t^\mathbb{Q}$
U^{RA}	frictionless benchmark $\lambda^* \mathbb{E}_f[\phi]$ (correct-belief value, at $g_t \equiv 1$)
κ_t	belief multiplier U_t/U^{RA} ; the bridge is $U_t = U^{\text{RA}} \kappa_t$

Appendix A. Proofs for Section 3

Proof of the example proposition (Section 2). (i) κ_0 is (1) at the common s_0 . (ii) Three timing checks make the window clean. The complacent model's exact quiet-spell deadline, ≈ 72 years, lies beyond every horizon here, so only the fearful cohorts can revise; before either does, both economies run the replicator (2), a function of the cross-section alone, from the common s_0 . In Aged the fearful deadline $\Delta^* - a \approx 3.9$ years precedes the window (any adoption age in $(\Delta^* - 5, \Delta^*) \approx (6.9, 11.9)$ years gives the same split): its wealth relocates to the posterior mode—complacent under the even public prior after any quiet spell—so the economy is degenerate at λ_{lo} from year four on and $\kappa_5 = \lambda_{lo}/\lambda^* = 0.333$. In Fresh the fearful deadline $\Delta^* \approx 11.9$ years lies beyond the window, only the replicator acts, and $s_5 = 0.237$, $\kappa'_5 = 0.558$. (iii) A functional of the date-0 state takes one value on the two economies, which share that state. $\square$

This appendix states and proves Lemma 2, Lemma 3, and Lemma IA.1. The first is a direct computation from the change-of-measure density; the second is the Kullback–Leibler rate calculation; and the third invokes the high-frequency threshold theory of Mancini (2009), whose hypotheses we verify by name before assembling the lemma.

We record the identification and regularity conditions on the belief family deferred from Section 3.B, verified below for the families used in the paper.

Assumption 4 (The belief family): *The size family $\{F_\theta : \theta \in \Theta\}$ of Section 3.B, with Lebesgue densities f_θ on $\mathcal{J}$ ($\Theta \subset \mathbb{R}$ a compact interval, and $\Lambda = [\underline{\lambda}, \bar{\lambda}] \subset (0, \infty)$ containing λ^* in its interior), satisfies:*

- (i) Full support: $(x, \theta) \mapsto f_\theta(x)$ is jointly continuous with $f_\theta(x) > 0$ on $\mathcal{J} \times \Theta$, hence bounded and bounded away from zero on that compact set.
- (ii) Regularity: $\theta \mapsto \log f_\theta(x)$ is continuously differentiable with score $\dot{\ell}_\theta(x) := \partial_\theta \log f_\theta(x)$, and the Fisher information $I(\theta) := \int_{\mathcal{J}} \dot{\ell}_\theta(x)^2 F_\theta(dx)$ is finite and strictly positive on Θ .

(iii) Identifiability: $\theta \neq \theta' \Rightarrow F_\theta \neq F_{\theta'}$.

(iv) Strict KL identifiability: $\theta \mapsto \text{KL}(F_J \| F_\theta)$ has a unique minimizer θ^* on Θ .

(v) Spanning: for every $\alpha > 0$, the family $\{(f_\theta/f)^{1/\alpha} : \theta \in \Theta\}$ is linearly independent in $L^1(\mathcal{J}, f)$ —no nontrivial finite linear combination vanishes f -a.e.

(vi) Smooth pseudo-true point: $\theta^* \in \text{int } \Theta$; for each $x \in \mathcal{J}$, $\theta \mapsto \log f_\theta(x)$ is twice continuously differentiable on a neighbourhood of θ^* , with $|\dot{\ell}_\theta(x)|$ and $|\partial_\theta^2 \log f_\theta(x)|$ dominated there by an envelope $\Psi(x)$ satisfying $\mathbb{E}_{F_J}[\Psi] < \infty$ and $\mathbb{E}_{F_J}[\Psi^2] < \infty$; and, writing $M(\theta) := \mathbb{E}_{F_J}[\log f_\theta(J)]$, the pseudo-Hessian $H := -M''(\theta^*)$ is strictly positive and $B := \mathbb{E}_{F_J}[\dot{\ell}_{\theta^*}(J)^2] < \infty$.

The conditions invoked in Assumption 4 play the following roles.

Condition (iv) is a separate identification requirement from (iii), neither implying the other: where (iii) asks injectivity of $\theta \mapsto F_\theta$, (iv) asks uniqueness of the Kullback–Leibler projection of the truth onto the family. It holds, for instance, whenever $\{F_\theta\}$ is a regular exponential family in θ (where $\theta \mapsto \text{KL}(F_J \| F_\theta)$ is strictly convex), and it is what makes the pseudo-true model a point rather than a set. It is invoked wherever a single pseudo-true point is needed—the pseudo-true model ϕ^* of (A3), the smooth pseudo-true/asymptotic-normality setup of Assumption 4(vi), and the concentration limits of Remark 6 and Proposition 17; without it each such statement comes with $\arg \min_\theta \text{KL}(F_J \| F_\theta)$ in place of a point. Condition (v) is the priced-aggregator analogue of identifiability: it ensures distinct *finite* size-belief mixtures leave distinct imprints on the risk-neutral tail (finite-mixture separation; it does not by itself give injectivity of the full mixture map over arbitrary continuum-supported states, which would need a completeness or transform-uniqueness condition). It holds for the families used here by distinct-exponent independence, not by analyticity alone. For the power-law size family $f_\theta(x) \propto x^{-(\theta+1)}$ on $\mathcal{J}$, $(f_\theta/f)^{1/\alpha} = c_\theta x^{-(\theta+1)/\alpha} f(x)^{-1/\alpha}$, which reduces to the monomial $c_\theta x^{(\theta_0-\theta)/\alpha}$ when $f = f_{\theta_0}$ lies in the family; in the misspecified case the common factor $f^{-1/\alpha}$ is shared across θ and so leaves independence

intact—distinct indices give distinct real exponents, linearly independent on any non-degenerate interval (their Wronskian is a nonzero Vandermonde determinant in the exponents); the same holds for any family reducible to distinct-exponent or regular-exponential form. It is used only in Proposition 3 (the infinite-dimensionality of the pricing state). Condition (vi) is the standard regularity under which a possibly-misspecified maximum-likelihood estimator is $\sqrt{n}$ -asymptotically normal (White, 1982; van der Vaart, 1998); it strengthens (ii) and, under correct specification, has its sandwich collapse ($H = B = I(\theta^*)$), recovering the Fisher-information form of (ii). It is used only in Lemma IA.1(iii).

Conventions carried from Section 3.B. The rejection criterion is the surprise $\bar{\varepsilon}$ (10): the held model is rejected iff $\bar{\varepsilon} \leq \varepsilon^i$, the boundary counted as a rejection. The closed-threshold form $\{D_{\phi,\Delta} \geq r_{\phi,\Delta}(\varepsilon)\}$ used in later expressions is shorthand for this p-value criterion; the operative criterion is always the p-value rule $\bar{\varepsilon} \leq \varepsilon^i$. Along a disaster-free stretch $\bar{\varepsilon}$ is continuous and non-increasing in the age, so the calm deadline Δ^* is its attained first crossing; the predictive law of D carries an atom at the zero-count value ($e^{-\lambda\Delta} > 0$), so the shorthand’s agreement with a strict quantile cutoff is a convention, not a continuity fact, and at any atom the p-value rule governs. The test is two-sided: the deviance is large whether disasters run too frequent or too rare, and whether the realized sizes pull $\hat{\theta}$ above or below θ . Two divergence rates also stay distinct: the full KL (A2) fixes second-order posterior concentration, the pseudo-true model (A3), and survival (Lemma 3), while the since-adoption rejection deviance that triggers revision drifts at the Poisson KL in timing and at the *excess* in-family size divergence relative to the pseudo-true fit, which can vanish at θ^* even when the full size KL is positive—so the two coincide only when the size family is correctly specified.

Why parameters, not outcomes. The disputed primitives are parameters of laws, not realized outcomes: a Dirac mass on a fixed disaster size is falsified by the first continuous draw, whereas disputing the size *law* and the arrival *rate* through their parameters is the faithful object, the one the data reveal only slowly (full support, Assumption 4(i)).

The rule in the literature. The rule is [Ortoleva](#)’s, applied in real time to the accumulating record as in the model-validation learning of [Cho and Kasa \(2015\)](#), so the rejection time is a first exit; by Sanov’s theorem ([Dembo and Zeitouni, 1998](#); [Cover and Thomas, 2006](#)) the acceptance region is the empirical law lying within a Kullback–Leibler ball of the held model. Fixed-tolerance testing makes recurrence the *foregone conclusion* of repeated significance testing under continuous monitoring ([Anscombe, 1954](#); [Armitage et al., 1969](#); [Robbins, 1970](#); [Robbins and Siegmund, 1970](#)): the standardized record crosses a fixed band infinitely often by the law of the iterated logarithm. [Cho and Kasa \(2015\)](#), who share the test-and-reselect primitive, instead engineer this away with a time-varying threshold and vanishing gain; recurrent regime change also arises as constant-gain escape dynamics ([Kasa, 2004](#); [Cho and Kasa, 2008](#)). The consolidating side is classical: under countable additivity a Bayesian cannot be driven to a foregone conclusion ([Kadane et al., 1996](#)) and the posterior merges with the truth ([Blackwell and Dubins, 1962](#); [Kalai and Lehrer, 1993](#)). That the since-adoption sufficient statistic—the count in the intensity channel; with size disagreement the (count, marks) pair, low-dimensional only for particular exponential-family size laws—stays sufficient for the held model at the *endogenous* revision time follows from the exponential-family (Wald) factorization of the disaster-count likelihood, the disaster channel being the last-observation-sufficient member of the families of [Küchler and Lauritzen \(1989\)](#) and [Küchler and Sørensen \(1989\)](#), for which sufficiency survives optional stopping ([Küchler and Sørensen, 1998](#)). The architecture itself—testing the held model on its since-adoption record while re-selecting on the full-history posterior—is the axiomatized primitive of the companion paper ([Tegua, 2026](#)): the acceptance region’s interval form is forced (an essentially complete class), the test size is the single behavioral scalar, the per-test reading carries a per-shift consistency control ([Weinstein, 2017](#)), and stakes-invariance separates the surprise-governed hazard from cost-based rivals. We import the rule; the equilibrium is this paper’s object.

We first record the change-of-measure and divergence facts.

Lemma 2 (Equivalence of held and physical laws on finite horizons): *Under Assumptions 1 and 4, for every $\phi = (\lambda, \theta) \in \Phi$ and every $t > 0$, $\mathbb{P}|_{\mathcal{F}_t} \sim \mathbb{P}^\phi|_{\mathcal{F}_t}$, with*

$$\left. \frac{d\mathbb{P}^\phi}{d\mathbb{P}} \right|_{\mathcal{F}_t} = \exp(-(\lambda - \lambda^*)t) \prod_{k: T_k \leq t} \frac{\lambda f_\theta(J_k)}{\lambda^* f(J_k)} \in (0, \infty) \quad \mathbb{P}\text{-a.s.} \quad (\text{A1})$$

Proof of Lemma 2. Imported result (marked-Poisson change of measure; Brémaud, 1981, Ch. VI; Jacod and Shiryaev, 2003, III.3). If $\mathbb{P}, \mathbb{P}^\phi$ are the laws on $(\Omega, \mathcal{F}_t)$ of marked Poisson processes with (intensity, mark law) (λ^*, F_J) and (λ, F_θ) , $\lambda^*, \lambda \in (0, \infty)$, and if $F_J \sim F_\theta$, then $\mathbb{P}|_{\mathcal{F}_t} \sim \mathbb{P}^\phi|_{\mathcal{F}_t}$ with Radon–Nikodym density (A1). *Hypotheses in our model.* (1) $\lambda^* > 0$ (Assumption 1) and $\lambda \in [\underline{\lambda}, \bar{\lambda}] \subset (0, \infty)$ (Assumption 4); (2) f and f_θ are continuous and strictly positive on the common compact support $\mathcal{J}$ (Assumption 1; Assumption 4(i)), so F_J and F_θ have identical null sets, i.e. $F_J \sim F_\theta$. Both hypotheses hold, so the import applies and yields (A1). Finally N has finitely many jumps on $[0, t]$ $\mathbb{P}$ -a.s. (Assumption 1), so the product is a finite product of strictly positive factors and the exponential is strictly positive; hence the density lies in $(0, \infty)$ $\mathbb{P}$ -a.s., and $\mathbb{P}, \mathbb{P}^\phi$ share their null sets.

On the full filtration. Lemma 2 is stated on the consumption-path filtration $\mathcal{F}_t$, which carries both the Brownian component σW and the marked-Poisson disaster component. The diffusion (μ_0, σ) is undisputed (common knowledge, Assumption 1): every held law $\mathbb{P}^\phi$ shares the law of the continuous part with $\mathbb{P}$ and differs only in the disaster law (λ, F_θ) . Because the Brownian and marked-Poisson components are independent under each law, the Radon–Nikodym density on $\mathcal{F}_t$ factors as the (identity) density of the unchanged Brownian law times the marked-Poisson density (A1); the change of measure acts only on the jump factor, and the equivalence $\mathbb{P}|_{\mathcal{F}_t} \sim \mathbb{P}^\phi|_{\mathcal{F}_t}$ holds on the full filtration. $\square$

Definition 5 (Per-period Kullback–Leibler divergence rate): *For a held model $\phi = (\lambda, \theta) \in \Phi$, let $\mathbb{P}^\phi$ be its law—the marked Poisson process of intensity λ with i.i.d. marks F_θ (Assumption 4)—and $\mathbb{P}$ the physical law (Assumption 1). The per-period Kullback–Leibler rate is the divergence*

of the truth from the held model, measured on the unit interval $\mathcal{F}_1$,

$$\mathcal{K}(\phi) := \text{KL}(\mathbb{P} \parallel \mathbb{P}^\phi) \big|_{\mathcal{F}_1} = \mathbb{E}_{\mathbb{P}} \left[\log \frac{d\mathbb{P}}{d\mathbb{P}^\phi} \bigg|_{\mathcal{F}_1} \right],$$

the $\mathbb{P}$ -expectation of the log-likelihood ratio of the two laws restricted to $\mathcal{F}_1$.

Lemma 3 (Divergence rate: decomposition and pseudo-true model): *Under Assumptions 1 and 4, $\mathcal{K}(\phi) \in [0, \infty)$ for every $\phi \in \Phi$ and decomposes additively into a timing part and a size part,*

$$\mathcal{K}(\phi) = \underbrace{\left[\lambda^* \log \frac{\lambda^*}{\lambda} - \lambda^* + \lambda \right]}_{\text{timing}} + \underbrace{\lambda^* \text{KL}(F_J \parallel F_\theta)}_{\text{size}}, \quad \text{KL}(F_J \parallel F_\theta) = \int_{\mathcal{J}} \log \frac{f(x)}{f_\theta(x)} F_J(dx). \quad (\text{A2})$$

Each term is nonnegative; the timing term is zero only at $\lambda = \lambda^*$, and the size term only when $f_\theta = f$, F_J -a.e. $\mathcal{K}$ is continuous on the compact Φ and attains its minimum, and because the two coordinates separate the minimizer factorizes,

$$\phi^* = (\lambda^*, \theta^*), \quad \theta^* = \arg \min_{\theta \in \Theta} \text{KL}(F_J \parallel F_\theta), \quad (\text{A3})$$

with θ^* unique by Assumption 4(iv). Finally $\mathcal{K}$ is a genuine per-unit-time rate: on any horizon $\text{KL}(\mathbb{P} \parallel \mathbb{P}^\phi) \big|_{\mathcal{F}_t} = t \mathcal{K}(\phi)$.

Proof of Lemma 3. Step 0 (the log-density is well defined, finite, and integrable). By Lemma 2, $\mathbb{P}|_{\mathcal{F}_1} \sim \mathbb{P}^\phi|_{\mathcal{F}_1}$, and evaluating (A1) at $t = 1$ and inverting,

$$\frac{d\mathbb{P}}{d\mathbb{P}^\phi} \bigg|_{\mathcal{F}_1} = \exp((\lambda - \lambda^*)) \prod_{k: T_k \leq 1} \frac{\lambda^* f(J_k)}{\lambda f_\theta(J_k)}.$$

Each factor is strictly positive: $\lambda, \lambda^* \in (0, \infty)$ gives $e^{(\lambda - \lambda^*)} > 0$; $f(J_k) > 0$ and $f_\theta(J_k) > 0$ (Assumption 4(i)) give $\lambda^* f(J_k) / (\lambda f_\theta(J_k)) > 0$; and $N_1 < \infty$ $\mathbb{P}$ -a.s. (Assumption 1) makes the product a finite product of strictly positive factors. Hence $d\mathbb{P}/d\mathbb{P}^\phi|_{\mathcal{F}_1} \in (0, \infty)$ $\mathbb{P}$ -a.s. For

integrability, the integrand of $\mathcal{K}(\phi)$ is, by (A4) below, $\log \frac{d\mathbb{P}}{d\mathbb{P}^\phi}|_{\mathcal{F}_1} = (\lambda - \lambda^*) + \sum_{k: T_k \leq 1} \log \frac{\lambda^* f(J_k)}{\lambda f_\theta(J_k)}$, so, with $B := \sup_{(x, \theta) \in \mathcal{J} \times \Theta} \left| \log \frac{f(x)}{f_\theta(x)} \right| < \infty$ (finite because $(x, \theta) \mapsto \log \frac{f(x)}{f_\theta(x)}$ is continuous on the compact $\mathcal{J} \times \Theta$, Assumption 4(i), by Weierstrass) and $C := \left| \log \frac{\lambda^*}{\lambda} \right| + B < \infty$, the triangle inequality gives

$$\left| \log \frac{d\mathbb{P}}{d\mathbb{P}^\phi} \Big|_{\mathcal{F}_1} \right| \leq |\lambda - \lambda^*| + \sum_{k: T_k \leq 1} \left(\left| \log \frac{\lambda^*}{\lambda} \right| + \left| \log \frac{f(J_k)}{f_\theta(J_k)} \right| \right) \leq |\lambda - \lambda^*| + C N_1.$$

Taking $\mathbb{P}$ -expectations and using $\mathbb{E}[N_1] = \lambda^* < \infty$,

$$\mathbb{E}_{\mathbb{P}} \left| \log \frac{d\mathbb{P}}{d\mathbb{P}^\phi} \Big|_{\mathcal{F}_1} \right| \leq |\lambda - \lambda^*| + C \mathbb{E}[N_1] = |\lambda - \lambda^*| + C \lambda^* < \infty,$$

so the integrand is $\mathbb{P}$ -integrable and $\mathcal{K}(\phi) = \mathbb{E}_{\mathbb{P}} \left[\log \frac{d\mathbb{P}}{d\mathbb{P}^\phi} \Big|_{\mathcal{F}_1} \right] = \text{KL}(\mathbb{P} \parallel \mathbb{P}^\phi |_{\mathcal{F}_1})$ is finite and well defined—the Kullback–Leibler divergence of the physical law from the held-model law, written throughout with the single ratio $d\mathbb{P}/d\mathbb{P}^\phi$.

Step 1 (the integrand). By Lemma 2, inverting (A1) at $t = 1$ and taking logs, the integrand of $\mathcal{K}$ is, with each jump factor split into a timing and a size piece,

$$\log \frac{d\mathbb{P}}{d\mathbb{P}^\phi} \Big|_{\mathcal{F}_1} = (\lambda - \lambda^*) + \sum_{k: T_k \leq 1} \left(\log \frac{\lambda^*}{\lambda} + \log \frac{f(J_k)}{f_\theta(J_k)} \right). \quad (\text{A4})$$

Step 2 (expectation of a jump sum). For any g with $\mathbb{E}_{F_J} |g| < \infty$, condition on $N_1 = n$; under $\mathbb{P}$ the marks are then i.i.d. F_J and independent of the count (Assumption 1), so

$$\begin{aligned} \mathbb{E}_{\mathbb{P}} \left[\sum_{k: T_k \leq 1} g(J_k) \Big| N_1 = n \right] &= n \mathbb{E}_{F_J} [g(J)], \\ \mathbb{E}_{\mathbb{P}} \left[\sum_{k: T_k \leq 1} g(J_k) \right] &= \sum_{n \geq 0} \mathbb{P}(N_1 = n) n \mathbb{E}_{F_J} [g] = \mathbb{E}[N_1] \mathbb{E}_{F_J} [g] = \lambda^* \mathbb{E}_{F_J} [g], \end{aligned}$$

using $\mathbb{E}[N_1] = \lambda^*$ on the unit interval.

Step 3 (apply Step 2 to the split term). With $g(x) = \log \frac{\lambda^*}{\lambda} + \log \frac{f(x)}{f_\theta(x)}$ (integrable by Step

0),

$$\mathbb{E}_{\mathbb{P}} \left[\sum_{k: T_k \leq 1} \log \frac{\lambda^* f(J_k)}{\lambda f_{\theta}(J_k)} \right] = \lambda^* \log \frac{\lambda^*}{\lambda} + \lambda^* \int_{\mathcal{J}} \log \frac{f(x)}{f_{\theta}(x)} F_J(dx) = \lambda^* \log \frac{\lambda^*}{\lambda} + \lambda^* \text{KL}(F_J \| F_{\theta}). \quad (\text{A5})$$

Step 4 (assemble). Taking $\mathbb{E}_{\mathbb{P}}$ of (A4) ($\lambda - \lambda^*$ is its own expectation) and substituting (A5), then regrouping the timing terms,

$$\mathcal{K}(\phi) = (\lambda - \lambda^*) + \lambda^* \log \frac{\lambda^*}{\lambda} + \lambda^* \text{KL}(F_J \| F_{\theta}) = \left[\lambda^* \log \frac{\lambda^*}{\lambda} - \lambda^* + \lambda \right] + \lambda^* \text{KL}(F_J \| F_{\theta}),$$

which is (A2).

Signs. Write $T(\lambda) = \lambda^* \log \frac{\lambda^*}{\lambda} - \lambda^* + \lambda = \lambda^* \log \lambda^* - \lambda^* \log \lambda - \lambda^* + \lambda$. Differentiating term by term in λ ($\lambda^* \log \lambda^*$ and $-\lambda^*$ constant),

$$T'(\lambda) = -\lambda^* \cdot \frac{1}{\lambda} + 1 = 1 - \frac{\lambda^*}{\lambda}, \quad T''(\lambda) = \frac{d}{d\lambda} \left(1 - \frac{\lambda^*}{\lambda} \right) = \frac{\lambda^*}{\lambda^2}. \quad (\text{A6})$$

By (A6) T' vanishes only at $\lambda = \lambda^*$, and $T''(\lambda) = \lambda^*/\lambda^2 > 0$ on $\Lambda \subset (0, \infty)$, so T is strictly convex with unique minimiser $\lambda = \lambda^*$ and $T(\lambda^*) = 0$; hence the timing term is ≥ 0 , zero only at $\lambda = \lambda^*$. By Gibbs' inequality $\text{KL}(F_J \| F_{\theta}) \geq 0$, zero iff $f_{\theta} = f$, F_J -a.e.; hence the size term is ≥ 0 with the stated equality. Both terms being ≥ 0 , $\mathcal{K} \geq 0$.

Continuity, minimiser, factorisation. T is continuous on Λ (bounded away from 0). For the size term, the KL integrand $x \mapsto \log(f(x)/f_{\theta}(x))$ is continuous in θ (Assumption 4(i)) and dominated by the integrable envelope B of Step 0 (which bounds the *log-ratio*, not $\log f_{\theta}$ alone), so by dominated convergence $\theta \mapsto \text{KL}(F_J \| F_{\theta})$ is continuous on the compact Θ ; thus $\mathcal{K}$ is continuous on the compact Φ and attains its minimum (extreme-value theorem). Since $\mathcal{K}(\lambda, \theta) = T(\lambda) + \lambda^* \text{KL}(F_J \| F_{\theta})$ separates additively, the joint minimiser factorises into $\arg \min_{\lambda} T = \{\lambda^*\}$ and $\arg \min_{\theta} \text{KL}(F_J \| F_{\theta}) = \{\theta^*\}$, the latter a singleton by Assumption 4(iv); this is (A3).

Rate. The marked Poisson record has stationary independent increments (Assump-

tion 1), so the densities multiply across disjoint unit blocks: on $\mathcal{F}_t$ with integer t , $\log \frac{d\mathbb{P}}{d\mathbb{P}^\phi} = \sum_{j=1}^t \log \frac{d\mathbb{P}}{d\mathbb{P}^\phi}|_{(j-1,j]}$, each block term i.i.d. with mean $\mathcal{K}(\phi)$ by Steps 0–4; hence $\text{KL}(\mathbb{P}||\mathbb{P}^\phi)|_{\mathcal{F}_t} = t\mathcal{K}(\phi)$, and the same expectation computation with $\mathbb{E}[N_t] = \lambda^*t$ extends it to all $t > 0$. Thus $\mathcal{K}$ is a per-unit-time rate. $\square$

The imported results, and our hypotheses. Mancini (2009, Thm. 3.1 and Cor. 3.2) consider $dX_t = a_t dt + \sigma_t dW_t + dJ_t$ with J a finite-activity jump process $J_t = \sum_{j=1}^{N_t} \gamma_j$, observed on a grid of mesh h over a fixed horizon. Under (1) a pathwise bound on the drift integral, which holds whenever a is càdlàg; (2) $\int_0^T \sigma_s^2 ds < \infty$ with σ pathwise bounded, which holds whenever σ is càdlàg; (3) a deterministic threshold $r(h)$ with $r(h) \rightarrow 0$ and $h \log(1/h)/r(h) \rightarrow 0$; and (4) $\mathbb{P}\{\Delta N_t \neq 0, \gamma_{N_t} = 0\} = 0$, the threshold a.s. isolates the jump-free intervals, $\mathbf{1}\{\Delta_i N = 0\} = \mathbf{1}\{(\Delta_i X)^2 \leq r(h)\}$ for small h , and the truncated realized variation is consistent for the integrated variance,

$$\text{plim}_{h \rightarrow 0} \sum_i (\Delta_i X)^2 \mathbf{1}\{(\Delta_i X)^2 \leq r(h)\} = \int_0^T \sigma_s^2 ds.$$

Mancini (2009, Thm. 3.7) adds that if J is a *compound Poisson* process, if σ is adapted, continuous, and *independent of W and N* (no leverage), with $\mathbb{E}[\int_0^T \sigma_s^2 ds] < \infty$, then the truncation estimator of each realized size, $\hat{\gamma}^{(i)} = \Delta_i X \mathbf{1}\{(\Delta_i X)^2 > r(h)\}$, satisfies

$$\sqrt{n} \sum_i (\hat{\gamma}^{(i)} - \gamma^{(i)}) \mathbf{1}\{\Delta_i N \geq 1\} \xrightarrow{d} MN\left(0, T \int_0^T \sigma_s^2 dN_s\right). \quad (\text{A7})$$

Our endowment (4) is the special case $a_t \equiv \mu_0 - \frac{1}{2}\sigma^2$, $\sigma_t \equiv \sigma$, $J_t = -\sum_{k:T_k \leq t} J_k$ with N $\text{Poisson}(\lambda^*)$ and marks $\{J_k\}$ i.i.d. F_J . We verify each hypothesis. The drift is constant, hence càdlàg, so (1) holds; σ is constant, hence càdlàg with $\int_0^T \sigma^2 ds = \sigma^2 T < \infty$, so (2) holds; take $r(h) = h^{0.99}$, for which $r(h) \rightarrow 0$ and $h \log(1/h)/r(h) = h^{0.01} \log(1/h) \rightarrow 0$, so (3) holds; the marks satisfy $J_k \in [\underline{J}, \overline{J}]$ with $\underline{J} > 0$, so $\gamma_{N_t} = -J_{N_t} \neq 0$ a.s. and (4) holds. For (A7): J is compound Poisson by construction; σ is constant, hence adapted, continuous, and (being

deterministic) independent of W and N , so the no-leverage hypothesis holds; and $\mathbb{E}[\int_0^T \sigma^2 ds] = \sigma^2 T < \infty$. All hypotheses of both results hold in our model.

What the path identifies, and what it does not. A model could in principle differ from the truth in the diffusion coefficient—an admissibility counterfactual outside the maintained space $\phi = (\lambda, \theta)$ —in the arrival intensity, or in the size law, and these have different fates. The diffusion coefficient and the realized disaster sizes are *pathwise functionals* of the record: recovered, in the high-frequency limit, from the path on any interval, so a wrong diffusion is refuted on every interval. The intensity and the size *law* are not functionals of any finite record; they are properties of the generating law, learned only from the disasters that accrue at the rare-disaster rate. That asymmetry is what makes the joint (λ, θ) the admissible object of persistent disagreement.

In the high-frequency limit the path delivers the diffusion coefficient and every realized disaster size exactly (Mancini, 2009; Aït-Sahalia and Jacod, 2009), leaving no room to disagree past the first interval; in the long-horizon limit the intensity and the size law are still being learned, at the common rate $(\lambda^* t)^{-1/2}$ in relative (log-intensity) error—the effective sample is the disaster count—because both are inferred from a sample that grows only as fast as disasters arrive (Lemma IA.1). A disagreement about σ cannot survive a single interval; a disagreement about (λ, θ) survives for a length of time set by $1/\lambda^*$. That is the property a disputed primitive must have. The intensity coordinate is the object of the companion foundation paper (Tegui, 2026); here it is carried jointly with the size law, the two learned in parallel from the same rare disasters. The identification asymmetry between the physical law and the held cross-section is stated and proved in Internet Appendix IA.3.

Appendix B. Proofs for Section 4

Sufficiency of the since-adoption record. For an exponential-family size law with natural sufficient statistic T , the pair $(n, \sum_j T(x_j))$ is jointly sufficient for (λ, θ) by the Fisher–Neyman factorisation criterion, so the deviance (9) depends on the record only through it (the predictive density (8) additionally carries the marks’ base-measure factors, which cancel in the deviance ratio); for a general family the test uses a goodness-of-fit statistic of (8) and no sufficiency is claimed. It is recorded for completeness—it is why exponential families are computationally special—but no result in the main text relies on it.

Proof of Proposition 4. By Proposition 9, $m_t = \mathsf{T}_{t\#}G$ with $\mathsf{T}_t(\cdot)$ deterministic, so the held-model marginal is the image of G under the threshold-to-trajectory map. Under (a): at T_k the rejecting cohort relocates to the common $\phi^{(k)}$ (Proposition 8) while the non-rejecting cohort of positive G -mass retains ϕ_0 ; since the destination differs in intensity by hypothesis, $\lambda(\phi^{(k)}) \neq \lambda(\phi_0)$, the held-model marginal places positive mass on two distinct intensities, whence $\text{Var}_{S_{T_k}}(\lambda) > 0$. (We invoke only distinctness: the revealed-preference monotonicity of Lemma 4 is a fixed-count comparison and does not by itself sign $\lambda(\phi^{(k)}) - \lambda(\phi_0)$ once the elapsed calm time at T_k enters.) Under (b): along the zero-count history the reviser of threshold ε adopts the calm mode $\lambda_{\Delta^*(\phi_0, \varepsilon)}^m$ at her deadline; with the deadline map non-constant on a positive-mass set and $\tau \mapsto \lambda_\tau^m$ strictly decreasing, two positive-mass groups of thresholds adopt distinct intensities before the first disaster, whence $\text{Var}_{S_t}(\lambda) > 0$ for t past the earliest such deadline. *From head count to consumption shares.* Each case produces a split of positive G -mass, hence positive head-count mass m_t (Proposition 9); the variance claim is in the consumption-share measure S_t . The two are mutually absolutely continuous: with the symmetric allocation (Assumption 3(iv)) the share is $s_t^i = (M_t^i)^{1/\alpha} / \int (M_t^j)^{1/\alpha} dj$, and M_t^i is bounded and bounded away from zero on the compact menu: by (C1), with $N_t < \infty$ (Assumption 1) the density is a finite product of reweights $r_i = \lambda_i f_{\theta_i} / (\lambda^* f)$ —each continuous and strictly positive on the compact $\Phi \times \mathcal{J}$ (Assumption 4), hence in a fixed positive range—times the intensity

factors accrued segment-by-segment over the full history—the density is the product across held-model segments—each of the form $e^{-(\lambda_i(u)-\lambda^*)\times\text{length}}$ and bounded for $\lambda_i \in [\underline{\lambda}, \bar{\lambda}]$ (a revision only relabels ϕ_i and restarts the segment, preserving the two-sided bound); so there are $0 < \underline{c} \leq \bar{c} < \infty$ with $\underline{c}m_t(A) \leq S_t(A) \leq \bar{c}m_t(A)$ for every Borel A . A positive- m_t -mass split into two intensity classes is therefore a positive- S_t -mass split. In either case the dispersion is strictly positive at a finite t , from the homogeneous start. $\square$

Lemma 4 (The second-order posterior is common): *For every pair of investors i, j and every t , $\rho_t^i = \rho_t^j =: \rho_t$. The common posterior is continuous in t between disasters and jumps at each disaster epoch; during a disaster-free spell its maximum-a-posteriori intensity $\lambda_t^{\text{m}} := \min \arg \max_{\lambda} \max_{\theta} \rho_t(\lambda, \theta)$ (the intensity coordinate of the tie-broken mode $\mathbf{m}(\rho_t)$ of Definition 1) is non-increasing in t .*

Proof of Lemma 4. By (6), $\rho_t^i(\phi) \propto \rho_0(\phi) L_t(\phi)$ with $L_t(\phi) = e^{-\lambda t} \lambda^{N_t} \prod_{k:T_k \leq t} f_{\theta}(J_k)$. The prior ρ_0 is common across investors (Assumption 2) and L_t is a function of the common path $(t, N_t, \{J_k\})$ alone, with no dependence on i or ε^i ; hence $\rho_t^i = \rho_t^j =: \rho_t$. Continuity in t follows from L_t and its normalizer: $L_t(\phi)$ is continuous in t off the disaster epochs and jumps at them, and the normalizing constant $Z_t = \int_{\Phi} \rho_0(\phi) L_t(\phi) d\phi$ is continuous and strictly positive *on disaster-free intervals* (a continuous, strictly positive integrand on the compact Φ) and jumps at each disaster epoch. At a disaster of size J_k , $L_t(\lambda, \theta)$ is multiplied pointwise by the model-specific factor $\lambda f_{\theta}(J_k)$, while the normalizer Z_t is multiplied by its scalar posterior-predictive average $\int_{\Phi} \lambda f_{\theta}(J_k) \rho_{t-}(d\phi)$, *not* by any model-dependent factor; the posterior $\rho_t = \rho_0 L_t / Z_t$ therefore jumps precisely because the numerator updates model by model and the denominator by this common scalar average, and is continuous between disasters. This gives the continuity-between-disasters and disaster-jump claims of the statement.

For the monotonicity, fix a disaster-free interval, on which N_t and the sizes are constant.

By the product prior $\rho_0 = \rho_0^\Lambda \otimes \rho_0^\Theta$ (Assumption 2), the log-posterior separates,

$$\log \rho_t(\lambda, \theta) = \underbrace{\left[\log \rho_0^\Lambda(\lambda) - \lambda t + N_t \log \lambda \right]}_{=: \ell_t^\Lambda(\lambda)} + \underbrace{\left[\log \rho_0^\Theta(\theta) + \sum_k \log f_\theta(J_k) \right]}_{\theta\text{-only}} + \text{const.}$$

The θ -only term does not depend on λ , so $\max_\theta \log \rho_t(\lambda, \theta) = \ell_t^\Lambda(\lambda) + \text{const}$ and the maximum-a-posteriori intensity is $\lambda_t^m = \min \arg \max_{\lambda \in \Lambda} \ell_t^\Lambda(\lambda)$. The maximizer set is non-empty because ℓ_t^Λ is continuous (hence upper semicontinuous) on the compact interval Λ , so $\lambda_t^m = \min \arg \max_{\lambda \in \Lambda} \ell_t^\Lambda(\lambda)$ is well defined. We prove monotonicity directly, by revealed preference. Fix $t_2 > t_1$ in the disaster-free interval and write $\lambda_1 := \lambda_{t_1}^m$, $\lambda_2 := \lambda_{t_2}^m$. Each is optimal at its own date:

$$\ell_{t_1}^\Lambda(\lambda_1) \geq \ell_{t_1}^\Lambda(\lambda_2), \quad \ell_{t_2}^\Lambda(\lambda_2) \geq \ell_{t_2}^\Lambda(\lambda_1).$$

Adding the two inequalities and regrouping,

$$\left[\ell_{t_2}^\Lambda(\lambda_2) - \ell_{t_1}^\Lambda(\lambda_2) \right] - \left[\ell_{t_2}^\Lambda(\lambda_1) - \ell_{t_1}^\Lambda(\lambda_1) \right] \geq 0. \quad (\text{B1})$$

Since N_t and the sizes are constant on the interval (write N for the common count), the prior and count terms of ℓ^Λ cancel in the difference at any fixed λ :

$$\ell_{t_2}^\Lambda(\lambda) - \ell_{t_1}^\Lambda(\lambda) = \left[\log \rho_0^\Lambda(\lambda) - \lambda t_2 + N \log \lambda \right] - \left[\log \rho_0^\Lambda(\lambda) - \lambda t_1 + N \log \lambda \right] = -\lambda(t_2 - t_1). \quad (\text{B2})$$

Evaluating (B2) at $\lambda = \lambda_2$ and $\lambda = \lambda_1$, the left-hand side of (B1) equals

$$\left[-\lambda_2(t_2 - t_1) \right] - \left[-\lambda_1(t_2 - t_1) \right] = -(t_2 - t_1)(\lambda_2 - \lambda_1).$$

Thus

$$-(t_2 - t_1)(\lambda_2 - \lambda_1) \geq 0,$$

and since $t_2 > t_1$ this forces $\lambda_2 \leq \lambda_1$, i.e. $\lambda_{t_2}^m \leq \lambda_{t_1}^m$. Hence λ_t^m is non-increasing in t through

the disaster-free spell. □

Proposition 7 (Law of motion; scheduled intensity relief): *Fix investor i . Between adoptions the held model ϕ_i is constant and the regime age advances at unit rate, $\dot{\tau}_t^i = 1$; the revision time is the first exit of the record from the typical set,*

$$\tau_i^{\text{rev}} = \inf\{t > a^i : \bar{\varepsilon}(\mathcal{R}_t^i; \phi_i, \tau_t^i) \leq \varepsilon^i\}. \quad (\text{B3})$$

Two exit channels generate it. (Calm.) If no disaster occurs after adoption, the record is the zero-count history, whose surprise $\bar{\varepsilon}$ tends to 0 as the age grows—the zero-count record becomes atypical of a process expected to deliver disasters; hence there is a finite, deterministic calm deadline

$$\Delta^*(\phi_i, \varepsilon^i) = \inf\{\Delta > 0 : \bar{\varepsilon}((0, \emptyset); \phi_i, \Delta) \leq \varepsilon^i\} < \infty, \quad (\text{B4})$$

at which the held model is rejected absent any disaster; the two-argument $\Delta^(\phi, \varepsilon)$ denotes this zero-count case. (Disaster.) At a disaster the record jumps—count +1 and a new size—and $\bar{\varepsilon}$ is re-evaluated, so a disaster is an exit if it lands the record outside the typical set. The revision time is the first exit of the updated record—the earlier of the current calm deadline and the next disaster that lands outside the typical set; after a non-rejecting disaster the count is $n \geq 1$, so the relevant calm deadline is the first passage of the deviance for that updated record, not the original zero-count deadline. At a calm rejection reached along a spell that is disaster-free since adoption, the re-selected intensity satisfies $\lambda_{\tau_i^{\text{rev}}}^{\text{m}} \leq \lambda_i$; an intervening non-rejecting disaster updates the record to count $n \geq 1$ and can raise the mode, so this scheduled relief is the disaster-free statement, relative to the start of the current disaster-free spell.*

Proof of Proposition 7. By Definition 1 the held model equals the second-order mode fixed at the last adoption and is unchanged until the next rejection, while the regime age advances at unit rate; this gives constancy and $\dot{\tau}_t^i = 1$. By Definition 1 a rejection is the first time the deviance exceeds its threshold, equivalently the first time $\bar{\varepsilon} \leq \varepsilon^i$, which is (B3).

Calm channel. Suppose no disaster occurs after adoption, so the record at age $\Delta = \tau_t^i$ is the zero-count history $(0, \emptyset)$. Its deviance is the pure count term, by (9) and $d_P(0||\lambda) = \lambda$,

$$D_{\phi_i, \Delta}((0, \emptyset)) = \Delta d_P(0||\lambda_i) = \lambda_i \Delta,$$

and by (10) its surprise is the predictive mass of records at least as deviant, which Markov's inequality (valid as $D \geq 0$) bounds by the mean deviance under the held model,

$$\bar{\varepsilon}((0, \emptyset); \phi_i, \Delta) = Q_{\phi_i, \Delta}(\{\mathcal{R}' : D_{\phi_i, \Delta}(\mathcal{R}') \geq \lambda_i \Delta\}) \leq \frac{\mathbb{E}_{Q_{\phi_i, \Delta}}[D_{\phi_i, \Delta}]}{\lambda_i \Delta}. \quad (\text{B5})$$

We show the bound vanishes as $\Delta \rightarrow \infty$. Under $Q_{\phi_i, \Delta}$ the count is $n \sim \text{Poisson}(m)$ with $m := \lambda_i \Delta$, and given n the sizes are i.i.d. F_{θ_i} ; by (9) the deviance splits as $D = c(n) + \Lambda_n$ into the count deviance $c(n) = n \log(n/m) - n + m \geq 0$ (with $c(0) = m$) and the size deviance $\Lambda_n = \sum_{k=1}^n \log \frac{f_{\hat{\theta}}(J_k)}{f_{\theta_i}(J_k)} = \sup_{\theta \in \Theta} \sum_{k=1}^n \log \frac{f_{\theta}(J_k)}{f_{\theta_i}(J_k)} \geq 0$, and we bound each mean.

Count mean: bounded. Write the count deviance as $c(n) = n \log(n/m) - n + m = n \log n - n \log m - n + m$. Taking the Q -expectation with $n \sim \text{Poisson}(m)$, so $\mathbb{E}[n] = m$ and hence $\mathbb{E}[-n + m] = 0$,

$$\mathbb{E}[c(n)] = \mathbb{E}[n \log n] - (\log m) \mathbb{E}[n] + \mathbb{E}[-n + m] = \mathbb{E}[n \log n] - m \log m. \quad (\text{B6})$$

For the first term, the $n = 0$ summand is zero and $n/n! = 1/(n-1)!$, so shifting the index by $j = n - 1$ (with $M \sim \text{Poisson}(m)$),

$$\begin{aligned} \mathbb{E}[n \log n] &= \sum_{n \geq 1} e^{-m} \frac{m^n}{n!} n \log n = \sum_{n \geq 1} e^{-m} \frac{m^n}{(n-1)!} \log n \\ &= m \sum_{j \geq 0} e^{-m} \frac{m^j}{j!} \log(j+1) = m \mathbb{E}[\log(M+1)]. \end{aligned} \quad (\text{B7})$$

Substituting (B7) into (B6) and combining the logarithms,

$$\mathbb{E}[c(n)] = m \mathbb{E}[\log(M+1)] - m \log m = m \mathbb{E}\left[\log \frac{M+1}{m}\right] \leq m \log \frac{m+1}{m} \leq m \cdot \frac{1}{m} = 1,$$

the first inequality by Jensen (log concave, $\mathbb{E}[M+1] = m+1$) and the second by $\log(1+x) \leq x$. Thus $\mathbb{E}[c(n)] \leq 1$ for every $\Delta > 0$.

Size mean: $o(m)$. The map $(\theta, x) \mapsto \log f_\theta(x)$ is continuous on the compact $\Theta \times \mathcal{J}$ (Assumption 4(i)), hence bounded, so $B := \sup_{\theta, \theta' \in \Theta, x \in \mathcal{J}} \left| \log \frac{f_\theta(x)}{f_{\theta'}(x)} \right| < \infty$ and $0 \leq \Lambda_n \leq Bn$. We invoke the uniform strong law of large numbers in the form underlying maximum-likelihood consistency (Wald, 1949): if Θ is compact, $\theta \mapsto \log f_\theta(x)$ is continuous for each x , and the envelope is F_{θ_i} -integrable, then

$$\sup_{\theta \in \Theta} \left| \frac{1}{n} \sum_{k \leq n} \log \frac{f_\theta(J_k)}{f_{\theta_i}(J_k)} - (-\text{KL}(F_{\theta_i} \| F_\theta)) \right| \xrightarrow{\text{a.s.}} 0 \quad (n \rightarrow \infty). \quad (\text{B8})$$

Each hypothesis holds in our model: Θ is compact (Assumption 4), $\log f_\theta(x)$ is continuous in θ (Assumption 4(i)–(ii)), and the constant envelope B is F_{θ_i} -integrable, $\int B dF_{\theta_i} = B < \infty$. Since $\sup_\theta (-\text{KL}(F_{\theta_i} \| F_\theta)) = 0$, attained at $\theta = \theta_i$, (B8) gives $\Lambda_n/n = \sup_\theta \frac{1}{n} \sum_{k \leq n} \log \frac{f_\theta}{f_{\theta_i}}(J_k) \rightarrow 0$ a.s., with $0 \leq \Lambda_n/n \leq B$. As $\Delta \rightarrow \infty$, $m \rightarrow \infty$ and $n \rightarrow \infty$ a.s., and $n/m \rightarrow 1$ a.s. with $\{n/m\}$ uniformly integrable ($\mathbb{E}[(n/m)^2] = 1 + 1/m$ is bounded). Hence $\Lambda_n/m = (\Lambda_n/n)(n/m) \rightarrow 0$ a.s. and is dominated by the uniformly integrable $B(n/m)$, so by Vitali's convergence theorem $\mathbb{E}[\Lambda_n]/m \rightarrow 0$.

Conclusion. Combining the two means, $\mathbb{E}_{Q_{\phi_i, \Delta}}[D]/(\lambda_i \Delta) \leq 1/m + \mathbb{E}[\Lambda_n]/m \rightarrow 0$, so by (B5) $\bar{\varepsilon}((0, \emptyset); \phi_i, \Delta) \rightarrow 0$ as $\Delta \rightarrow \infty$. As $\varepsilon^i > 0$, the set $\{\Delta > 0 : \bar{\varepsilon}((0, \emptyset); \phi_i, \Delta) \leq \varepsilon^i\}$ is non-empty (it contains the non-empty strict sublevel set), so its infimum—the calm deadline (B4)—is finite, $\Delta^*(\phi_i, \varepsilon^i) < \infty$. By definition of the infimum, $\bar{\varepsilon} \geq \varepsilon^i$ for every $\Delta < \Delta^*$, so the held model survives up to Δ^* and is rejected there: Δ^* is the first crossing, and no monotonicity of $\bar{\varepsilon}$ in the age is needed. The infimum is attained: on a disaster-free stretch $\Delta \mapsto \bar{\varepsilon}((0, \emptyset); \phi, \Delta)$

is continuous (the exact Poisson tail is continuous in the mean $\lambda\Delta$), so the sublevel set is closed and $\bar{\varepsilon}(\Delta^*) \leq \varepsilon$; a disaster before Δ^* updates the record and the deadline is re-evaluated on it, the first-exit convention of Definition 1. The count contributes atoms to the deviance, so $\Delta \mapsto \bar{\varepsilon}((0, \emptyset); \phi_i, \Delta)$ may step downward across ε^i and need not attain it; the closed-threshold convention of Definition 1—reject at the first age with $\bar{\varepsilon} \leq \varepsilon^i$ —fixes the boundary, and rejection occurs exactly at the infimum Δ^* . It is deterministic, since $\bar{\varepsilon}((0, \emptyset); \phi_i, \cdot)$ is a function of (ϕ_i, Δ) alone. Finally the deadline is non-increasing in ε^i : a larger threshold enlarges the rejection set $\{\Delta : \bar{\varepsilon} \leq \varepsilon^i\}$, lowering its infimum—which, as all investors share ϕ_0 at the start, is what staggers calm revision across the population.

Disaster channel. At a disaster the record gains a count and a size, so the deviance jumps and is re-evaluated; the record exits the typical set there iff the post-disaster deviance exceeds the threshold. The revision time is the earlier of the calm deadline and the first such disaster.

Relief (disaster-free spell). Consider a calm rejection reached along a spell disaster-free since adoption a^i , with re-selected intensity the mode $\lambda_{\tau_i}^{\text{m rev}}$ and incumbent the mode at adoption $\lambda_i = \lambda_{a^i}^{\text{m}}$. By Lemma 4 the mode is non-increasing *through a disaster-free spell*, so $\lambda_{\tau_i}^{\text{m rev}} \leq \lambda_i$. If instead a non-rejecting disaster intervenes it appends a mark and can raise the mode—Lemma 4 bounds the mode only along disaster-free motion—so the bound then holds relative to the mode at the start of the *current* disaster-free spell, not necessarily to λ_i . This is the scheduled relief that erodes the held intensity *between* disasters; disasters themselves enter through the disaster reweight and relocation, not the calm relief. $\square$

Proposition 8 (Synchronization at disasters; staggered calm): *At a disaster epoch T_k every investor's record receives the same increment at the same instant. Within the cohort of investors sharing a held model ϕ and adoption date a at T_k^- (hence the same record and predictive law), the post-disaster surprise depth is a common value $\bar{\varepsilon}_k(\phi, a)$, so the investors of that cohort who revise are exactly those with $\varepsilon^i \geq \bar{\varepsilon}_k(\phi, a)$; and every investor who revises at T_k , across all*

cohorts, adopts the common mode $\phi^{(k)} = \mathbf{m}(\rho_{T_k})$ (Definition 1). Revision at a disaster is thus synchronized in timing and destination, with a cohort-specific trigger. Away from disasters, investor i revises at her calm deadline $a^i + \Delta^*(\phi_i, \varepsilon^i)$, evaluated on the record accrued since her current adoption (a post-adoption disaster that does not itself trigger rejection resets the deadline on the updated, positive-count record; the zero-count expression is the disaster-free special case); we write $\Delta^*(\phi, \varepsilon)$ for the zero-count deadline and leave the record argument implicit on positive-count records—a time non-increasing in ε^i (Proposition 7), so calm revisions are staggered across thresholds wherever that deadline map is non-constant. Starting from the homogeneous configuration of Assumption 2, both channels generate dispersion rather than assume it—the synchronized disaster split and the staggered calm relief each disperse the held intensity, under conditions made precise in Proposition 4 (see also Remark 4).

Proof of Proposition 8. At a disaster T_k the path increments by a single count and the common size J_k , which enters every investor’s record simultaneously. Two investors of the same cohort—same held ϕ and same adoption date a at T_k^- —have, by construction, the identical record on $(a, T_k]$ and hence the identical post-disaster record; since the deviance (9) and the surprise (10) are deterministic functions of $(\mathcal{R}, \phi, \Delta)$, both share a common post-disaster surprise $\bar{\varepsilon}_k(\phi, a)$. By Definition 1 cohort member i rejects iff $\varepsilon^i \geq \bar{\varepsilon}_k(\phi, a)$. Every investor who rejects at T_k , of any cohort, re-selects $\arg \max_{\phi} \rho_{T_k}^i(\phi)$, which by Lemma 4 is the common posterior ρ_{T_k} ; the deterministic tie-break $\mathbf{m}$ of Definition 1 then selects the single common point $\phi^{(k)} = \mathbf{m}(\rho_{T_k}) \in \arg \max_{\phi} \rho_{T_k}(\phi)$. The destination depends on the common full-history posterior alone—not on the reviser’s held model or adoption date—so it is common to all revisers, while the trigger $\bar{\varepsilon}_k(\phi, a)$ is constant only within a cohort.

In calm, by Proposition 7 investor i revises at $a^i + \Delta^*(\phi_i, \varepsilon^i)$, a deadline *non-increasing* in ε^i ; investors of a common cohort therefore revise at times ordered (weakly) by ε^i —staggered across thresholds with *distinct* deadlines, and simultaneous on any level set where $\Delta^*(\phi, \cdot)$ is flat. From the homogeneous start (Assumption 2) all hold ϕ_0 at $t = 0$ with common second-

order prior; wherever the deadline map $\Delta^*(\phi_0, \cdot)$ is non-constant across thresholds (Remark 4) the population begins to disperse in calm before any disaster, and each disaster induces a synchronized split—dispersion is generated, not assumed. $\square$

Remark 4 (When the held-model marginal has continuum support). The size-only design produced a finitely supported held-model marginal because all revision was synchronized at disasters. The calm channel can enlarge the support: if, on a sub-interval of $(0, 1)$, the calm deadline $\varepsilon \mapsto \Delta^*(\phi_0, \varepsilon)$ is continuous and strictly monotone and the calm intensity mode $\tau \mapsto \lambda_\tau^m$ is strictly decreasing, then the first-rejection cohort adopts a continuum of distinct intensities and the held-model marginal is not finitely supported. In pure calm $\ell_\tau^\Lambda(\lambda) = \log \rho_0^\Lambda(\lambda) - \lambda\tau$ has strictly decreasing differences in (λ, τ) —for $\lambda' > \lambda$, $\ell_\tau^\Lambda(\lambda') - \ell_\tau^\Lambda(\lambda) = \log \frac{\rho_0^\Lambda(\lambda')}{\rho_0^\Lambda(\lambda)} - (\lambda' - \lambda)\tau$ is strictly decreasing in τ from the linear $-\lambda\tau$ term alone—so by Topkis’s theorem the maximiser λ_τ^m is non-increasing in τ , with no condition on ρ_0^Λ . The *strict* interior decline is the separate, stronger property: it holds when ρ_0^Λ is strictly log-concave (the first-order condition then has a strictly decreasing interior root), while at the boundary λ_τ^m can plateau (no interior maximizer is assumed). The strict monotonicity of $\Delta^*(\phi_0, \cdot)$ requires, in *addition*, that the zero-count surprise $\bar{\varepsilon}$ be strictly decreasing in the age—a deadline-regularity condition not implied by log-concavity alone, which we impose where this remark is invoked. Absent either regularity the support may remain finite; the claim is scoped to these conditions and is not used downstream.

Proposition 9 (The cross-section is a pushforward of G): *Fix the common path. Let $\mathsf{T}_t(\varepsilon)$ be the held model and regime age that Definition 1 produces at t for an investor of threshold ε , started from ϕ_0 at age 0. Then T_t is a deterministic measurable map and the cross-sectional distribution of (held model, regime age) is the pushforward*

$$m_t = \mathsf{T}_{t\#} G, \tag{B9}$$

with no law of large numbers invoked: G is the primitive continuum type law of Assumption 2, not a random sample, so the transport is exact. The held-model marginal jumps at disasters,

where all revisers move to the common $\phi^{(k)}$, and changes at the staggered calm deadlines $a^i + \Delta^*(\phi_i, \varepsilon^i)$, where the reviser of threshold ε moves to the common mode prevailing at her deadline.

Proof of Proposition 9. Fix the common path. By Definition 1 and Proposition 7, given the path an investor's revision times are determined by her threshold ε alone—the calm deadline (B4) and the disaster rejections $\{T_k : \varepsilon > \bar{\varepsilon}_k\}$ are deterministic functions of ε —and the re-selected models are the common modes prevailing at those instants (Lemma 4), also path-determined. Hence $\varepsilon \mapsto \mathsf{T}_t(\varepsilon) = (\phi(t; \varepsilon), \tau(t; \varepsilon))$ is a deterministic map of the threshold. It is Borel measurable. On $[0, t]$ there are a.s. finitely many disasters; *between* consecutive disasters an investor may undergo several calm rejections, each resetting her regime, but for a fixed $\varepsilon > 0$ the calm deadline is bounded below, $\Delta^*(\phi, \varepsilon) \geq \delta(\varepsilon) > 0$ (the fresh record itself is the deepest point of its own deviance tail, so its surprise is at least $e^{-\bar{\lambda}\Delta}$, exceeding ε for $\Delta < \bar{\lambda}^{-1} \log(1/\varepsilon)$, uniformly over Φ —a short no-rejection window after every reset), so only finitely many calm resets occur in $[0, t]$. Thus $\varepsilon \mapsto \mathsf{T}_t(\varepsilon)$ is measurable by countable partition on the reset count: on $\{K_t = k\}$ it is the k -fold composition of the calm flow and the jump update, alternating calm and disaster steps—each measurable: the calm deadline $\varepsilon \mapsto \Delta^*(\phi, \varepsilon)$ is monotone hence Borel (Proposition 7), and the re-selected mode is a measurable function of the record—while $\{K_t \leq k\}$ is a finite intersection of measurable deadline and trigger comparisons; each fixed ε has finitely many resets before t by the calm-deadline floor $\delta(\varepsilon) > 0$, so the cells exhaust $(0, 1)$ and T_t is Borel on their union (the piecewise-deterministic construction of Davis, 1993; Jacobsen, 2006): the calm deadline $\varepsilon \mapsto \Delta^*(\phi, \varepsilon)$ is monotone in ε (Proposition 7), hence Borel, and the calm re-selected mode is a measurable function of the path-determined record at the deadline; the disaster-rejection threshold at T_k is not a population constant but the state-dependent surprise depth $\bar{\varepsilon}_k(\phi(T_k^-; \varepsilon), a(\varepsilon))$, a measurable function of the (already measurable) pre-disaster held model and adoption date, so $\{\varepsilon : \varepsilon > \bar{\varepsilon}_k(\cdot)\}$ is Borel as the preimage of a measurable map even though it need not be a fixed sub-interval of $(0, 1)$; and the post-disaster mode $\phi^{(k)} = \mathbf{m}(\rho_{T_k})$ is a measurable function of the common record. A finite alternating composition of these measurable maps is Borel—without asserting that $\phi(t; \cdot)$ is constant on

threshold intervals, which the staggered calm deadlines and the state-dependent $\bar{\varepsilon}_k$ may violate. Since investors differ only in the permanent threshold, whose population distribution is G (Assumption 2), and all share the one path, each investor's state is the deterministic image $\mathbb{T}_t(\varepsilon^i)$ of her type; the cross-sectional distribution of states is therefore the image measure $m_t = \mathbb{T}_{t\#}G$ by construction—there is no idiosyncratic randomness to average, so no law of large numbers is invoked. This is (B9). The held-model marginal jumps at disasters, where all revisers move to the common $\phi^{(k)}$ (Proposition 8), and changes at the calm deadlines $a^i + \Delta^*(\phi_i, \varepsilon^i)$, where the reviser of threshold ε moves to the common mode prevailing at her deadline (Lemma 4). $\square$

Appendix C. Proofs for the equilibrium kernel and the disaster-fear bridge

The belief density is a positive martingale. We verify the claim after Definition 6. Under $\mathbb{P}$ the disaster measure μ has $(\mathcal{F}_t, \mathbb{P})$ -compensator $\nu(dt, dx) = \lambda^* dt F_J(dx)$ (5). The held model $\phi_i(t-) = (\lambda_i(t-), \theta_i(t-))$ is predictable (left-continuous and piecewise constant, Definition 1) and proposes the compensator $\nu^i(dt, dx) = \lambda_i(t-) dt F_{\theta_i(t-)}(dx)$. The process (C1) is the Radon–Nikodym density of the marked-point-process law with compensator ν^i against that with ν (Brémaud, 1981, VI, Theorems 2–4), which is a positive $(\mathcal{F}_t, \mathbb{P})$ -local martingale solving $dM_t^i = M_{t-}^i \int_{\mathcal{J}} (r_i(x; t) - 1) (\mu - \nu)(dt, dx)$. We check the two hypotheses of Brémaud (1981, VI, T4) under which it is a true martingale with $\mathbb{E}[M_t^i] = 1$. (a) *Bounded intensity.* $\Lambda = [\underline{\lambda}, \bar{\lambda}] \subset (0, \infty)$ (Assumption 4), so $\int_0^t \lambda_i(s-) ds \leq \bar{\lambda} t < \infty$. (b) *Bounded likelihood ratio.* The map $(x, \phi) \mapsto r(x; \phi) = \lambda f_{\theta}(x) / (\lambda^* f(x))$ is continuous and strictly positive on the compact $\mathcal{J} \times \Phi$ (full support, Assumption 4(i)), hence $0 < \underline{r} := \min_{\mathcal{J} \times \Phi} r \leq r_i(x; t) \leq \bar{r} := \max_{\mathcal{J} \times \Phi} r < \infty$. Conditions (a)–(b) give $\mathbb{E}[\int_0^t \int_{\mathcal{J}} |r_i(x; s) - 1| \nu(ds, dx)] \leq (\bar{r} + 1) \bar{\lambda} t < \infty$, so the stochastic integral is a true martingale and $\mathbb{E}[M_t^i] = 1$ (Brémaud, 1981, VI, T4). Positivity is immediate from $r_i > 0$. $\square$

Independent increments and the realized-density representation. The product form $\prod_k r_i(J_k; T_k)$ is the independent-increment factorization of the Poisson record—each disaster contributing its own likelihood ratio, with no cross-disaster interaction *given the held-model path*; revisions, which change ϕ_i , are the only channel coupling disasters. The same structure makes the since-adoption segment likelihood the *conditional* Radon–Nikodym increment given the pre-adoption past, so testing the held model on its since-adoption record (Definition 1) is exact as conditional testing—a property of the Poisson record’s independent increments, not of general locally equivalent laws. The ratio M_T^i/M_t^i is built from the predictable held-model process $\phi_i(s-)$ over $(t, T]$, the models adopted at the agent’s own future revisions included, which makes M^i a $\mathbb{P}$ -martingale and feeds the Negishi construction of Appendix F. The valuation primitive is *sophisticated with respect to the rule*: the investor evaluates consumption plans under this realized-rule density, her own future revisions included, so her problem is time-consistent, the martingale method applies without qualification, and her wealth at each revision is the equilibrium continuation value by definition rather than by convention. Behaviorally she remains the paper’s tester—at every date she prices the instantaneous objects under her currently held model, dogmatically, and at a rejection she re-solves under the re-selected model. Lemma 1 establishes that the sophistication is not load-bearing for prices: the realized path satisfies the naive Euler equation on every inter-revision interval, and the concatenation of those first-order conditions along the realized re-adoptions is exactly the realized-density construction—the state-price density a market of naive revisers supports, with no anticipation of one’s own revisions entering the price system. Levels are where the sophistication has bite: a fully self-financing *naive* sequence would price its frozen tail rather than the equilibrium one (the two coincide only at logarithmic utility), while the sophisticated investor’s wealth at each revision is the equilibrium continuation value—exactly the level the naive re-solve needs. On the naive reading, the departures from dynamic consistency are confined to the rejection epochs, each gated by the tolerance (Weinstein, 2017)—a per-shift control, with no lifetime bound claimed. The two readings agree on every *instantaneous* object—the short rate and

the fear factor depend on the current model $\phi_i(t-)$ alone—and the forward-looking objects are equilibrium prices, pinned by ξ , not by any single investor's first-person forward valuation.

Definition 6 (Belief density): *Investor i 's belief density is the Radon–Nikodym derivative of her held-model law $\mathbb{P}^{\phi_i}$ against the physical law $\mathbb{P}$ on $\mathcal{F}_t$,*

$$M_t^i = \frac{d\mathbb{P}^{\phi_i}}{d\mathbb{P}} \Big|_{\mathcal{F}_t} = \exp\left(-\int_0^t (\lambda_i(s-) - \lambda^*) ds\right) \prod_{k: T_k \leq t} r_i(J_k; T_k), \quad r_i(x; t) := \frac{\lambda_i(t-) f_{\theta_i(t-)}(x)}{\lambda^* f(x)}, \quad (\text{C1})$$

with $M_0^i = 1$: the intensity-wedge drift times the per-disaster likelihood ratios r_i . Investor i 's conditional expectation is $\mathbb{E}_t^i[X] = \mathbb{E}_t[(M_T^i/M_t^i)X]$.

Remark 5 (The allocation is selected, not derived). Under complete markets with heterogeneous beliefs, equal initial wealth and a common multiplier cannot both hold. The CRRA optimum is $c_t^i = \zeta_i^{-1/\alpha} (M_t^i)^{1/\alpha} (e^{-\delta t}/\xi_t)^{1/\alpha}$, so ζ_i is pinned by the budget $\zeta_i^{-1/\alpha} \mathbb{E}[\int_0^\infty \xi_t^{1-1/\alpha} e^{-\delta t/\alpha} (M_t^i)^{1/\alpha} dt] = W_0^i$, whose value depends on the law of $\{M_t^i\}$, hence on the surprise threshold ε^i . Common wealth W_0^i forces type-dependent ζ_i , while common ζ_i forces type-dependent wealth—a genuine fork. We take the second branch: ε indexes a revision *rule*, not a preference or endowment we wish to price, so equal weights isolate the belief channel from Pareto-weight dispersion. This setup is the standard perpetual-youth reduced form (Yaari, 1965; Blanchard, 1985; Gârleanu and Panageas, 2015); the general heterogeneous- ζ kernel of Proposition 1 remains available.

Definition 7 (Competitive equilibrium): *A competitive equilibrium is a state-price density $(\xi_t)_{t \geq 0}$ and a consumption allocation (c_t^i) such that (i) each c^i maximizes $\mathbb{E}^i[\int_0^\infty e^{-\delta t} u(c_t^i) dt]$ subject to the static budget $\mathbb{E}[\int_0^\infty \xi_t c_t^i dt] \leq \mathbb{E}[\int_0^\infty \xi_t e_t^i dt]$, the expectation under the physical law $\mathbb{P}$ and e^i investor i 's endowment—the supporting endowment that implements the selected Pareto weights of Remark 5 (under heterogeneous beliefs a common multiplier implies type-dependent e^i), not a separate primitive of the economy; and (ii) the goods market clears, $\int_0^1 c_t^i di = C_t$ for every t .*

Proof of Proposition 1. Step 1 (the aggregator is finite and positive). Fix a path with $N_t < \infty$

disasters on $[0, t]$ (a.s., Assumption 1). By (C1) and the bounds $\underline{r} \leq r_i \leq \bar{r}$ and $\lambda_i \in [\underline{\lambda}, \bar{\lambda}]$,

$$e^{-(\bar{\lambda}-\lambda^*)t} \underline{r}^{N_t} \leq M_t^i \leq e^{(\lambda^*-\underline{\lambda})t} \bar{r}^{N_t} \quad \text{for every } i, \quad (\text{C2})$$

so M_t^i is bounded above and below by path-dependent constants uniform in i . Hence $(M_t^i)^{1/\alpha}$ is bounded uniformly in i , and with $\int_0^1 \zeta_i^{-1/\alpha} di < \infty$ (Assumption 3),

$$0 < \Xi_t = \int_0^1 \zeta_i^{-1/\alpha} (M_t^i)^{1/\alpha} di \leq (e^{(\lambda^*-\underline{\lambda})t} \bar{r}^{N_t})^{1/\alpha} \int_0^1 \zeta_i^{-1/\alpha} di < \infty.$$

Step 2 (individual optimum). Markets are complete (Assumption 3), so investor i 's dynamic problem is the static budget problem under the unique state-price density ξ (Cox and Huang, 1989; Karatzas et al., 1990). Writing the objective under $\mathbb{P}$ via $\mathbb{E}^i[\cdot] = \mathbb{E}[M_t^i \cdot]$, the Lagrangian $\mathbb{E}[\int_0^\infty (e^{-\delta t} u(c_t^i) M_t^i - \zeta_i \xi_t c_t^i) dt]$ is concave in c^i (as u is concave), so its pointwise (in (t, ω)) stationarity condition is necessary and sufficient:

$$e^{-\delta t} u'(c_t^i) M_t^i = \zeta_i \xi_t. \quad (\text{C3})$$

With $u'(c) = c^{-\alpha}$, (C3) inverts to

$$c_t^i = \zeta_i^{-1/\alpha} (\xi_t e^{\delta t})^{-1/\alpha} (M_t^i)^{1/\alpha}. \quad (\text{C4})$$

Step 3 (market clearing fixes the kernel). Integrate (C4) over i and impose $\int_0^1 c_t^i di = C_t$.

The factor $(\xi_t e^{\delta t})^{-1/\alpha}$ is constant in i , so

$$\begin{aligned} (\xi_t e^{\delta t})^{-1/\alpha} \int_0^1 \zeta_i^{-1/\alpha} (M_t^i)^{1/\alpha} di &= C_t \\ (\xi_t e^{\delta t})^{-1/\alpha} \Xi_t &= C_t \\ (\xi_t e^{\delta t})^{-1/\alpha} &= C_t / \Xi_t \end{aligned} \tag{C5}$$

$$\begin{aligned} \xi_t e^{\delta t} &= (C_t / \Xi_t)^{-\alpha} = C_t^{-\alpha} \Xi_t^\alpha \\ \xi_t &= e^{-\delta t} C_t^{-\alpha} \Xi_t^\alpha, \end{aligned} \tag{C6}$$

which is (12); $\Xi_t \in (0, \infty)$ by Step 1, so ξ_t is well-defined and positive.

Step 4 (shares). Divide (C4) by C_t and substitute $(\xi_t e^{\delta t})^{-1/\alpha} = C_t / \Xi_t$ from (C5):

$$s_t^i = \frac{c_t^i}{C_t} = \frac{\zeta_i^{-1/\alpha} (C_t / \Xi_t) (M_t^i)^{1/\alpha}}{C_t} = \frac{\zeta_i^{-1/\alpha} (M_t^i)^{1/\alpha}}{\Xi_t}, \tag{C7}$$

which is (13); $\int_0^1 s_t^i di = \Xi_t / \Xi_t = 1$ and $s_t^i > 0$, so s_t is a probability density in i .

Step 5 (existence and uniqueness). Equations (C6) and (C7) construct (ξ, c^i) ; by Step 2 each c^i is the (unique) individual optimum given ξ , and by Step 3 the goods market clears, so (ξ, c^i) is a competitive equilibrium (Definition 7). Uniqueness: at any equilibrium the FOC (C3) and clearing hold, and Steps 2–3 derive (12)–(13) from them with no freedom left once the ζ_i are fixed. Uniqueness is stated *given* the selected weights ζ_i : the supporting endowments of the equilibrium definition are chosen to implement them, so each budget holds by construction rather than pinning the weights. In a fixed-endowment economy the weights would in turn be determined by the strictly monotone Negishi map from Pareto weights to excess demands (Cox and Huang, 1989; Karatzas et al., 1990); nothing below requires that step. Under the symmetric allocation $\zeta_i \equiv \zeta = 1$ (Assumption 3(iv); a weight normalization, not equal initial wealth, which under heterogeneous beliefs would make ζ_i type-dependent, Remark 5) ζ cancels from (C7).

Step 6 (the power mean is not the arithmetic mean). From (C7), $\zeta_i^{-1/\alpha}(M_t^i)^{1/\alpha} = s_t^i \Xi_t$, so $M_t^i = \zeta_i (s_t^i \Xi_t)^\alpha$. Substituting into the share-weighted arithmetic mean $\langle M \rangle_t = \int_0^1 s_t^i M_t^i di$,

$$\langle M \rangle_t = \int_0^1 s_t^i \zeta_i (s_t^i \Xi_t)^\alpha di = \Xi_t^\alpha \int_0^1 \zeta_i (s_t^i)^{1+\alpha} di.$$

Under the equal-Pareto-weight selection $\zeta_i \equiv \zeta$ (the case carried downstream), the shares are price-irrelevantly ζ -free and

$$\langle M \rangle_t = \zeta \Xi_t^\alpha \int_0^1 (s_t^i)^{1+\alpha} di \geq \zeta \Xi_t^\alpha,$$

the inequality by Jensen applied to the convex map $x \mapsto x^{1+\alpha}$ ($\alpha > 0$) and the probability measure di on $[0, 1]$, since $\int_0^1 (s_t^i)^{1+\alpha} di \geq (\int_0^1 s_t^i di)^{1+\alpha} = 1$, with equality iff s_t^i is constant in i —equivalently iff the M_t^i are equal across the cross-section. Hence the priced power-mean factor Ξ_t^α and the descriptive arithmetic mean $\langle M \rangle_t$ coincide (up to the price-irrelevant constant ζ) only at cross-sectional degeneracy, and otherwise differ, the arithmetic mean strictly larger; the kernel prices the former. $\square$

Proof of Lemma 1. (i) With $c_{i,s} = \psi_i^{-1/\alpha}(M_s^i)^{1/\alpha} C_s / \Xi_s$ (Proposition 1) and $\xi_s = e^{-\delta s} C_s^{-\alpha} \Xi_s^\alpha$, the held model is constant on $[\tau_k, \tau_{k+1})$, so $M_s^i / M_t^i = M_{t \rightarrow s}^{(k)}$ there and

$$e^{-\delta(s-t)} \left(\frac{c_{i,s}}{c_{i,t}} \right)^{-\alpha} M_{t \rightarrow s}^{(k)} = e^{-\delta(s-t)} \frac{M_t^i}{M_s^i} \left(\frac{C_s}{C_t} \right)^{-\alpha} \left(\frac{\Xi_s}{\Xi_t} \right)^\alpha M_{t \rightarrow s}^{(k)} = \frac{\xi_s}{\xi_t},$$

which is (14); equivalently the path satisfies the frozen problem's first-order condition $e^{-\delta(s-\tau_k)} c_{i,s}^{-\alpha} M_{\tau_k \rightarrow s}^{(k)} = \eta_k \xi_s / \xi_{\tau_k}$ with multiplier $\eta_k = \psi_i \xi_{\tau_k} e^{\delta \tau_k} / M_{\tau_k}^i$. For local optimality, let Δc be supported in $[\tau_k, \tau_{k+1})$ and self-financing at market prices, $\mathbb{E}_{\tau_k} \int \xi_s \Delta c_s ds = 0$ (state prices are objective; beliefs enter the objective, not the budget). Its first-order effect on the frozen objective is $\mathbb{E}_{\tau_k}^{(k)} \int e^{-\delta(s-\tau_k)} u'(c_{i,s}) \Delta c_s ds = \eta_k \mathbb{E}_{\tau_k} \int (\xi_s / \xi_{\tau_k}) \Delta c_s ds = 0$, since on the support of Δc the frozen density coincides with the realized one; the second-order effect is strictly negative by strict concavity. (ii) Rearranging (14) recovers ξ_s / ξ_{τ_k} on each interval from the naive first-order condition

alone; multiplying the interval densities $M^{(k)}$ along the realized re-adoptions is precisely the realized-density construction of Appendix C, and dynamic completeness (Assumption 3) prices every claim from this ξ by absence of arbitrage. (iii) By strict concavity, the naive optimum from wealth W coincides with the equilibrium path on $[\tau_k, \tau_{k+1})$ iff its budget-pinned multiplier equals η_k , i.e. iff W equals the market value of the plan with multiplier η_k —the frozen continuation $c_s^{(k)} = \psi_i^{-1/\alpha} (M_{\tau_k}^i M_{\tau_k \rightarrow s}^{(k)})^{1/\alpha} C_s / \Xi_s$. Self-financing hands the τ_{k+1} problem the value of plan (k) 's tail, while coincidence there requires the value of the *equilibrium* tail; the two are

$$\psi_i^{-1/\alpha} \xi_{\tau_{k+1}}^{-1} \mathbb{E}_{\tau_{k+1}} \int_{\tau_{k+1}}^{\infty} e^{-\delta s} C_s^{1-\alpha} \Xi_s^{\alpha-1} D_s^{1/\alpha} ds, \quad D_s \in \{ M_{\tau_k}^i M_{\tau_k \rightarrow s}^{(k)}, M_s^i \},$$

frozen versus realized density. At $\alpha = 1$ the integrand is $e^{-\delta s} D_s$, and both densities are $\mathbb{P}$ -martingales with the common value $M_{\tau_{k+1}}^i$ at τ_{k+1} (M^i is continuous at a calm re-adoption, its new segment starting at ratio one), so the two integrals coincide identically. For $\alpha \neq 1$ the factor $C_s^{1-\alpha} \Xi_s^{\alpha-1}$ weights the two densities by state, and their equality is an additional restriction that equilibrium does not impose; in equilibrium the investor's wealth at each revision is the equilibrium continuation value, which supplies exactly the required level. At a disaster-triggered revision the same argument runs from the post-jump state; transversality (T) makes every displayed integral finite, and markets clear at ξ because the allocation is the equilibrium one throughout. $\square$

Proof of Proposition 2. Step 1 (the kernel's jump ratio). At a disaster at t of size $x = J_k$, the endowment falls by $C_t = C_{t-} e^{-x}$ (Assumption 1), so

$$\frac{C_t^{-\alpha}}{C_{t-}^{-\alpha}} = (e^{-x})^{-\alpha} = e^{\alpha x}.$$

Each density jumps by its likelihood ratio, $M_t^i = M_{t-}^i r_i(x; t)$ (C1), so $(M_t^i)^{1/\alpha} = (M_{t-}^i)^{1/\alpha} r_i(x; t)^{1/\alpha}$,

and by (12) and (C7),

$$\begin{aligned}
\frac{\Xi_t}{\Xi_{t-}} &= \frac{1}{\Xi_{t-}} \int_0^1 \zeta_i^{-1/\alpha} (M_t^i)^{1/\alpha} di \\
&= \frac{1}{\Xi_{t-}} \int_0^1 \zeta_i^{-1/\alpha} (M_{t-}^i)^{1/\alpha} r_i(x; t)^{1/\alpha} di \\
&= \int_0^1 \frac{\zeta_i^{-1/\alpha} (M_{t-}^i)^{1/\alpha}}{\Xi_{t-}} r_i(x; t)^{1/\alpha} di \\
&= \int_0^1 s_{t-}^i r_i(x; t)^{1/\alpha} di = g_t(x),
\end{aligned} \tag{C8}$$

the third line moving the constant Ξ_{t-}^{-1} under the integral and the fourth using $s_{t-}^i = \zeta_i^{-1/\alpha} (M_{t-}^i)^{1/\alpha} / \Xi_{t-}$ (C7); this is (15). Multiplying, the state-price density jumps by

$$\frac{\xi_t}{\xi_{t-}} = \frac{C_t^{-\alpha}}{C_{t-}^{-\alpha}} \left(\frac{\Xi_t}{\Xi_{t-}} \right)^\alpha = e^{\alpha x} g_t(x)^\alpha, \tag{C9}$$

the continuous parts of $C^{-\alpha}$ and Ξ^α contributing no jump.

Step 2 (risk-neutral compensator). Markets are complete (Assumption 3), so there is a unique equivalent martingale measure $\mathbb{Q}$ with density process $Z_t = \xi_t \exp(\int_0^t r_s ds) / \xi_0$. We verify the hypotheses of the change-of-measure theorem for marked point processes (Brémaud, 1981, VI, T2–T4). (a) Z is a local martingale. The equilibrium short rate is the process that deflates the state-price density to a local martingale, and this fixes the drift cancellation. Write the stochastic logarithm of the SDF (12) as

$$\frac{d\xi_t}{\xi_{t-}} = b_t dt + dm_t,$$

with b_t the predictable finite-variation drift coefficient—the Itô drift of $\xi_t = e^{-\delta t} C_t^{-\alpha} \Xi_t^\alpha$, a functional of the cross-section through Ξ —and m a local martingale collecting the Brownian and compensated-jump parts. The equilibrium short rate is, by its defining no-arbitrage property, $r_t := -b_t$; this property is logically prior to its closed form (which we do not need here). Since $e^{\int_0^t r_s ds}$ is continuous and of finite variation it has zero covariation with ξ , so integration by parts

gives

$$d(\xi_t e^{\int_0^t r_s ds}) = e^{\int_0^t r_s ds} (d\xi_t + \xi_{t-} r_t dt) = e^{\int_0^t r_s ds} \xi_{t-} ((b_t + r_t) dt + dm_t) = e^{\int_0^t r_s ds} \xi_{t-} dm_t,$$

the drift vanishing by $r_t = -b_t$. Hence $\xi_t e^{\int_0^t r_s ds}$ is a positive $\mathbb{P}$ -local-martingale, and $Z_t = \xi_t e^{\int_0^t r_s ds} / \xi_0$ is its unit normalization, with $Z_0 = 1$ and $\xi > 0$. The remaining and substantive task is to upgrade this to a *true* $(\mathcal{F}_t, \mathbb{P})$ -martingale with $\mathbb{E}[Z_t] = 1$ (hence the density of $\mathbb{Q}$), which requires controlling *both* factors of Z , since ξ carries a continuous (Brownian) part as well as the disaster jumps. *Continuous part:* the diffusive loading of $\log \xi$ is $-\alpha \sigma dW_t$, constant because σ is constant (Assumption 1); the continuous factor of Z is then the Doléans exponential $\mathcal{E}(-\alpha \sigma W)_t$, a true martingale by Novikov's condition $\mathbb{E}[\exp(\frac{1}{2} \int_0^t (\alpha \sigma)^2 ds)] = \exp(\frac{1}{2} \alpha^2 \sigma^2 t) < \infty$. *Jump part:* the bounded predictable jump ratio of (b)–(c) and the integrability bound $\int_0^t \int_{\mathcal{J}} |Y_s(x) - 1| \nu(ds, dx) \leq (e^{\alpha \bar{r}} + 1) \lambda^* t < \infty$ place the jump factor within Brémaud (1981, VI, T4). The two factors are driven by independent sources (W and μ), so their product Z is a true martingale; completeness independently yields a unique equivalent martingale measure, of which Z is the density. (b) The bank account $\exp(\int_0^t r_s ds)$ is continuous and of finite variation and ξ_0 is constant, so Z and ξ have the same jump ratio: at a disaster of size x , $Z_t/Z_{t-} = \xi_t/\xi_{t-} = e^{\alpha x} g_t(x)^\alpha =: Y_t(x)$ by (C9). (c) Y_t is strictly positive and predictable (it is built from the left limits s_{t-}^i and g_{t-}) and bounded on the compact $\mathcal{J}$, since $e^{\alpha x} \leq e^{\alpha \bar{r}}$ and $0 < g_t(x) \leq \bar{r}^{1/\alpha}$ by (15). Under (a)–(c) the theorem gives the $\mathbb{Q}$ -compensator of μ as $\nu^{\mathbb{Q}}(dt, dx) = Y_t(x) \nu(dt, dx)$; with $\nu(dt, dx) = \lambda^* dt F_J(dx)$ (5),

$$\nu^{\mathbb{Q}}(dt, dx) = \lambda^* e^{\alpha x} g_t(x)^\alpha dt F_J(dx), \quad (\text{C10})$$

which is (16); Y_t bounded makes $\nu^{\mathbb{Q}}$ a finite measure, so the change of measure is admissible.

Step 3 (size law). Using $F_J(dx) = f(x) dx$ (Assumption 1), the finite measure (C10) is $\nu^{\mathbb{Q}}(dt, dx) = \lambda^* e^{\alpha x} g_t(x)^\alpha f(x) dt dx$. Factor it as intensity times a normalized size density,

$\nu^{\mathbb{Q}}(dt, dx) = \lambda_t^{\mathbb{Q}} dt f_t^{\mathbb{Q}}(x) dx$, by integrating out x for the intensity and dividing for the density:

$$\begin{aligned}\lambda_t^{\mathbb{Q}} &= \int_{\mathcal{J}} \frac{\nu^{\mathbb{Q}}(dt, dx)}{dt} = \lambda^* \int_{\mathcal{J}} e^{\alpha x} g_t(x)^{\alpha} f(x) dx = \lambda^* \mathbb{E}_f[e^{\alpha J} g_t(J)^{\alpha}], \\ f_t^{\mathbb{Q}}(x) &= \frac{\lambda^* e^{\alpha x} g_t(x)^{\alpha} f(x)}{\lambda_t^{\mathbb{Q}}} = \frac{e^{\alpha x} g_t(x)^{\alpha} f(x)}{\mathbb{E}_f[e^{\alpha J} g_t(J)^{\alpha}]},\end{aligned}\tag{C11}$$

where $\mathbb{E}_f[h] = \int_{\mathcal{J}} h f$; the denominator is finite and positive by the bounds in Step 2, so $f_t^{\mathbb{Q}}$ is a proper density on $\mathcal{J}$. This is (17) and proves (i).

Step 4 (the bridge). Multiply and divide $U_t = \lambda_t^{\mathbb{Q}}$ by $\mathbb{E}_f[e^{\alpha J}]$:

$$U_t = \lambda^* \mathbb{E}_f[e^{\alpha J} g_t(J)^{\alpha}] = \underbrace{\lambda^* \mathbb{E}_f[e^{\alpha J}]}_{U^{\text{RA}}} \cdot \underbrace{\frac{\mathbb{E}_f[e^{\alpha J} g_t(J)^{\alpha}]}{\mathbb{E}_f[e^{\alpha J}]}}_{\kappa_t},\tag{C12}$$

which is (18). The factor $U^{\text{RA}} = \lambda^* \mathbb{E}_f[e^{\alpha J}]$ contains only the physical (λ^*, F_J) and α ; the factor κ_t contains the cross-section only, through g_t . This proves (ii).

Step 5 (belief-driven variation). Holding (α, λ^*, F_J) fixed, U^{RA} is constant in t , so taking logs of (C12) and differentiating,

$$d \log U_t = d \log U^{\text{RA}} + d \log \kappa_t = d \log \kappa_t,$$

which is (19). In (C11) the only cross-section-dependent object is g_t , so $f_t^{\mathbb{Q}}$, $\bar{U}_t$, and U_t are functionals of g_t alone—of the held models ϕ_i and shares s_{t-}^i of the cross-section (Proposition 9)—while (λ^*, F_J) stay fixed. This proves (iii). $\square$

Appendix D. Proofs for the revision hazard and the law of motion of the pricing state

Proposition 10 (Selection dynamics of the kernel and shares): *In the equilibrium of Proposition 1, between revisions (held models fixed):*

(i) *in calm (no disaster) the aggregator and the shares drift by selection,*

$$\frac{d \log \Xi_t}{dt} = -\frac{1}{\alpha}(\bar{\lambda}_t^S - \lambda^*), \quad \frac{d \log s_t^i}{dt} = \frac{1}{\alpha}(\bar{\lambda}_t^S - \lambda_i(t)), \quad (\text{D1})$$

so the share of a below-average-intensity (complacent) investor grows and that of an above-average-intensity (fearful) investor shrinks;

(ii) *at a disaster of size x the aggregator and the shares jump multiplicatively,*

$$\frac{\Xi_t}{\Xi_{t-}} = g_t(x), \quad \frac{s_t^i}{s_{t-}^i} = \frac{r_i(x; t)^{1/\alpha}}{g_t(x)}, \quad (\text{D2})$$

so an investor whose held model fits the realized disaster better than the aggregate, $r_i(x; t)^{1/\alpha} > g_t(x)$, gains share.

Proof of Proposition 10. (i) *Calm.* Between disasters the product in (C1) is constant, so $M_t^i = \exp(-\int_0^t (\lambda_i(s-) - \lambda^*) ds) \times (\text{const})$ and, with the held model fixed,

$$\frac{d \log M_t^i}{dt} = -(\lambda_i(t) - \lambda^*), \quad \frac{d \log (M_t^i)^{1/\alpha}}{dt} = -\frac{1}{\alpha}(\lambda_i(t) - \lambda^*). \quad (\text{D3})$$

Differentiating $\Xi_t = \int_0^1 \zeta_i^{-1/\alpha} (M_t^i)^{1/\alpha} di$ in t : the interchange of $\frac{d}{dt}$ and $\int_0^1 \cdot di$ is justified by dominated convergence, since on any compact t -interval both the integrand $\zeta_i^{-1/\alpha} (M_t^i)^{1/\alpha}$ and its t -derivative $\zeta_i^{-1/\alpha} (M_t^i)^{1/\alpha} (-\frac{1}{\alpha}(\lambda_i - \lambda^*))$ are bounded uniformly in i by $\zeta_i^{-1/\alpha}$ times path-constants (by (C2) and $\lambda_i \in [\underline{\lambda}, \bar{\lambda}]$), with $\int_0^1 \zeta_i^{-1/\alpha} di < \infty$ (Assumption 3). Using (D3) for

$\frac{d}{dt}(M_t^i)^{1/\alpha} = (M_t^i)^{1/\alpha}(-\frac{1}{\alpha}(\lambda_i - \lambda^*))$ and then $\zeta_i^{-1/\alpha}(M_t^i)^{1/\alpha} = s_t^i \Xi_t$ from (C7),

$$\begin{aligned} \frac{d\Xi_t}{dt} &= \int_0^1 \zeta_i^{-1/\alpha}(M_t^i)^{1/\alpha} \left(-\frac{1}{\alpha}(\lambda_i - \lambda^*)\right) di \\ &= -\frac{1}{\alpha} \Xi_t \int_0^1 s_t^i (\lambda_i - \lambda^*) di = -\frac{1}{\alpha} \Xi_t (\bar{\lambda}_t^S - \lambda^*), \end{aligned} \quad (\text{D4})$$

the last equality using $\int_0^1 s_t^i \lambda_i di = \bar{\lambda}_t^S$ and $\int_0^1 s_t^i di = 1$ (Definition 3); this gives the first equation of (D1). For the share, take logs of (C7): $\log s_t^i = -\frac{1}{\alpha} \log \zeta_i + \frac{1}{\alpha} \log M_t^i - \log \Xi_t$; the first term is constant in t , so

$$\frac{d \log s_t^i}{dt} = \frac{1}{\alpha} \frac{d \log M_t^i}{dt} - \frac{d \log \Xi_t}{dt}. \quad (\text{D5})$$

By (D3), $\frac{1}{\alpha} d \log M_t^i / dt = -\frac{1}{\alpha}(\lambda_i - \lambda^*)$; and dividing (D4) by Ξ_t , $d \log \Xi_t / dt = \Xi_t^{-1} d\Xi_t / dt = -\frac{1}{\alpha}(\bar{\lambda}_t^S - \lambda^*)$. Substituting both into (D5),

$$\frac{d \log s_t^i}{dt} = -\frac{1}{\alpha}(\lambda_i - \lambda^*) + \frac{1}{\alpha}(\bar{\lambda}_t^S - \lambda^*) = \frac{1}{\alpha}(\bar{\lambda}_t^S - \lambda_i), \quad (\text{D6})$$

the λ^* terms cancelling; this is the second equation of (D1).

(ii) *Disaster*. At a disaster of size x , $M_t^i = M_{t-}^i r_i(x; t)$ (C1), so $(M_t^i)^{1/\alpha} = (M_{t-}^i)^{1/\alpha} r_i(x; t)^{1/\alpha}$ and $\Xi_t / \Xi_{t-} = g_t(x)$ exactly as in (C8). Then from $s_t^i = \zeta_i^{-1/\alpha} (M_t^i)^{1/\alpha} / \Xi_t$, the constant $\zeta_i^{-1/\alpha}$ cancels in the ratio:

$$\frac{s_t^i}{s_{t-}^i} = \frac{(M_t^i)^{1/\alpha} / \Xi_t}{(M_{t-}^i)^{1/\alpha} / \Xi_{t-}} = \frac{(M_t^i)^{1/\alpha}}{(M_{t-}^i)^{1/\alpha}} \cdot \frac{\Xi_{t-}}{\Xi_t} = r_i(x; t)^{1/\alpha} \cdot \frac{1}{g_t(x)} = \frac{r_i(x; t)^{1/\alpha}}{g_t(x)},$$

which is (D2); the share grows iff $r_i(x; t)^{1/\alpha} > g_t(x)$. $\square$

Proof of Proposition 3, the no-finite-state claim. That the calm drift (D1) depends on S_t only through $\bar{\lambda}_t^S = \int_{\Phi} \lambda dS_t$ is read off (D4), whose right-hand side involves the cross-section only through $\bar{\lambda}_t^S$. For the disaster tilt, (15) written against S_t is (24), a linear map $S \mapsto g(\cdot) = \int_{\Phi} H(\cdot, \phi) S(d\phi)$ on signed measures, with kernel $H(x, \phi) = (\lambda / \lambda^*)^{1/\alpha} (f_{\theta}(x) / f(x))^{1/\alpha}$. Since the factor $(\lambda / \lambda^*)^{1/\alpha} > 0$ is a θ -independent scalar, $\text{span}\{H(\cdot, \phi) : \phi \in \Phi\} = \text{span}\{(f_{\theta}/f)^{1/\alpha} :$

$\theta \in \Theta\}$, which by Assumption 4(v) is infinite-dimensional in $L^1(\mathcal{J}, f)$; hence the linear map $T : \eta \mapsto \int_{\Phi} H(\cdot, \phi) \eta(d\phi)$ has infinite-dimensional range.

Fix any finite collection of moment functionals $L_1, \dots, L_n$ of the pricing state (each $L_j(\eta) = \int_{\Phi} \ell_j d\eta$ for a bounded ℓ_j), and adjoin the total-mass functional $L_0(\eta) = \int_{\Phi} d\eta$. Work in the space $\mathcal{V}$ of signed measures on Φ with bounded Lebesgue density, and let $\mathcal{N} = \{\eta \in \mathcal{V} : L_0(\eta) = L_1(\eta) = \dots = L_n(\eta) = 0\}$, the intersection of $n+1$ hyperplanes, so $\text{codim}_{\mathcal{V}} \mathcal{N} \leq n+1$.

There is $\eta \in \mathcal{N}$ with $T\eta \not\equiv 0$. Suppose not, i.e. $T(\mathcal{N}) = \{0\}$, so $\mathcal{N} \subseteq \ker T$. Then $\text{codim } \ker T \leq \text{codim } \mathcal{N} \leq n+1$, and by rank-nullity $\dim \text{range}(T|_{\mathcal{V}}) = \text{codim } \ker T \leq n+1$. But T has infinite-dimensional range already on bounded-density measures (the functions $(f_{\theta}/f)^{1/\alpha}$ are approximated in $L^1(\mathcal{J}, f)$ by T of bounded-density measures concentrated near each θ , and are linearly independent by (v)), a contradiction. Hence such an η exists; fix it, with bounded density and $\|\eta\|_{\infty} < \infty$.

Admissibility of the perturbation. Let $\mathcal{S}$ denote the *admissible instantaneous pricing states*—probability measures on Φ with a density bounded below on the compact Φ , the valid wealth-weighted held-model cross-sections. The directions η constructed above keep the perturbation inside $\mathcal{S}$ (next paragraph; the explicit m -parameter version is Proposition IA.1), so the statement is about the *instantaneous pricing map* $S \mapsto (U_t, f_t^{\mathbb{Q}})$ over $\mathcal{S}$: no finite collection of moments is a sufficient statistic for it, two states in $\mathcal{S}$ sharing $L_1, \dots, L_n$ yet pricing disasters differently. The dynamically reachable cross-sections—wealth-weighted pushforwards of the primitive type law G (Proposition 9)—need not lie in $\mathcal{S}$ —they can be atomic, as at the homogeneous start or on a finite menu; the claim is no finite sufficient statistic for the pricing map on $\mathcal{S}$, not that the economy visits any particular separating pair.

The perturbation is a probability measure. For $0 < \varepsilon \leq c/\|\eta\|_{\infty}$, the density of $S'_t = S_t + \varepsilon\eta$ is $\geq c - \varepsilon\|\eta\|_{\infty} \geq 0$, and its total mass is $1 + \varepsilon L_0(\eta) = 1$; so S'_t is a probability measure on Φ .

By construction $L_j(S'_t) = L_j(S_t)$ for $j = 1, \dots, n$ (as $L_j(\eta) = 0$), yet

$$g'_t(\cdot) - g_t(\cdot) = T(S'_t - S_t) = \varepsilon T\eta \neq 0 \quad \text{in } L^1(\mathcal{J}, f).$$

Distinct g_t give distinct prices. The risk-neutral compensator obeys $\nu_t^{\mathbb{Q}}(dx) \propto e^{\alpha x} g_t(x)^\alpha f(x) dx$ (16). The map $y \mapsto y^\alpha$ is strictly increasing on $(0, \infty)$, so $g'_t \neq g_t$ on a set of positive f -measure forces $g_t'^\alpha \neq g_t^\alpha$ there, hence $e^{\alpha x} g_t'^\alpha f \neq e^{\alpha x} g_t^\alpha f$ on that set and $\nu_t^{\mathbb{Q}'} \neq \nu_t^{\mathbb{Q}}$ as measures; two finite measures that differ as measures differ in at least one of total mass or normalized shape, and U_t is the total mass while $f_t^{\mathbb{Q}}$ is the normalized shape, so $(U'_t, f_t^{\mathbb{Q}'}) \neq (U_t, f_t^{\mathbb{Q}})$. Thus S_t and S'_t agree on $L_1, \dots, L_n$ but price disaster risk differently: no finite collection of moments is a sufficient statistic for $(U_t, f_t^{\mathbb{Q}})$. $\square$

Corollary 2 (Intensity channel: a scalar fear state): *If the size law is undisputed and correct— $f_{\theta_i} = f$ for all i , the held size densities equalling the objective f —then $r_i(x; t) = \lambda_i(t-)/\lambda^*$ is independent of x ,*

$$g_t(x) \equiv G_t := \int_{\Phi} \left(\frac{\lambda}{\lambda^*}\right)^{1/\alpha} S_t(d\phi) = \int_0^1 s_t^i\left(\frac{\lambda_i(t)}{\lambda^*}\right)^{1/\alpha} di, \quad (\text{D7})$$

displayed at the generic state S_t ; as a jump compensator G_t is read at the predictable left limit S_{t-} , the two agreeing off disasters. The risk-neutral size law then collapses to the fixed risk-aversion tilt $f_t^{\mathbb{Q}}(x) = e^{\alpha x} f(x)/\mathbb{E}_f[e^{\alpha J}]$, and the fear factor is the single scalar

$$U_t = G_t^\alpha U^{\text{RA}}. \quad (\text{D8})$$

The size dispute is precisely what makes $f_t^{\mathbb{Q}}$ move; absent it, beliefs price only the disaster frequency U_t , through the one scalar G_t .

Proof of Corollary 2. With $f_{\theta_i} = f$ the likelihood ratio (C1) simplifies:

$$\begin{aligned} r_i(x; t) &= \frac{\lambda_i(t-) f_{\theta_i}(x)}{\lambda^* f(x)} \\ &= \frac{\lambda_i(t-) f(x)}{\lambda^* f(x)} \end{aligned} \tag{D9}$$

$$= \frac{\lambda_i(t-)}{\lambda^*}, \tag{D10}$$

line (D9) substituting $f_{\theta_i} = f$ and (D10) cancelling $f(x)$; r_i is independent of x . Substituting $r_i(x; t) = \lambda_i(t-)/\lambda^*$ into (15),

$$g_t(x) = \int_0^1 s_{t-}^i \left(\frac{\lambda_i(t-)}{\lambda^*} \right)^{1/\alpha} di =: G_t,$$

independent of x ; this is (D7), the predictable aggregator evaluated at the left-limit S_{t-} (equal to S_t off disasters and off the predictable deadline dates, at which the calm switch flow can move S ; (D7) displays it at the generic state). For the size law, start from (17) and use $g_t \equiv G_t$:

$$\begin{aligned} f_t^{\mathbb{Q}}(x) &= \frac{e^{\alpha x} g_t(x)^{\alpha} f(x)}{\mathbb{E}_f[e^{\alpha J} g_t(J)^{\alpha}]} \\ &= \frac{e^{\alpha x} G_t^{\alpha} f(x)}{\mathbb{E}_f[e^{\alpha J} G_t^{\alpha}]} \end{aligned} \tag{D11}$$

$$= \frac{e^{\alpha x} G_t^{\alpha} f(x)}{G_t^{\alpha} \mathbb{E}_f[e^{\alpha J}]} \tag{D12}$$

$$= \frac{e^{\alpha x} f(x)}{\mathbb{E}_f[e^{\alpha J}]}, \tag{D13}$$

line (D11) substituting $g_t = G_t$, (D12) pulling the constant G_t^{α} out of the f -expectation, and (D13) cancelling it; the law is free of G_t . For the fear factor, from $U_t = \lambda_t^{\mathbb{Q}} = \lambda^* \mathbb{E}_f[e^{\alpha J} g_t(J)^{\alpha}]$

(17),

$$U_t = \lambda^* \mathbb{E}_f[e^{\alpha J} G_t^\alpha] \quad (\text{D14})$$

$$= G_t^\alpha \lambda^* \mathbb{E}_f[e^{\alpha J}] \quad (\text{D15})$$

$$= G_t^\alpha U^{\text{RA}}, \quad (\text{D16})$$

line (D14) substituting $g_t = G_t$, (D15) pulling the constant G_t^α out of the expectation, and (D16) using $U^{\text{RA}} = \lambda^* \mathbb{E}_f[e^{\alpha J}]$ (Definition 2); this is (D8). $\square$

Corollary 3 (Finite menu: finite-dimensional pricing functionals): *If the menu is finite, $\Phi = \{\phi_1, \dots, \phi_K\}$, then $S_t = (S_t^1, \dots, S_t^K)$ lies in the simplex Δ^{K-1} and the instantaneous pricing objects $\bar{\lambda}_t^S$, $g_t(\cdot)$, f_t^Q , and U_t are functions of this finite vector. (The vector is a sufficient statistic for the instantaneous pricing objects above; it is not a sufficient statistic for forward-looking prices such as the price–dividend ratio $v(\Psi_t)$, nor by itself a Markov state for the dynamics—see the proof.) Under the intensity channel of Corollary 2 with a two-point menu $\{\lambda_{lo}, \lambda_{hi}\}$, the instantaneous fear factor is a function of the single share S_t^{hi} ,*

$$U_t = \left[S_t^{hi} \left(\frac{\lambda_{hi}}{\lambda^*} \right)^{1/\alpha} + (1 - S_t^{hi}) \left(\frac{\lambda_{lo}}{\lambda^*} \right)^{1/\alpha} \right]^\alpha U^{\text{RA}}, \quad (\text{D17})$$

recovering the scalar, intensity-only fear factor as a doubly-restricted special case.

Proof of Corollary 3. With $\Phi = \{\phi_1, \dots, \phi_K\}$ finite the pricing state is the atomic measure $S_t = \sum_{k=1}^K S_t^k \delta_{\phi_k}$ with $(S_t^1, \dots, S_t^K) \in \Delta^{K-1}$. Each aggregate is the integral of a fixed function against S_t , so it collapses to a finite sum of the shares:

$$\begin{aligned} \bar{\lambda}_t^S &= \int_{\Phi} \lambda S_t(d\phi) = \sum_{k=1}^K S_t^k \lambda_k, \\ g_t(x) &= \int_{\Phi} r(\phi, x)^{1/\alpha} S_t(d\phi) = \sum_{k=1}^K S_t^k r(\phi_k, x)^{1/\alpha}, \end{aligned} \quad (\text{D18})$$

with $r(\phi, x) = \lambda f_\theta(x)/(\lambda^* f(x))$ as in (15). Through (17), $f_t^\mathbb{Q}(x) = e^{\alpha x} g_t(x)^\alpha f(x)/\mathbb{E}_f[e^{\alpha J} g_t(J)^\alpha]$ and $U_t = \lambda^* \mathbb{E}_f[e^{\alpha J} g_t(J)^\alpha]$ depend on the cross-section only through $g_t(\cdot)$, which by (D18) is a function of (S_t^k) ; hence $\bar{\lambda}_t^S, g_t(\cdot), f_t^\mathbb{Q}, U_t$ are functions of $(S_t^k) \in \Delta^{K-1}$.

Under the intensity channel (Corollary 2) with $K = 2$ and $\Phi = \{\lambda_{lo}, \lambda_{hi}\}$, $f_\theta = f$ gives $r(\phi_k, x)^{1/\alpha} = (\lambda_k/\lambda^*)^{1/\alpha}$ (independent of x), so (D18) becomes

$$\begin{aligned} g_t(x) \equiv G_t &= S_t^{hi} \left(\frac{\lambda_{hi}}{\lambda^*} \right)^{1/\alpha} + S_t^{lo} \left(\frac{\lambda_{lo}}{\lambda^*} \right)^{1/\alpha} \\ &= S_t^{hi} \left(\frac{\lambda_{hi}}{\lambda^*} \right)^{1/\alpha} + (1 - S_t^{hi}) \left(\frac{\lambda_{lo}}{\lambda^*} \right)^{1/\alpha}, \end{aligned} \quad (\text{D19})$$

line (D19) substituting $S_t^{lo} = 1 - S_t^{hi}$ (the simplex constraint). Substituting (D19) into $U_t = G_t^\alpha U^{\text{RA}}$ (D8) gives (D17).

The reduction is of the *instantaneous* pricing map only. The vector (S_t^k) summarizes current prices, but it is not by itself a Markov state: its own law of motion is not autonomous, because the calm switch flow out of each class depends on the within-class distribution of regime ages and record statistics, not on the shares alone (Proposition 3). The full Markov state therefore carries that within-class record structure, and the valuation operator acts on it rather than on (S_t^k) alone—a point made precise where the price level is computed. $\square$

Proposition 11 (The revision hazard as a deviance first-passage): *Fix an investor holding $\phi = (\lambda, \theta)$ with tolerance ε and since-adoption record $\mathcal{R} = (n, (x_1, \dots, x_n))$ at age τ ; let $D_{\phi, \tau}(\mathcal{R})$ be the deviance (9) and $r_{\phi, \tau}(\varepsilon)$ the acceptance threshold of (10). The revision time $\tau^{\text{rev}} = \inf\{t : D_{\phi_i, \tau_t^i}(\mathcal{R}_t^i) \geq r_{\phi_i, \tau_t^i}(\varepsilon^i)\}$ is the first passage of the deviance across r (the closed-threshold form is shorthand for the p-value criterion $\bar{\varepsilon} \leq \varepsilon^i$, boundary counted; the conventions preceding Definition 5 record exactly when the two coincide), with:*

(i) (calm drift) *between disasters the record is fixed and the deviance drifts deterministically,*

$$\frac{\partial}{\partial \tau} D_{\phi, \tau}(\mathcal{R}) = \lambda - \frac{n}{\tau}, \quad (\text{D20})$$

rising—the held model increasingly contradicted—iff the realized rate n/τ is below the held λ ; with $\mathcal{R}$ fixed, D meets the moving threshold $r_{\phi,\tau}(\varepsilon)$ at a deterministic calm deadline, the record- $\mathcal{R}$ generalization of $\Delta^*(\phi, \varepsilon)$, whose $n = 0$ slice is exactly Proposition 7;

(ii) (disaster trigger) a disaster arrives at the true rate λ^* (Assumption 1) with size $x \sim f$, appends x to the record, and triggers rejection iff the updated deviance leaves the typical set, so the disaster-triggered rejection intensity is

$$\lambda^* \pi(\phi, \tau, \mathcal{R}; \varepsilon), \quad \pi(\phi, \tau, \mathcal{R}; \varepsilon) = \int_{\mathcal{J}} \mathbf{1}\{D_{\phi,\tau}(\mathcal{R} \oplus x) \geq r_{\phi,\tau}(\varepsilon)\} f(x) dx, \quad (\text{D21})$$

a functional of the full record $\mathcal{R}$, not of the age τ alone;

(iii) (no closed form) τ^{rev} is the first passage of the piecewise-deterministic Markov process on the record-resolved state $(\phi, \mathcal{R}, \tau)$ —the scalar deviance alone is not Markov in the size-dispute model—with calm drift (D20), mark updates $\mathcal{R} \mapsto \mathcal{R} \oplus x$ at rate $\lambda^* f(x) dx$, and rejection exits on the trigger set (D21); for a general regular family $\{F_\theta\}$ neither the threshold $r_{\phi,\tau}(\varepsilon)$ nor the first-passage law has an elementary form, and both are computed numerically from the process.

Two special cases calibrate the generality. A hazard in the low-dimensional state (n, τ) —the since-adoption count $n = n_t^i = N_t - N_{a^i}$ (7) and the age, rather than the full mark record—and an exactly computable passage law—finite Poisson sums over the (n, τ) grid—obtain under the restriction of Corollary 2 (intensity-only), where $f_\theta \equiv f$ makes the deviance depend on the record only through the count n (so the rejection still reads the since-adoption count n , not the age alone). The finite menu of Corollary 3 reduces the pricing functionals to finite dimension but leaves the rejection hazard record-dependent—the marks x_j still enter the deviance through f_θ —so in general the rejection is governed by the full record.

Proof of Proposition 11. (i) With the held model ϕ fixed and no disaster on the interval ending at age τ , the realized sizes and their MLE $\hat{\theta}$ are constant, so the size term of (9) is constant in

τ and only the count term $C(\tau) := \tau d_P(n/\tau \parallel \lambda)$ varies. Substituting $a = n/\tau$ into $d_P(a \parallel \lambda) = a \log \frac{a}{\lambda} - a + \lambda$,

$$\begin{aligned} C(\tau) &= \tau \left[\frac{n}{\tau} \log \frac{n/\tau}{\lambda} - \frac{n}{\tau} + \lambda \right] \\ &= n \log \frac{n/\tau}{\lambda} - n + \lambda\tau \end{aligned} \tag{D22}$$

$$= n \log \frac{n}{\tau\lambda} - n + \lambda\tau, \tag{D23}$$

line (D22) distributing the factor τ and (D23) using $\frac{n/\tau}{\lambda} = \frac{n}{\tau\lambda}$; with the convention $0 \log 0 = 0$ this holds also at $n = 0$, where $C(\tau) = \lambda\tau$. To differentiate, expand the logarithm $\log \frac{n}{\tau\lambda} = \log n - \log \tau - \log \lambda$:

$$C(\tau) = n \log n - n \log \tau - n \log \lambda - n + \lambda\tau.$$

Only the $-n \log \tau$ and $\lambda\tau$ terms depend on τ , and $D_{\phi,\tau}(\mathcal{R})$ differs from $C(\tau)$ by a τ -constant, so

$$\frac{\partial}{\partial \tau} D_{\phi,\tau}(\mathcal{R}) = C'(\tau) = -\frac{n}{\tau} + \lambda \tag{D24}$$

$$= \lambda - \frac{n}{\tau}, \tag{D25}$$

(for $n \geq 1$; at $n = 0$ the count deviance is $\lambda\tau$ outright and its τ -derivative is λ , no $\log n$ expansion entering) line (D24) using $\partial_\tau(-n \log \tau) = -n/\tau$ and $\partial_\tau(\lambda\tau) = \lambda$ and (D25) reordering; this is (D20). The drift is positive—the held model increasingly contradicted—iff $\lambda > n/\tau$, i.e. the realized rate n/τ lies below the held λ . With $\mathcal{R}$ fixed, $D_{\phi,\tau}(\mathcal{R})$ meets the moving threshold $r_{\phi,\tau}(\varepsilon)$ at a deterministic calm deadline (the record- $\mathcal{R}$ generalization of $\Delta^*(\phi, \varepsilon)$); at the $n = 0$ slice $C(\tau) = \lambda\tau$ and $\partial_\tau D = \lambda > 0$, recovering the deterministic deadline $\Delta^*(\phi, \varepsilon)$ of Proposition 7.

(ii) Disasters arrive as a marked Poisson process with $(\mathcal{F}_t, \mathbb{P})$ -compensator $\nu(dt, dx) = \lambda^* dt f(x) dx$ (5) (Assumption 1). A disaster of size x at age τ appends x to the record, $\mathcal{R} \mapsto \mathcal{R} \oplus x = (n+1, (x_1, \dots, x_n, x))$, and by the pass/fail rule (10) the test rejects at that

instant iff the updated record leaves the typical set,

$$D_{\phi,\tau}(\mathcal{R} \oplus x) \geq r_{\phi,\tau}(\varepsilon).$$

The triggering marks form the set $B_{\phi,\tau,\mathcal{R}} := \{x \in \mathcal{J} : D_{\phi,\tau}(\mathcal{R} \oplus x) \geq r_{\phi,\tau}(\varepsilon)\}$, matching the closed-threshold convention of Definition 1; its boundary $\{x : D_{\phi,\tau}(\mathcal{R} \oplus x) = r_{\phi,\tau}(\varepsilon)\}$ under the closed-threshold convention of Definition 1 the trigger set and its intensity π are unambiguous; the open convention can differ—in the intensity channel $D_{\phi,\tau}(\mathcal{R} \oplus x)$ is mark-independent, so the equality set can carry full mark mass—and is not used. Restricting the disaster random measure to $B_{\phi,\tau,\mathcal{R}}$ predictably thins it to a point process whose (record-dependent) intensity is the ν -mass of $B_{\phi,\tau,\mathcal{R}}$:

$$\int_{B_{\phi,\tau,\mathcal{R}}} \frac{\nu(dt, dx)}{dt} = \lambda^* \int_{\mathcal{J}} \mathbf{1}\{x \in B_{\phi,\tau,\mathcal{R}}\} f(x) dx \quad (\text{D26})$$

$$= \lambda^* \int_{\mathcal{J}} \mathbf{1}\{D_{\phi,\tau}(\mathcal{R} \oplus x) \geq r_{\phi,\tau}(\varepsilon)\} f(x) dx \quad (\text{D27})$$

$$= \lambda^* \pi(\phi, \tau, \mathcal{R}; \varepsilon), \quad (\text{D28})$$

line (D26) writing the restricted mass against $\nu = \lambda^* dt f dx$, (D27) substituting the definition of $B_{\phi,\tau,\mathcal{R}}$, and (D28) the definition of π ; this is the disaster-triggered rejection intensity (D21). It depends on $\mathcal{R}$ through the map $x \mapsto D_{\phi,\tau}(\mathcal{R} \oplus x)$, not on the age τ alone.

(iii) Between disasters D evolves by (D20); at disasters it jumps as in (ii); τ^{rev} is the first time D exceeds $r_{\phi,\tau}(\varepsilon)$ —the first passage of this piecewise-deterministic Markov process. The threshold is the $(1 - \varepsilon)$ -quantile of $D_{\phi,\tau}$ under $Q_{\phi,\tau}$ (8), a Poisson–mark mixture with no elementary quantile for general $\{F_\theta\}$, and the first-passage law of a PDMP across a moving boundary is likewise not elementary; both are computed numerically. The state collapses to (n, τ) only when the size law is undisputed, and to a finite vector under a finite menu; in general the rejection is governed by the full record. $\square$

Lemma 5 (Rejection hazard, switch hazard, and the switch flow): *Fix the public posterior ρ_t and its mode $\phi_t^* = \mathbf{m}(\rho_t)$ (11). For an investor in state $(\phi, \mathcal{R}, \tau, \varepsilon)$, let the rejection hazard $H_t^{\text{rej}}(d\tau' \mid \phi, \mathcal{R}, \tau, \varepsilon)$ be the hazard (intensity) measure of the first-passage time τ^{rev} of Proposition 11—a disaster-triggered rate $\lambda^* \pi(\phi, \tau, \mathcal{R}; \varepsilon)$ (D21) together with certain rejection at the deterministic calm deadline Δ^* , so that τ^{rev} is the earlier of Δ^* and the first disaster-triggered rejection, so along the fixed record the compensator of τ^{rev} is $\lambda^* \pi dt$ plus a unit atom at Δ^* ; the probability that the record itself survives unchanged to u is $\mathbf{1}\{u < \Delta^*\} e^{-\lambda^*(u-\tau)}$ and the marginal law of τ^{rev} is the recursive piecewise-deterministic object, not this local factor, dropping to zero at Δ^* (the deterministic deadline is not represented as a unit atom of an ordinary exponential hazard). Write $\beta_t(d\phi)$ for the wealth-weighted rate at which holders of ϕ reach their scheduled calm deadlines at t . Then:*

(i) *the switch hazard is the rejection hazard restricted to the switch event $\{\phi \neq \phi_t^*\}$,*

$$H_t^{\text{sw}}(d\tau' \mid \phi, \mathcal{R}, \tau, \varepsilon) = \mathbf{1}\{\phi \neq \phi_t^*\} H_t^{\text{rej}}(d\tau' \mid \phi, \mathcal{R}, \tau, \varepsilon); \quad (\text{D29})$$

(ii) *the switch flow has two parts on different timescales: a continuous calm switch flow*

$$\Phi_t^{\text{sw}}(A) = \int_A \mathbf{1}\{\phi \neq \phi_t^*\} \beta_t(d\phi), \quad A \subseteq \Phi, \quad (\text{D30})$$

with β_t non-atomic where deadlines stagger (and carrying a predictable atom where a positive G -mass synchronizes, which by Lemma 6 moves no traded price), and a disaster-triggered component whose predictable rejection intensity (compensator, integrating over marks) is $\lambda^ \pi(\phi, \tau, \mathcal{R}; \varepsilon)$ (D21), the relocations among the rejectors being those for whom the held model differs from the post-jump mode (a self-re-adoption moves nothing); the realized state jump at a disaster epoch is set by the realized mark x and its triggered set, not by a continuous rate. Every law-of-motion term runs against the switch set $\{\phi \neq \phi_t^*\}$, never the raw rejection rate, so a self-re-adoption ($\phi = \phi_t^*$) renews the test window but*

leaves ϕ , hence S_t , unchanged.

On the hazard's reading: every disaster, rejecting or not, moves the state; a non-rejecting disaster appends a mark, updating both π and the next deadline Δ^* , after which the first passage continues as the piecewise-deterministic law of Proposition 11(iii). The indicator is an instantaneous-at- t restriction defining the *current* flow; at a future rejection τ' the relevant mode is $\mathbf{m}(\rho_{\tau'})$, and the hazard is re-read there.

Proof of Lemma 5. (i) Switch hazard. By the re-selection map (11), after a rejection the held model is $\phi_t^* = \mathbf{m}(\rho_t)$, which differs from the incumbent ϕ exactly on the switch event $\{\phi \neq \phi_t^*\}$. A rejection resets the record $\mathcal{R} \rightarrow \emptyset$ and age $\tau \rightarrow 0$ whether or not the held model moves, but the held model itself moves only on a switch; hence the held-model part of the rejection hazard is its restriction to $\{\phi \neq \phi_t^*\}$, which is (D29). The current-mode indicator is exact for the *calm* (deadline) part of H^{rej} ; on the disaster-triggered part the switch event is evaluated at the post-jump mode, so the triggered *switch* hazard is $\lambda^* \int_{\mathcal{J}} \mathbf{1}\{D_{\phi,\tau}(\mathcal{R} \oplus x) \geq r_{\phi,\tau}(\varepsilon)\} \mathbf{1}\{\phi \neq \mathbf{m}(\rho_{t-} \oplus x)\} f(x) dx$, by splitting the trigger indicator over the destination event—the law of motion, Proposition 3(iii), reads the relocation exactly there, and a triggered self-re-adoption moves nothing. A self-re-adoption $\phi = \phi_t^*$ renews the test window but leaves ϕ , and therefore S_t , unchanged.

(ii) *Two timescales.* For a single investor the calm-deadline first passage Δ^* is deterministic—an atom of her revision law. Across investors the induced calm-deadline law is atomless on strictly monotone ranges: the tolerance law G is continuous (Assumption 2) and the calm deadline $\varepsilon \mapsto a + \Delta^*(\phi, \varepsilon)$ is non-increasing in ε , and *strictly* so wherever the continuous size-deviance component of (9) renders the threshold $r_{\phi,\Delta}(\varepsilon)$ strictly monotone (Proposition 11); on such ranges the deadline map has G -null level sets, its pushforward places no positive mass at any single date, the wealth-weighted deadline-arrival rate $\beta_t(d\phi)$ is non-atomic, and the calm switch flow (D30) is a *continuous* finite-variation flow. On a flat level set of $\Delta^*(\phi, \cdot)$ of positive G -mass—possible where the discrete count deviance dominates—the deadline-coincident

revisers re-select simultaneously, contributing a *predictable* finite-variation jump rather than a Brownian increment; either way the flow is finite-variation, and the *traded* price is continuous across it (Lemma 6), so the no-arbitrage decomposition is unaffected. The disaster-triggered component, whose predictable intensity (compensator, integrating over marks) is $\lambda^* \pi(\phi, \tau, \mathcal{R}; \varepsilon)$ (D21), can fire only at a disaster epoch; the realized jump is then set by the realized mark x , simultaneously for the whole triggered set, and is not a continuous rate. Both parts are restricted to $\{\phi \neq \phi_t^*\}$, so a self-re-adoption moves no held-model mass and leaves S_t unchanged; it does reset the reviser’s record and age, a movement of the record-resolved Ψ_t that the forward-looking valuation tracks (Proposition 13). $\square$

Remark 6 (Why a stationary fear cross-section needs demographic refresh). The re-selection (11) reads the second-order posterior ρ_t^i , updated on the *full* history (6), whereas the test reads only the since-adoption window: the two memories differ. Under the true DGP and the identifiable family of Assumption 4, ρ_t^i is consistent for the Kullback–Leibler model ϕ^* , so $\text{MAP}(\rho_t^i) \rightarrow \phi^*$; once an infinitely-lived investor holds ϕ^* she re-selects near ϕ^* at each rejection, and the held-model cross-section concentrates on ϕ^* —the fear dispersion dies out. A *nondegenerate* stationary fear cross-section is therefore not a property of the present primitives; it is sustained by demographic refresh—newborns who enter holding a *drawn* model $\phi_0 \sim \Pi_\phi$ with a reset test clock (Assumption 5(iv)), held until their first rejection—which an overlapping-generations birth–death structure supplies, added as an explicit primitive where the stationary cross-section is required. The refresh is in the *held models*, not the posteriors: each newborn inherits the (concentrated) common posterior ρ_t , so $\mathbf{m}(\rho_t)$ stays common across cohorts, but holds her drawn $\phi_0 \neq \mathbf{m}(\rho_t)$ until she revises, keeping a positive mass of wealth away from ϕ^* . The drawn ϕ_0 is an exogenous status-quo endowment, not a birth-time MAP re-initialization: a newborn carries her inherited held model until her own test rejects it, and the MAP rule (11) governs only re-selection at rejection, exactly as for an incumbent. This departure from MAP-initialization at birth is the OLG primitive on which the stationary dispersion rests. We do not assert stationarity from the present primitives. (*Not proved here; the posterior-consistency*

step and the stationary cross-section are formalized with the demographic primitive.)

Proof of Proposition 3, the law of motion. (i) Between disasters and switches every held model ϕ_i is fixed. Writing S_t as the pushforward of the wealth measure $s_t^i di$ under $i \mapsto \phi_i$ (Definition 3), $\langle h, S_t \rangle = \int_0^1 h(\phi_i) s_t^i di$. The interchange of $\frac{d}{dt}$ and $\int_0^1 \cdot di$ is justified by dominated convergence: h is bounded and, on compact t -intervals, s_t^i and its t -derivative are bounded uniformly in i (as in Proposition 10, via (C2) and $\lambda_i \in [\underline{\lambda}, \bar{\lambda}]$). Then

$$\begin{aligned} \frac{d}{dt} \langle h, S_t \rangle &= \int_0^1 h(\phi_i) \frac{ds_t^i}{dt} di \\ &= \int_0^1 h(\phi_i) s_t^i \frac{d \log s_t^i}{dt} di \end{aligned} \tag{D31}$$

$$= \int_0^1 h(\phi_i) s_t^i \frac{1}{\alpha} (\bar{\lambda}_t^S - \lambda_i) di \tag{D32}$$

$$= \frac{1}{\alpha} \int_{\Phi} (\bar{\lambda}_t^S - \lambda) h(\phi) S_t(d\phi), \tag{D33}$$

line (D31) using $ds_t^i/dt = s_t^i d \log s_t^i/dt$, (D32) substituting the calm share drift (D6), and (D33) pulling out the constant $\frac{1}{\alpha}$ and rewriting the i -integral as a ϕ -integral against S_t (Definition 3); this is (21).

(ii) At a calm-deadline switch, investor i 's held model jumps from ϕ_i to $\phi_i' = \mathbf{m}(\rho_t^i)$ (11). By Lemma 4 the second-order posterior is common, $\rho_t^i = \rho_t$, so $\phi_i' = \mathbf{m}(\rho_t) = \phi_t^*$ for every investor revising at t . Her contribution to $\langle h, S_t \rangle$ changes by $s_t^i[h(\phi_t^*) - h(\phi_i)]$; the aggregate rate of switching out of ϕ is the calm switch flow $\Phi_t^{\text{sw}}(d\phi)$ of Lemma 5 (D30) (restricted to $\{\phi \neq \phi_t^*\}$). Where the deadline pushforward of the tolerance law is atomless the flow is continuous in time; on a flat deadline level set of positive G -mass it carries a predictable finite-variation atom, and (22) is then read in the Stieltjes sense, integrating $h(\phi_t^*) - h(\phi)$ against $d\Phi^{\text{sw}}$ —a predictable step in S_t that by Lemma 6 moves no traded price. Aggregating gives (22). Disaster-triggered switches do not enter here—they fire only at disaster epochs, in (iii).

(iii) At a disaster of size x , Proposition 10(ii) gives $s_t^i = s_{t-}^i r(\phi_i, x)^{1/\alpha} / g_t(x)$ (with

$r_i(x; t) = r(\phi_i, x)$). Hence for any bounded h ,

$$\begin{aligned}\langle h, S_t \rangle &= \int_0^1 h(\phi_i) s_t^i di \\ &= \int_0^1 h(\phi_i) s_{t-}^i \frac{r(\phi_i, x)^{1/\alpha}}{g_t(x)} di\end{aligned}\tag{D34}$$

$$= \frac{1}{g_t(x)} \int_0^1 h(\phi_i) r(\phi_i, x)^{1/\alpha} s_{t-}^i di\tag{D35}$$

$$= \frac{1}{g_t(x)} \int_{\Phi} h(\phi) r(\phi, x)^{1/\alpha} S_{t-}(d\phi),\tag{D36}$$

line (D34) substituting the share jump, (D35) pulling out the constant $g_t(x)^{-1}$, and (D36) rewriting the i -integral against S_{t-} (Definition 3). As this holds for every bounded h , it is the measure identity (23)—but only the *reweight* step (a) of Definition 4, in which h is still read at the pre-disaster model ϕ . The full post-disaster state applies the relocation step (d) on top: the holders the disaster triggers move to ϕ_t^* at the disaster epoch (a jump, not the continuous flow of (ii)), so for them h is read at ϕ_t^* . The triggered investors form the region $\mathcal{T}_t^x := \{(\phi, \mathcal{R}, \tau, \varepsilon) : D_{\phi, \tau}(\mathcal{R} \oplus x) \geq r_{\phi, \tau}(\varepsilon)\}$ (D21) in the *record-resolved* state—not a subset of Φ , since the trigger depends on the full state and two investors sharing ϕ can differ in it—so the relocation is read against Ψ_{t-} , with post-disaster destination $\phi_t^*(x) := \mathbf{m}(\rho_{t-} \oplus x)$ reading the mark-updated posterior (independent of x in the intensity channel):

$$\langle h, S_t^x \rangle = \frac{1}{g_t(x)} \int r(\phi, x)^{1/\alpha} \left[h(\phi) \mathbf{1}\{(\phi, \mathcal{R}, \tau, \varepsilon) \notin \mathcal{T}_t^x\} + h(\phi_t^*(x)) \mathbf{1}\{(\phi, \mathcal{R}, \tau, \varepsilon) \in \mathcal{T}_t^x\} \right] \Psi_{t-}(d(\phi, \mathcal{R}, \tau, \varepsilon));\tag{D37}$$

the reweight (D36) is the special case $\mathcal{T}_t^x = \emptyset$ (no triggered switchers). Since $\mathcal{T}_t^x$ and $\phi_t^*(x)$ are functionals of the record cross-section and the public posterior, the disaster map (D37) reads Ψ_{t-} and not S_{t-} alone; that no finite collection of moments of S_t suffices even for the instantaneous prices is the separate claim proved above. $\square$

Appendix E. Proofs for Section 5

A. Where each instantaneous price sits in the decomposition

The short rate, the price–dividend ratio, and the disaster premium are derived below; here we record where each sits in the decomposition of Proposition 5.

The short rate is pure $B_t(S_t)$. The instantaneous safe rate is $r_t = \delta + \alpha\mu_0 - \frac{1}{2}\alpha(1+\alpha)\sigma^2 + \bar{\lambda}_t^S - U_t$ (Proposition 12): a constant power-utility diffusive part plus the belief-driven disaster part $\bar{\lambda}_t^S - U_t$, both functionals of S_t , so the rate carries no survival-history residual and a Markov-in-cross-section benchmark matches it exactly. Beliefs enter only through the difference, so a fear wave lowers the rate precisely when the priced-tilt response dominates the mean-intensity response, $dU_t > d\bar{\lambda}_t^S$ —the flight-to-safety of the lazy economy, purely belief-driven since $(\mu_0, \sigma, \lambda^*, F_J)$ never move. The same fear factor that lowers the rate raises the disaster premium, so one channel delivers a low safe rate and a high premium together, the pairing the rare-disaster literature uses to address both puzzles (Barro, 2006). Under power utility this passes a volatile fear wave straight into r_t , against smooth real rates; recursive-preference disaster models keep r_t smooth (Wachter, 2013), a device we forgo, so the belief block must be disciplined (Section 6).

The price–dividend ratio carries the residual. The ratio is $v(\Psi_t)$ (Proposition 13), a Feynman–Kac functional of the full record-resolved state—not autonomous in S_t in any channel and, under size disagreement, with no finite-dimensional instantaneous reduction (Proposition 3); in the decomposition it splits as $v_B(S_t) + \mathcal{H}_t^v(\Psi_t)$, the residual driven by the same staggered-revision channels as the forward-fear residual below and zero for any Markov benchmark. Its comparative static is *two-signed*: the valuation-growth rate loads on the cross-section through two non-interchangeable aggregates—the *arithmetic* mean $\bar{\lambda}_t^S$ in the reallocation drift and the *power* mean $G_t^\alpha \leq \bar{\lambda}_t^S/\lambda^*$ in the priced-jump term—so whether a more fearful cross-section raises valuation growth and *expands* the price–dividend ratio, against the naive “higher discount, lower price” reading, is a quantitative condition met at our calibration, not a consequence

of Jensen. A finite-menu Feynman–Kac equation and a first-order belief-dispersion expansion follow (Corollaries IA.1 and IA.2).

The premium loads on the fear factor, and predicts returns. The instantaneous excess return is $\alpha\sigma^2 + \text{DRP}_t$ (Proposition 14): the diffusive premium $\alpha\sigma^2$ is constant, so all variation is the disaster premium, which loads on the fear factor $U_t = U^{\text{RA}}\kappa_t$. Priced disaster fear therefore forecasts excess returns while carrying no information about diffusive volatility, pinned at the constant σ —the return-premium leg of the Andersen et al. (2015b) unspanned-tail finding, with the caveat that, σ constant, the non-prediction of volatility is an identity rather than a match. The premium’s *level* runs through the price–dividend jump ratio ψ_t , hence $v(\Psi_t)$, so it too carries the survival-history residual.

Lemma 6 (Predictable revisions carry no price jump): *Let τ be a calm-deadline revision time—a predictable time with $\Delta N_\tau = 0$ at which the revising cohort’s held model jumps from ϕ to the common destination $\phi_\tau^* = \mathbf{m}(\rho_\tau)$ (Lemma 4), possibly for a positive G -mass synchronizing on a flat level set of $\Delta^*(\phi, \cdot)$. Then (i) every belief density is continuous at τ , $M_\tau^i = M_{\tau-}^i$, so Ξ_t , the kernel ξ_t , and every consumption share s_t^i are continuous at τ and each reviser carries its pre-jump share with no instantaneous wealth transfer; (ii) the share-weighted held-model distribution S_t jumps in total variation, so the instantaneous functionals $g_t(\cdot)$, U_t , the short rate, and the level of the disaster premium carry a predictable jump at τ ; (iii) the traded price is continuous, $P_{\tau-} = P_\tau$, equivalently $v(\Psi_{\tau-}, \rho_{\tau-}) = v(\Psi_\tau, \rho_\tau)$ on the record-resolved state of Definition 4, and P jumps only at the totally inaccessible disaster epochs. The conclusion uses no atomlessness of the deadline law.*

Proof. (i). In (C1) the factor $\exp(-\int_0^t (\lambda_i(s-) - \lambda^*) ds)$ is absolutely continuous in t , and the product $\prod_{T_k \leq t} r_i(J_k; T_k)$ acquires a factor only when t crosses a disaster epoch T_k ; since $\Delta N_\tau = 0$, none is acquired at τ , and the revision changes $\lambda_i(\cdot-)$ and the forward likelihood ratios only on $\{s \geq \tau\}$ —a Lebesgue-null contribution to M_τ^i . Hence $M_\tau^i = M_{\tau-}^i$: the held-model switch moves the predictable forward drift of M^i , not its level. (Equivalently, M^i is a true $(\mathcal{F}_t, \mathbb{P})$ -

martingale built from the predictable, left-continuous held-model compensator (Brémaud, 1981, VI, T4), and the driving Brownian-plus-Poisson filtration is quasi-left-continuous, so $\mathcal{F}_\tau = \mathcal{F}_{\tau-}$ and $M_\tau^i = \mathbb{E}[M_\tau^i \mid \mathcal{F}_{\tau-}] = M_{\tau-}^i$.) On a disaster-free neighbourhood of τ the ratio M_t^i/M_τ^i is bounded above and below uniformly in i (Λ compact), so dominated convergence gives $\Xi_t \rightarrow \Xi_\tau$ in (12); with C_t continuous in calm, $\xi_t = e^{-\delta t} C_t^{-\alpha} \Xi_t^\alpha$ and the shares (13) are continuous, so $s_\tau^i = s_{\tau-}^i$.

(ii). A positive mass relocates $\phi \rightarrow \phi_\tau^*$ while s^i is continuous, so $S_t = \int_0^1 s_t^i \delta_{\phi_i(t)} di$ jumps in total variation; the belief aggregator $g_t(x) = \int_\Phi r(\phi, x)^{1/\alpha} S_t(d\phi)$ and the functionals it determines inherit a predictable jump.

(iii). The equilibrium gain process is a $\mathbb{P}$ -martingale (Proposition 1), so $\xi_t P_t = \mathbb{E}_t[\int_t^\infty \xi_s C_s ds]$. The dividend carries no mass at the single time τ , and by quasi-left-continuity $\mathcal{F}_\tau = \mathcal{F}_{\tau-}$ with $\rho_\tau = \rho_{\tau-}$ (the public posterior is continuous in calm), so the two conditional expectations coincide and $\xi_{\tau-} P_{\tau-} = \xi_\tau P_\tau$; dividing by the continuous, strictly positive ξ gives $P_{\tau-} = P_\tau$. Writing $P_t = C_t v(\Psi_t, \rho_t)$ with $C_\tau = C_{\tau-}$ yields the value-matching form. The destination $\phi_\tau^* = \mathbf{m}(\rho_\tau)$ is read from ρ_τ , so the relocation map is $\mathcal{F}_{\tau-}$ -measurable precisely because the forward-looking state carries ρ_t ; and $\Psi_{\tau-}$ already encodes the imminent deadline (the ages and records that pin Δ^*), so the forward value already incorporates the scheduled relocation. The disaster epochs are totally inaccessible, so P jumps only there. $\square$

Proof of Proposition 5. The decomposition (25) itself is self-contained: $B_t = \mathbb{E}[F_t \mid S_t]$ is the L^2 projection of F_t onto the σ -algebra generated by S_t , so it is S_t -measurable, and the residual $\mathcal{H}_t = F_t - B_t$ has $\mathbb{E}[\mathcal{H}_t \mid S_t] = 0$ by the tower property and is measurable in the finer Ψ_t , since $S_t = \pi_\phi \Psi_t$. That each price object F_t is itself measurable in the pricing state—in (Ψ_t, ρ_t) , with ρ_t entering through re-selection—is the content of the state-sufficiency results (Propositions 15 and 13); the projection argument takes it as input. This holds for any equilibrium price object F_t and uses nothing beyond the definitions of the two states. The classification (i)–(iii) of which objects carry a zero versus a nonzero residual is read off the pricing results established later in

this appendix; we record it here only to fix the organizing split, and it carries no logical weight in the present argument. Throughout, the residual's leading term is the *centered* channel—the staggered-revision channels minus their S_t -conditional mean; the raw channels coincide with the residual only where their S_t -projection vanishes. (i) The fear factor and multiplier, the risk-neutral disaster law, and the short rate are S_t -measurable (Proposition 15), so each equals its own projection and $\mathcal{H}_t \equiv 0$. (ii) The price–dividend ratio and the premium level are functionals of the full Ψ_t , not autonomous in S_t (Proposition 13); and the forward fear factor has $F_t(0) = U_t$ measurable in S_t with right-hand slope $\partial_s^+ F_t(0) = b_S(S_t) + U_t c_{\text{sw}}(\Phi_t^{\text{sw}}) + \lambda^* \Theta_{\text{sw}}(\Psi_t)$, whose last two terms are measurable in Ψ_t and not in S_t (Theorem 1), so $\mathcal{H}_t = s (U_t c_{\text{sw}}(\Phi_t^{\text{sw}}) + \lambda^* \Theta_{\text{sw}}(\Psi_t) - \mathbb{E}[U_t c_{\text{sw}} + \lambda^* \Theta_{\text{sw}} \mid S_t]) + o(s)$ —the *centered* staggered-revision channels, the raw channels net of their S_t -conditional mean—zero at horizon zero and first order in s . (iii) For any benchmark whose pricing state is Markov in the current cross-section S_t , the conditional law of the future is by construction a functional of S_t , so every forward object $\mathbb{E}[\cdot \mid \mathcal{F}_t]$ equals its own S_t -projection and $\mathcal{H}_t \equiv 0$. The worked instance is Theorem 1(iv) with Internet Appendix IA.9: under Bayesian updating the per-investor state is the posterior, its own sufficient statistic, so the cross-section is Markov, the deadline flow Φ_t^{sw} and trigger propensity π are absent, and every forward object closes in S_t ; an affine latent-state specification and a representative filtering economy are Markov in their own states by construction, with the same conclusion. $\square$

Proposition 12 (The equilibrium short rate): *In the equilibrium of Proposition 1, the instantaneous risk-free rate is*

$$r_t = \delta + \alpha \mu_0 - \frac{1}{2} \alpha (1 + \alpha) \sigma^2 + \bar{\lambda}_t^S - U_t, \quad (\text{E1})$$

with $\bar{\lambda}_t^S = \int_{\Phi} \lambda dS_t$ the wealth-weighted held intensity (Definition 3) and U_t the disaster-fear factor (17). The diffusive part $\delta + \alpha \mu_0 - \frac{1}{2} \alpha (1 + \alpha) \sigma^2$ is the constant power-utility rate; the disaster part $\bar{\lambda}_t^S - U_t$ is belief-driven, and at correct beliefs ($\bar{\lambda}_t^S = \lambda^*$, $U_t = U^{\text{RA}}$) collapses to the rare-disaster precautionary term $-\lambda^*(\mathbb{E}_f[e^{\alpha J}] - 1) < 0$.

Proof of Proposition 12. Take logs of $\xi_t = e^{-\delta t} C_t^{-\alpha} \Xi_t^\alpha$ (12):

$$\log \xi_t = -\delta t - \alpha \log C_t + \alpha \log \Xi_t. \quad (\text{E2})$$

By Assumption 1 (the jump multiplies C by e^{-x} , so $\Delta \log C = -x$),

$$d \log C_t = (\mu_0 - \tfrac{1}{2} \sigma^2) dt + \sigma dW_t - \int_{\mathcal{J}} x \mu(dt, dx), \quad (\text{E3})$$

and by Proposition 10—the calm drift $d \log \Xi_t / dt = -\frac{1}{\alpha}(\bar{\lambda}_t^S - \lambda^*)$ (from (D4) divided by Ξ_t) and the disaster jump $\Xi_t / \Xi_{t-} = g_t(x)$ (D2), so $\Delta \log \Xi = \log g_t(x)$, with no diffusion since M^i carries none—

$$d \log \Xi_t = -\frac{1}{\alpha}(\bar{\lambda}_t^S - \lambda^*) dt + \int_{\mathcal{J}} \log g_t(x) \mu(dt, dx). \quad (\text{E4})$$

Differentiating (E2) and substituting (E3)–(E4),

$$\begin{aligned} d \log \xi_t &= -\delta dt - \alpha d \log C_t + \alpha d \log \Xi_t \\ &= [-\delta - \alpha(\mu_0 - \tfrac{1}{2} \sigma^2) - (\bar{\lambda}_t^S - \lambda^*)] dt - \alpha \sigma dW_t + \int_{\mathcal{J}} [\alpha x + \alpha \log g_t(x)] \mu(dt, dx) \end{aligned} \quad (\text{E5})$$

the dt , dW , and μ terms collected from (E3)–(E4).

Now pass from $\log \xi$ to $\xi = e^{\log \xi}$ by Itô's formula for the exponential of a semimartingale. Write $L = \log \xi$; by (E5) its continuous martingale part L^c satisfies $dL_t^c = -\alpha \sigma dW_t$, so, with $d\langle W \rangle_t = dt$, its quadratic variation and the resulting Itô correction are

$$d\langle L^c \rangle_t = (-\alpha \sigma)^2 d\langle W \rangle_t = \alpha^2 \sigma^2 dt, \quad \tfrac{1}{2} d\langle L^c \rangle_t = \tfrac{1}{2} \alpha^2 \sigma^2 dt, \quad (\text{E6})$$

and its jump at a disaster of size x is $\Delta L_t = \alpha x + \alpha \log g_t(x)$ (the μ -integrand of (E5)). Itô's exponential formula then gives

$$\frac{d\xi_t}{\xi_{t-}} = dL_t^c + (\text{drift of } L) dt + \tfrac{1}{2} d\langle L^c \rangle_t + \int_{\mathcal{J}} (e^{\Delta L_t(x)} - 1) \mu(dt, dx), \quad (\text{E7})$$

in which the jump factor evaluates as

$$e^{\Delta L_t} - 1 = e^{\alpha x + \alpha \log g_t(x)} - 1 = e^{\alpha x} e^{\alpha \log g_t(x)} - 1 = e^{\alpha x} g_t(x)^\alpha - 1$$

(consistent with (C9)). Collecting the dt terms of (E7)—the drift of L read from (E5) plus the correction (E6)—the continuous drift of $d\xi_t/\xi_t$ is therefore

$$\left(\frac{d\xi_t}{\xi_t}\right)_{\text{drift}}^{\text{cont}} = -\delta - \alpha(\mu_0 - \tfrac{1}{2}\sigma^2) - (\bar{\lambda}_t^S - \lambda^*) + \tfrac{1}{2}\alpha^2\sigma^2. \quad (\text{E8})$$

The mean jump rate is the jump factor integrated against the disaster compensator $\nu(dt, dx) = \lambda^* dt f(x) dx$ (5):

$$\begin{aligned} \frac{1}{dt} \mathbb{E}_t \left[(e^{\alpha x} g_t(x)^\alpha - 1) \mu \right] &= \lambda^* \int_{\mathcal{J}} (e^{\alpha x} g_t(x)^\alpha - 1) f(x) dx \\ &= \lambda^* \mathbb{E}_f [e^{\alpha J} g_t(J)^\alpha] - \lambda^* \end{aligned} \quad (\text{E9})$$

$$= U_t - \lambda^*, \quad (\text{E10})$$

line (E9) splitting the integral and (E10) using $U_t = \lambda^* \mathbb{E}_f [e^{\alpha J} g_t(J)^\alpha]$ (17). The safe rate is minus the total expected return on ξ , $r_t = -\frac{1}{dt} \mathbb{E}_t [d\xi_t/\xi_t]$, i.e. minus the sum of (E8) and (E10):

$$\begin{aligned} r_t &= \delta + \alpha(\mu_0 - \tfrac{1}{2}\sigma^2) - \tfrac{1}{2}\alpha^2\sigma^2 + (\bar{\lambda}_t^S - \lambda^*) - (U_t - \lambda^*) \\ &= \delta + \alpha\mu_0 - \tfrac{1}{2}\alpha\sigma^2 - \tfrac{1}{2}\alpha^2\sigma^2 + \bar{\lambda}_t^S - U_t \end{aligned} \quad (\text{E11})$$

$$= \delta + \alpha\mu_0 - \tfrac{1}{2}\alpha(1 + \alpha)\sigma^2 + \bar{\lambda}_t^S - U_t, \quad (\text{E12})$$

line (E11) expanding $\alpha(\mu_0 - \frac{1}{2}\sigma^2)$ and cancelling $\mp\lambda^*$, and (E12) factoring $-\frac{1}{2}\alpha\sigma^2 - \frac{1}{2}\alpha^2\sigma^2 = -\frac{1}{2}\alpha(1 + \alpha)\sigma^2$; this is (E1). At correct beliefs $\bar{\lambda}_t^S = \lambda^*$ and $U_t = U^{\text{RA}} = \lambda^* \mathbb{E}_f [e^{\alpha J}]$, so $\bar{\lambda}_t^S - U_t = -\lambda^*(\mathbb{E}_f [e^{\alpha J}] - 1)$. $\square$

Lemma 7 (Payoff-growth domination): *Under Assumption 1, for every horizon $h > 0$ and every*

$$K \geq 0, \mathbb{E}[\sup_{t \leq s \leq t+h} (C_s/C_t)^{1-\alpha} e^{KN_{(t,s]}}] < \infty.$$

Proof. Write $(C_s/C_t)^{1-\alpha} = e^{(1-\alpha)(\mu_0 - \sigma^2/2)(s-t)} e^{(1-\alpha)\sigma(W_s - W_t)} e^{-(1-\alpha)\sum_{t < u \leq s} x_u}$. The drift factor is bounded on $[0, h]$; $\mathbb{E}[\sup_{s \leq t+h} e^{(1-\alpha)\sigma(W_s - W_t)}] < \infty$ by the reflection principle, the running maximum of Brownian motion having Gaussian tails; and the jump factor is bounded by $e^{(|1-\alpha|\bar{J}+K)N_{(t,t+h]}}$, $\bar{J}$ the compact support bound, with finite compound-Poisson exponential moment $\exp\{\lambda^* h(e^{1-\alpha|\bar{J}+K} - 1)\}$. The factors are independent, so the product is integrable. $\square$

Lemma 8 (Pathwise total-variation stability of the flow): *Let Ψ, Ψ' be record-resolved states with $\|\Psi - \Psi'\|_{\text{TV}} = \varepsilon$, and couple the two economies on a common driving path—the same aggregate marked Poisson measure and, under a TV-optimal coupling of the initial states, the same idiosyncratic death clocks and type draws, and sharing the public record to date, hence the posterior ρ (the comparison varies the cross-section on a common public history)—so that investors of joint mass $1-\varepsilon$ are shared. Then for every horizon $h > 0$: (i) each shared investor’s state coincides in the two economies on $[t, t+h]$; (ii) pathwise, $\|S_s - S'_s\|_{\text{TV}} \leq \varepsilon q^{-N_{(t,s]}} e^{c(s-t)}$ for $s \leq t+h$, with $q = (\underline{r}/\bar{r})^{1/\alpha}$ and $c = 2\lambda_{hi}/\alpha + \Omega$; and (iii) under the margin $\delta > \kappa_L := \bar{m} + c + \bar{\Lambda}(1/q - 1)$ —with $\bar{m} := \sup_S m(S)$ the transversality bound and $\bar{\Lambda} := \lambda^* e^{(\alpha-1)\bar{J}} \bar{r}$ a crude bound on the payoff-tilted disaster intensity— $|v(\Psi) - v(\Psi')| \leq L\varepsilon$ for an explicit finite L ; κ_L is a crude sufficient constant, and parts (i)–(ii) need no condition.*

Proof. (i) A shared investor carries the same held model, record, age, and tolerance in both economies at t , faces the same disasters and death clock, and—the one aggregate input to her dynamics—relocates, when she revises, to the *public* destination $\mathbf{m}(\rho_s)$, a functional of the common driving path (Lemma 4). Her deadline, trigger evaluations, and destinations therefore coincide, and her state does, by induction over her revision times. (ii) Write the discrepancy as the unshared weight. In calm, each side’s shares move by the replicator at log-rates bounded by λ_{hi}/α , and turnover retires unshared mass at rate Ω while entrants are shared; so the unshared TV-mass grows at most at rate $2\lambda_{hi}/\alpha + \Omega$ through the relative renormalisation. At a disaster the reweight multiplies every relative weight by a factor in $[q, 1/q]$ (15), inflating the discrepancy

by at most q^{-1} ; relocations move mass to the common public destination and do not increase total variation. Compounding gives the display. (iii) Write $Y_\tau := e^{-\delta\tau}(C_{t+\tau}/C_t)^{1-\alpha}(\Xi_{t+\tau}/\Xi_t)^\alpha$ for the discounted payoff; on the common driving path its consumption factor is identical across the pair, so only the aggregator ratio differs. Pathwise,

$$A_\tau := \left| \log \frac{\Xi_{t+\tau}}{\Xi_t} - \log \frac{\Xi'_{t+\tau}}{\Xi'_t} \right| \leq \int_0^\tau \frac{\Delta_\lambda}{\alpha} \|S_u - S'_u\|_{\text{TV}} du + \sum_{\text{disasters} \leq \tau} L_g \|S_u - S'_u\|_{\text{TV}} \leq \varepsilon q^{-N_\tau} e^{c\tau} \left(\frac{\Delta_\lambda \tau}{\alpha} + L_g N_\tau \right)$$

using (ii) and the bounded-Lipschitz aggregate inputs on the compact $\Phi \times \mathcal{J}$ ($\Delta_\lambda := \bar{\lambda} - \underline{\lambda}$; $L_g := (\bar{r}^{1/\alpha} - \underline{r}^{1/\alpha})/\underline{r}^{1/\alpha}$). Since $|Y_\tau - Y'_\tau| \leq \min(Y_\tau, Y'_\tau)(e^{\alpha A_\tau} - 1)$ and also $\leq Y_\tau + Y'_\tau$, the elementary bound $1 \wedge (e^x - 1) \leq e x$ for $x \geq 0$ gives $|Y_\tau - Y'_\tau| \leq e\alpha (Y_\tau + Y'_\tau) A_\tau \leq e\alpha \varepsilon (Y_\tau + Y'_\tau) H_\tau$. Change measure to the payoff-tilted law at horizon τ , $d\mathbb{Q}^\tau \propto Y_\tau d\mathbb{P}$: under it the disaster count is dominated by a Poisson process of intensity $\bar{\Lambda}$ (the tilt weights each mark by $e^{-(1-\alpha)x} g(x)^\alpha \leq e^{(\alpha-1)\bar{J}\bar{r}}$), so $\mathbb{E}^{\mathbb{Q}^\tau}[q^{-N_\tau}(a + bN_\tau)] \leq (a + b\bar{\Lambda}\tau/q) e^{\bar{\Lambda}\tau(1/q-1)}$, while $\mathbb{E}[Y_\tau] \leq e^{-(\delta-\bar{m})\tau}$; the same bounds hold for Y' . Hence

$$\mathbb{E}[|Y_\tau - Y'_\tau|] \leq C_0 \varepsilon (1 + \tau) e^{-(\delta-\bar{m}-c-\bar{\Lambda}(1/q-1))\tau}, \quad C_0 := 2e\alpha \left(\frac{\Delta_\lambda}{\alpha} + L_g \left(1 + \frac{\bar{\Lambda}}{q} \right) \right),$$

and integrating over τ gives $|v(\Psi) - v(\Psi')| \leq L\varepsilon$ with $L = C_0 \int_0^\infty (1 + \tau) e^{-(\delta-\kappa_L)\tau} d\tau < \infty$ under $\delta > \kappa_L$. $\square$

Proposition 13 (The price–dividend ratio): *Let $v_t = P_t/C_t$ be the tree’s price–dividend ratio, and let Ψ_t denote the record-resolved pricing state—the wealth-weighted distribution over per-investor states $(\phi, \mathcal{R}, \tau, \varepsilon)$ (held model, since-adoption record, regime age, tolerance), whose ϕ -marginal is the pricing state S_t (Definition 3). Then $v_t = v(\Psi_t)$ with*

$$v(\Psi_t) = \mathbb{E}_t \left[\int_t^\infty e^{-\delta(s-t)} \left(\frac{C_s}{C_t} \right)^{1-\alpha} \left(\frac{\Xi_s}{\Xi_t} \right)^\alpha ds \right]. \quad (\text{E13})$$

The local expected growth rate m_t of $Y_t := C_t^{1-\alpha} \Xi_t^\alpha$ is a function of the marginal S_t alone,

$$m(S_t) = (1 - \alpha)\mu_0 - \frac{1}{2}\alpha(1 - \alpha)\sigma^2 - (\bar{\lambda}_t^S - \lambda^*) + \lambda^*(\mathbb{E}_f[e^{-(1-\alpha)J} g_t(J)^\alpha] - 1), \quad (\text{E14})$$

but the valuation v depends on the full Ψ_t : scale-freeness removes only the level C_t , and the conditional law of the future path $\{S_s\}_{s \geq t}$ —hence of $\{m(S_s)\}$ and of Ξ_s/Ξ_t —is governed by the record-resolved dynamics of Proposition 3, which are Markov in the pair (Ψ_t, ρ_t) —the calm re-selection destination $\phi_s^* = \mathbf{m}(\rho_s)$ being read from ρ_s —and not autonomous in S_t . Throughout, $v(\Psi_t)$ abbreviates $v(\Psi_t, \rho_t)$: re-selection reads ρ_t , which since-adoption records do not recover. Under the transversality condition

$$\delta > \sup m := \sup_S m(S) \quad (\text{T})$$

the integral (E13) converges, $0 < v(\Psi_t) \leq (\delta - \sup m)^{-1} < \infty$, and v is the unique bounded valuation. It is not semi-closed: v is a Feynman–Kac functional of the full record-resolved state Ψ_t , whose conditional law is Markov in (Ψ_t, ρ_t) and not autonomous in its marginal S_t (above), and even the instantaneous disaster tilt has no finite-dimensional sufficient statistic (Proposition 3); so v is not summarized by finitely many moments of S_t . In the decomposition of Proposition 5 the valuation splits as $v(\Psi_t) = v_B(S_t) + \mathcal{H}_t^v(\Psi_t)$: the cross-section part $v_B(S_t) = \mathbb{E}[v \mid S_t]$ satisfies $\mathbb{E}[\mathcal{H}_t^v \mid S_t] = 0$, and the survival-history residual $\mathcal{H}_t^v(\Psi_t)$ is generated by the record-resolved conditional law of the payoff path: its leading local component is the forward valuation-growth residual $\mathbb{E}_t[m(S_{t+\tau}) \mid \Psi_t] - \mathbb{E}[m(S_{t+\tau}) \mid S_t]$, entering through the discounted Feynman–Kac accumulation (E13), and the higher-order components enter through the same staggered-revision channels’ $(\Phi_t^{\text{sw}}, \pi)$ effect on the path law; it is zero for any Markov benchmark. In the intensity channel ($f_\theta \equiv f$, $g_t \equiv G_t = \mathbb{E}_{S_t}[(\lambda/\lambda^*)^{1/\alpha}]$), (E14) reads

$$m(S_t) = \text{const} - (\bar{\lambda}_t^S - \lambda^*) + \lambda^*(\mathbb{E}_f[e^{-(1-\alpha)J}] G_t^\alpha - 1), \quad (\text{E15})$$

which carries the cross-section through two distinct aggregates: the wealth-weighted arithmetic mean $\bar{\lambda}_t^S$ in the reallocation drift $-(\bar{\lambda}_t^S - \lambda^*)$, and the power mean G_t^α in the priced-jump term, with $G_t^\alpha \leq \bar{\lambda}_t^S / \lambda^*$, strict by Jensen for $\alpha > 1$ unless λ is S_t -a.s. constant (29).

Reading the proposition. The transversality margin is non-vacuous because each term of (E14) is a bounded functional of S_t over the compact menu (g_t bounded, $\mathcal{J}$ compact), so $-\infty < \inf_S m(S) \leq \sup_S m(S) < \infty$. The projection v_B is the best S_t -measurable summary of the valuation—the component any Markov-in-cross-section benchmark can at most reproduce, its own valuation being S_t -measurable, so the defining property $\mathbb{E}[\mathcal{H}_t^v \mid S_t] = 0$ makes the residual orthogonal to every such benchmark. A Markov-in- S_t benchmark prices forward objects by its *own* functional of S_t , which need not equal the projection B_t ; the decomposition says only that whatever that functional is, it leaves the same residual. Capitalization is at the discount net of growth $\delta - m(S_t)$, not at a gross return. The comparative static is two-signed. The two aggregates are not interchangeable, and a more fearful cross-section raises both; whether the priced-jump (marginal-utility) term dominates the frequency drag—so that m *rises* and the constant- m ratio *expands*—is the quantitative condition above, met at our calibration; compression of *levered* equity would in any case require a leverage channel, absent here. That S_t is genuinely insufficient—not merely non-autonomous—has a witness. Take the aged/fresh pair of Theorem 1(iii): a common cross-section S_t and a common public posterior ρ_t , calm deadline flows $\Phi_t^{\text{sw}} \geq \Phi_t^{\text{sw}}$ setwise with the difference charging the switch set. On the calm-conditional paths the dynamics are deterministic, and at the origin the selection terms cancel (both read the shared S_t), leaving only the flow difference: with $Z := (\lambda/\lambda^*)^{1/\alpha}$,

$$\frac{d}{du} [m(S_u) - m(S'_u)] \Big|_{u=t} = c_1 := -\Delta_1 + \alpha \lambda^* \mathbb{E}_f[e^{-(1-\alpha)J}] G_t^{\alpha-1} \Delta_G,$$

where $\Delta_1 := \int_\Phi (\lambda - \lambda(\phi_t^*)) d(\Phi' - \Phi)_t^{\text{sw}} \geq 0$ and $\Delta_G := \int_\Phi (Z - Z(\phi_t^*)) d(\Phi' - \Phi)_t^{\text{sw}} \geq 0$, both non-negative because monotone re-selection gives $\lambda(\phi_t^*) \leq \lambda$ pointwise on the switch set and Z is increasing in λ . The two terms of c_1 carry opposite signs—the frequency-drag relief against

the priced-jump decline, the two-signed comparative static of (E15)—so $m(S_u) - m(S'_u) = c_1(u - t) + o(u - t)$ and the calm-conditional growth paths separate at first order whenever $c_1 \neq 0$ —the conditional law of $\{m(S_s)\}_{s \geq t}$ given Ψ_t is not a function of S_t , which is the insufficiency claimed; the knife-edge is the exact offset $\Delta_1 = \alpha \lambda^* \mathbb{E}_f[e^{-(1-\alpha)J}] G_t^{\alpha-1} \Delta_G$, and the sign of c_1 is the same dominance comparison as (E15), met at our calibration: two states, one marginal, two values.

Proof of Proposition 13. The valuation and its state. By (12), $\xi_u = e^{-\delta u} C_u^{-\alpha} \Xi_u^\alpha$, so

$$\begin{aligned} \frac{\xi_s C_s}{\xi_t C_t} &= \frac{e^{-\delta s} C_s^{-\alpha} \Xi_s^\alpha C_s}{e^{-\delta t} C_t^{-\alpha} \Xi_t^\alpha C_t} \\ &= e^{-\delta(s-t)} \left(\frac{C_s}{C_t} \right)^{1-\alpha} \left(\frac{\Xi_s}{\Xi_t} \right)^\alpha, \end{aligned} \quad (\text{E16})$$

line (E16) using $C_s^{-\alpha} C_s = C_s^{1-\alpha}$ (likewise at t) and collecting the $e^{-\delta}$ and Ξ factors. The tree pays the dividend C_s , so by the pricing identity $P_t = \mathbb{E}_t[\int_t^\infty (\xi_s/\xi_t) C_s ds]$; inserting C_t/C_t and using (E16),

$$\begin{aligned} P_t &= C_t \mathbb{E}_t \left[\int_t^\infty \frac{\xi_s C_s}{\xi_t C_t} ds \right] \\ &= C_t \mathbb{E}_t \left[\int_t^\infty e^{-\delta(s-t)} \left(\frac{C_s}{C_t} \right)^{1-\alpha} \left(\frac{\Xi_s}{\Xi_t} \right)^\alpha ds \right]. \end{aligned}$$

Dividing by C_t gives $v_t = P_t/C_t =$ (E13). The integrand is homogeneous of degree zero in the level C_t , so v_t does not depend on C_t ; it depends on the conditional law given $\mathcal{F}_t$ of the path $(C_s/C_t, \Xi_s/\Xi_t)_{s \geq t}$. The ratios C_s/C_t are functions of the primitive increments only (Assumption 1), independent of the cross-section. The ratios Ξ_s/Ξ_t are driven by the future cross-section path, whose conditional law is Markov in the record-resolved state Ψ_t and *not* autonomous in its ϕ -marginal S_t (Proposition 3: the switch flow depends on the within-class record distribution). Hence $v_t = v(\Psi_t)$.

The growth rate $m(S_t)$. Take logs of $Y_t = C_t^{1-\alpha} \Xi_t^\alpha$, $\log Y_t = (1-\alpha) \log C_t + \alpha \log \Xi_t$, and

substitute (E3)–(E4):

$$\begin{aligned} d \log Y_t &= (1 - \alpha) d \log C_t + \alpha d \log \Xi_t \\ &= \left[(1 - \alpha)(\mu_0 - \tfrac{1}{2}\sigma^2) - (\bar{\lambda}_t^S - \lambda^*) \right] dt + (1 - \alpha)\sigma dW_t + \int_{\mathcal{J}} \left[-(1 - \alpha)x + \alpha \log g_t(x) \right] \mu(dt, dx). \end{aligned} \quad (\text{E17})$$

By Itô's formula for $Y = e^{\log Y}$, the continuous martingale part of (E17) is $(1 - \alpha)\sigma dW_t$, with quadratic variation and Itô correction

$$((1 - \alpha)\sigma)^2 d\langle W \rangle_t = (1 - \alpha)^2 \sigma^2 dt, \quad \tfrac{1}{2}(1 - \alpha)^2 \sigma^2 dt,$$

and the jump factor at a disaster of size x is $e^{-(1-\alpha)x + \alpha \log g_t(x)} - 1 = e^{-(1-\alpha)x} g_t(x)^\alpha - 1$; hence

$$\frac{dY_t}{Y_{t-}} = \left[(1 - \alpha)(\mu_0 - \tfrac{1}{2}\sigma^2) - (\bar{\lambda}_t^S - \lambda^*) + \tfrac{1}{2}(1 - \alpha)^2 \sigma^2 \right] dt + (1 - \alpha)\sigma dW_t + \int_{\mathcal{J}} (e^{-(1-\alpha)x} g_t(x)^\alpha - 1) \mu(dt, dx).$$

Take $\mathbb{E}_t[\cdot]/dt$ (conditioning on $\mathcal{F}_t$): the dW_t term has mean zero, and the random measure μ has compensator $\nu = \lambda^* dt f dx$ (5), so

$$m(S_t) = \tfrac{1}{dt} \mathbb{E}_t \left[\frac{dY_t}{Y_{t-}} \right] = (1 - \alpha)(\mu_0 - \tfrac{1}{2}\sigma^2) - (\bar{\lambda}_t^S - \lambda^*) + \tfrac{1}{2}(1 - \alpha)^2 \sigma^2 + \lambda^* \int_{\mathcal{J}} (e^{-(1-\alpha)x} g_t(x)^\alpha - 1) f(x) dx. \quad (\text{E18})$$

The diffusion terms in (E18) collect as

$$(1 - \alpha)(\mu_0 - \tfrac{1}{2}\sigma^2) + \tfrac{1}{2}(1 - \alpha)^2 \sigma^2 = (1 - \alpha)\mu_0 + \tfrac{1}{2}\sigma^2 [-(1 - \alpha) + (1 - \alpha)^2] \quad (\text{E19})$$

$$= (1 - \alpha)\mu_0 + \tfrac{1}{2}\sigma^2(1 - \alpha)[(1 - \alpha) - 1] \quad (\text{E20})$$

$$= (1 - \alpha)\mu_0 + \tfrac{1}{2}\sigma^2(1 - \alpha)(-\alpha) \quad (\text{E21})$$

$$= (1 - \alpha)\mu_0 - \tfrac{1}{2}\alpha(1 - \alpha)\sigma^2, \quad (\text{E22})$$

(E19) factoring $\tfrac{1}{2}\sigma^2$ from the two variance terms, (E20) factoring $(1 - \alpha)$ from the bracket,

(E21) using $(1 - \alpha) - 1 = -\alpha$, and (E22) multiplying out. The jump integral splits, using $\int_{\mathcal{J}} f dx = 1$, as

$$\begin{aligned} \lambda^* \int_{\mathcal{J}} (e^{-(1-\alpha)x} g_t(x)^\alpha - 1) f(x) dx &= \lambda^* \int_{\mathcal{J}} e^{-(1-\alpha)x} g_t(x)^\alpha f(x) dx - \lambda^* \int_{\mathcal{J}} f(x) dx \\ &= \lambda^* \mathbb{E}_f[e^{-(1-\alpha)J} g_t(J)^\alpha] - \lambda^* \\ &= \lambda^* (\mathbb{E}_f[e^{-(1-\alpha)J} g_t(J)^\alpha] - 1). \end{aligned} \quad (\text{E23})$$

Substituting (E22) and (E23) into (E18),

$$m(S_t) = (1 - \alpha)\mu_0 - \frac{1}{2}\alpha(1 - \alpha)\sigma^2 - (\bar{\lambda}_t^S - \lambda^*) + \lambda^* (\mathbb{E}_f[e^{-(1-\alpha)J} g_t(J)^\alpha] - 1), \quad (\text{E24})$$

which depends on the cross-section only through $\bar{\lambda}_t^S$ and g_t , i.e. through S_t ; this is (E14).

Convergence, bounds, uniqueness. Set $Z_s = e^{-\sup m(s-t)} Y_s / Y_t$. Its local drift rate is $(m(S_s) - \sup m)Z_s \leq 0$ (by the definition of $\sup m$), and its jumps have finite mean (g_t is bounded by $\bar{r}^{1/\alpha}$ (15) and $\mathcal{J}$ is compact), so Z is a supermartingale and $\mathbb{E}_t[Y_s / Y_t] \leq e^{\sup m(s-t)}$. Therefore, under (T),

$$v(\Psi_t) \leq \int_t^\infty e^{-\delta(s-t)} e^{\sup m(s-t)} ds = \int_0^\infty e^{-(\delta - \sup m)u} du = \frac{1}{\delta - \sup m} < \infty, \quad (\text{E25})$$

and $v > 0$ since the integrand is positive. *The lower bound is uniform.* Set $\inf m := \inf_S m(S) > -\infty$, finite because the jump integrand in (E24) is bounded ($g_t \in [\underline{r}^{1/\alpha}, \bar{r}^{1/\alpha}]$ by (15) and $\mathcal{J}$ is compact). Then $\tilde{Z}_s = e^{-\inf m(s-t)} Y_s / Y_t$ has local drift $(m(S_s) - \inf m)\tilde{Z}_s \geq 0$, a nonnegative-drift local submartingale that is uniformly integrable on compacts by Lemma 7, hence a true submartingale, so $\mathbb{E}_t[Y_s / Y_t] \geq e^{\inf m(s-t)}$ and

$$v(\Psi_t) \geq \int_t^\infty e^{-\delta(s-t)} e^{\inf m(s-t)} ds = \frac{1}{\delta - \inf m} > 0,$$

a positive lower bound independent of Ψ_t (it uses only $\inf m$ and δ). With (E25) this makes

$\psi_t(x) = v(\Psi_t^x)/v(\Psi_{t-})$ bounded above and bounded below away from zero—the bound used in Proposition 14 and the excess-volatility result of the Internet Appendix (whose Dynkin step is licensed by the PDMP extended-generator characterization, v absolutely continuous along the calm flow by its Feynman–Kac form; Davis 1993). *Uniqueness among bounded valuations.* If v_1, v_2 are bounded solutions of the pricing identity, then for every horizon $T > t$ their difference $D := v_1 - v_2$ satisfies $D(\Psi_t) = e^{-\delta(T-t)} \mathbb{E}_t[(Y_T/Y_t) D(\Psi_T)]$; bounding $|D| \leq \|D\|_\infty$ and using $\mathbb{E}_t[Y_T/Y_t] \leq e^{\sup m(T-t)}$ (the supermartingale bound above),

$$|D(\Psi_t)| \leq \|D\|_\infty e^{-(\delta - \sup m)(T-t)} \xrightarrow{T \rightarrow \infty} 0 \quad \text{under (T),}$$

so $D \equiv 0$ and the bounded valuation is unique. That the bounded valuation depends on the full record-resolved state Ψ_t , not on finitely many moments of S_t , has two sources, in order of generality: the law of motion is non-autonomous—the switch flow and revision deadlines are functionals of Ψ_t (Propositions 11, 3)—so the forward expectation cannot collapse to a function of S_t alone in *any* channel; and in the size channel the instantaneous disaster tilt is already infinite-dimensional in S_t (Proposition 3). $\square$

Proposition 14 (Equity premium and disaster premium): *In the equilibrium of Proposition 1, the tree’s instantaneous expected excess return decomposes into a diffusive and a disaster part,*

$$\mathbb{E}_t[dR_t^e] - r_t dt = \underbrace{\alpha \sigma^2}_{\text{diffusive}} dt + \text{DRP}_t dt, \quad (\text{E26})$$

where, with the tree’s disaster return $\Delta R_t^e(x) = e^{-x} \psi_t(x) - 1$ and $\psi_t(x) = v(\Psi_t^x)/v(\Psi_{t-})$ the price–dividend jump ratio (the record-resolved state moving from Ψ_{t-} to its post-disaster value Ψ_t^x), the disaster risk premium is the gap between the physical and risk-neutral expected disaster returns,

$$\text{DRP}_t = \lambda^* \mathbb{E}_f[\Delta R_t^e(J)] - U_t \mathbb{E}_{f^Q}[\Delta R_t^e(J)]. \quad (\text{E27})$$

The diffusive premium $\alpha \sigma^2$ is constant. The disaster premium is priced by the risk-neutral

disaster law $(U_t, f_t^{\mathbb{Q}})$ —the fear factor of Definition 2—so, holding (α, λ^*, F_J) fixed, its belief-driven variation runs through $(U_t, f_t^{\mathbb{Q}})$. In the intensity channel (shape fixed) the U_t -loading component of this variation runs through U_t alone (Proposition 2); in the general joint model the premium also moves with the shape $f_t^{\mathbb{Q}}$ and the state-dependent return jump. Even in the intensity channel, though, the premium's level and part of its variation run through the jump ratio ψ_t , hence through $v(\Psi_t)$ (Proposition 13), which varies with the full cross-section and is not summarized by U_t . In the decomposition of Proposition 5, $\text{DRP}_t = B_t^{\text{DRP}}(S_t) + \mathcal{H}_t^{\text{DRP}}(\Psi_t)$: the cross-section part carries the S_t -measurable fear-factor intensity, and the survival-history residual enters through the price–dividend jump ratio ψ_t and $v(\Psi_t)$ (Proposition 13), zero for any Markov benchmark.

Proof of Proposition 14. The no-arbitrage relation. The consumption claim priced by ξ has gain process $\xi_t P_t + \int_0^t \xi_s C_s ds$, a $\mathbb{P}$ -martingale (Proposition 1). Conditioning on $\mathcal{F}_t$, its drift is zero:

$$\mathbb{E}_t[d(\xi_t P_t)] + \xi_t C_t dt = 0. \quad (\text{E28})$$

By Itô's product rule for semimartingales,

$$d(\xi_t P_t) = \xi_{t-} dP_t + P_{t-} d\xi_t + d[\xi, P]_t. \quad (\text{E29})$$

Divide (E29) by $\xi_{t-} P_{t-}$:

$$\frac{d(\xi_t P_t)}{\xi_{t-} P_{t-}} = \frac{dP_t}{P_{t-}} + \frac{d\xi_t}{\xi_{t-}} + \frac{d[\xi, P]_t}{\xi_{t-} P_{t-}}. \quad (\text{E30})$$

The cum-dividend return is $dR_t^e = dP_t/P_{t-} + (C_t/P_t) dt$, so $dP_t/P_{t-} = dR_t^e - (C_t/P_t) dt$; substitute this into (E30), take $\mathbb{E}_t[\cdot]$, and use (E28) divided by the left limit $\xi_{t-} P_{t-}$ (i.e. $\mathbb{E}_t[d(\xi_t P_t)/(\xi_{t-} P_{t-})] = -(C_t/P_{t-}) dt$ to first order, the price jump entering the martingale part, not the drift):

$$\mathbb{E}_t[dR_t^e] - \frac{C_t}{P_{t-}} dt + \mathbb{E}_t\left[\frac{d\xi_t}{\xi_{t-}}\right] + \mathbb{E}_t\left[\frac{d[\xi, P]_t}{\xi_{t-} P_{t-}}\right] = -\frac{C_t}{P_{t-}} dt.$$

Cancel $-(C_t/P_t) dt$ from both sides, then use $\mathbb{E}_t[d\xi_t/\xi_{t-}] = -r_t dt$ (Proposition 12) and $\mathbb{E}_t[d[\xi, P]_t/(\xi_{t-} P_{t-})] =$

$\text{Cov}_t(d\xi_t/\xi_{t-}, dR_t^e)$ (the predictable covariation, the $(C_t/P_t)dt$ yield term being finite-variation and contributing none):

$$\mathbb{E}_t[dR_t^e] - r_t dt = -\text{Cov}_t\left(\frac{d\xi_t}{\xi_{t-}}, dR_t^e\right).$$

Predictability and the jump structure. Placing the price jump in the martingale part of (E28) requires the jumps of P to be *inaccessible*, and they are. Between disasters the belief densities M^i are continuous—they jump only at the points of the disaster measure N , by (C1)—so the kernel ξ and, by Lemma 6, the price P are continuous across every predictable revision time, including a synchronized calm deadline at which a positive G -mass re-selects on a flat level set. There the share-weighted distribution S_t carries a predictable finite-variation jump, but the *traded* price does not: the re-selecting cohort relocates to the public and *predictable* destination $\phi_t^* = \mathbf{m}(\rho_t)$, an anticipated move that leaves the forward-looking value unchanged (Lemma 6(iii)). The only jumps in P are at disasters, totally inaccessible stopping times of the Poisson filtration, so (E28) holds without any atomlessness restriction on the deadline law.

The two covariation channels. The martingale part of $d\xi_t/\xi_t$ is $-\alpha\sigma dW_t$ plus the compensated jump $\int_{\mathcal{J}}(e^{\alpha x}g_t(x)^\alpha - 1)(\mu - \nu)(dt, dx)$ ((E5), (C9)); the martingale part of dR_t^e is σdW_t (the record-resolved state Ψ_t has no Brownian part by Proposition 3, so the deterministic $v(\Psi_t)$ is diffusion-free and the sole Brownian source is σdW_t from C_t) plus the compensated jump $\int_{\mathcal{J}} \Delta R_t^e(x)(\mu - \nu)(dt, dx)$ with $\Delta R_t^e(x) = e^{-x}\psi_t(x) - 1$. The Brownian and Poisson sources are independent, so the covariation splits into a diffusion and a jump part. The diffusion part, using $d\langle W \rangle_t = dt$,

$$\text{Cov}_t(-\alpha\sigma dW_t, \sigma dW_t) = -\alpha\sigma \cdot \sigma d\langle W \rangle_t = -\alpha\sigma^2 dt.$$

For the jump part, the predictable covariation of two compensated integrals $\int H(\mu - \nu)$ and $\int K(\mu - \nu)$ against the same random measure is $\int HK\nu$, valid for predictable H, K with $\int H^2\nu, \int K^2\nu < \infty$ (Brémaud, 1981); here $H = e^{\alpha x}g_t^\alpha - 1$ and $K = \Delta R_t^e$ are bounded (g_t bounded by (15), ψ_t bounded since v is bounded above and below away from zero under (T),

$\mathcal{J}$ compact), so with $\nu = \lambda^* dt f dx$,

$$\text{Cov}_t^{\text{jump}} = \lambda^* \int_{\mathcal{J}} (e^{\alpha x} g_t(x)^\alpha - 1) \Delta R_t^e(x) f(x) dx dt.$$

Hence the premium is $-\text{Cov}_t = -(\text{diffusion}) - (\text{jump})$:

$$\frac{\mathbb{E}_t[dR_t^e]}{dt} - r_t = \alpha\sigma^2 - \lambda^* \int_{\mathcal{J}} (e^{\alpha x} g_t(x)^\alpha - 1) \Delta R_t^e(x) f(x) dx. \quad (\text{E31})$$

Split the integral over the factor $(e^{\alpha x} g_t^\alpha - 1)$:

$$\begin{aligned} -\lambda^* \int_{\mathcal{J}} (e^{\alpha x} g_t^\alpha - 1) \Delta R_t^e f dx &= \lambda^* \int_{\mathcal{J}} \Delta R_t^e f dx - \lambda^* \int_{\mathcal{J}} e^{\alpha x} g_t^\alpha \Delta R_t^e f dx \\ &= \lambda^* \mathbb{E}_f[\Delta R_t^e] - U_t \mathbb{E}_{f_t^\mathbb{Q}}[\Delta R_t^e], \end{aligned} \quad (\text{E32})$$

line (E32) using $\int \Delta R_t^e f dx = \mathbb{E}_f[\Delta R_t^e]$ and the risk-neutral disaster compensator $\lambda^* e^{\alpha x} g_t(x)^\alpha f(x) = \nu_t^\mathbb{Q}(dx)/dt = U_t f_t^\mathbb{Q}(x)$ (17), so that $\lambda^* \int e^{\alpha x} g_t^\alpha \Delta R_t^e f dx = U_t \int \Delta R_t^e f_t^\mathbb{Q} dx = U_t \mathbb{E}_{f_t^\mathbb{Q}}[\Delta R_t^e]$. Substituting (E32) into (E31) gives the diffusive term $\alpha\sigma^2$ and $\text{DRP}_t = \lambda^* \mathbb{E}_f[\Delta R_t^e] - U_t \mathbb{E}_{f_t^\mathbb{Q}}[\Delta R_t^e]$, which are (E26)–(E27). $\square$

Proposition 15 (State reduction to the wealth-weighted distribution): *In the equilibrium of Proposition 1, each of the following instantaneous equilibrium price quantities is a measurable functional of the pricing state S_t (Definition 3) alone: the belief aggregator $g_t(\cdot)$ (24), the kernel calm drift and disaster jump ratio (D1)–(D2) (jump quantities at a disaster epoch evaluated at the predictable left limit S_{t-}), the fear factor U_t and multiplier κ_t and the risk-neutral disaster law $f_t^\mathbb{Q}$ (17)–(18), the short rate r_t (E1), the local valuation growth $m(S_t)$ (E14), and the diffusive premium $\alpha\sigma^2$ (E26). The disaster-premium level is excluded: it carries ψ_t (Proposition 14). Equivalently, any two record-resolved states Ψ_t, Ψ'_t (Definition 4) with a common held-model marginal $\pi_\phi \Psi_t = \pi_\phi \Psi'_t = S_t$ yield identical values of all these quantities at t : the since-adoption records, regime ages, surprise thresholds, and second-order posteriors enter instantaneous prices*

only through S_t . These prices factor through the aggregator $g_t(\cdot)$, itself a functional of S_t coarser than the full measure; what fails under size disagreement is any finite-dimensional reduction—by Proposition 3 no finite collection of linear moments of S_t determines $g_t(\cdot)$; in the intensity channel $g_t \equiv G_t$ is a scalar, and on a finite menu S_t is itself a point of the simplex, so the non-reduction is a size-disagreement statement. The forward-looking objects do not reduce to S_t : the price–dividend ratio $v(\Psi_t)$ (E13), the price–dividend jump ratio ψ_t , and hence the level of the disaster premium DRP_t (E27) are functionals of the full record-resolved state Ψ_t (Proposition 13).

Proof of Proposition 15. Each instantaneous quantity is, by its defining display, a functional of S_t .

- (a) By (24), $g_t(x) = \int_{\Phi} (\lambda/\lambda^*)^{1/\alpha} (f_{\theta}(x)/f(x))^{1/\alpha} S_t(d\phi)$ is the integral of a fixed (t, Ψ) -independent kernel against S_t ; so the whole function $g_t(\cdot)$ depends on Ψ_t only through $S_t = \pi_{\phi}\Psi_t$.
- (b) By Definition 3, $\bar{\lambda}_t^S = \int_{\Phi} \lambda S_t(d\phi)$. Hence the calm drift (D1), $-\frac{1}{\alpha}(\bar{\lambda}_t^S - \lambda^*)$, and the disaster jump ratio (D2), $g_t(x)$, are functionals of S_t by (a).
- (c) By (17)–(18), $U_t = \lambda^* \mathbb{E}_f[e^{\alpha J} g_t(J)^{\alpha}]$, $\kappa_t = U_t/U^{\text{RA}}$, and $f_t^{\mathbb{Q}}(x) = e^{\alpha x} g_t(x)^{\alpha} f(x)/\mathbb{E}_f[e^{\alpha J} g_t(J)^{\alpha}]$ are fixed functionals of $g_t(\cdot)$, hence of S_t by (a).
- (d) By (E1), $r_t = \delta + \alpha\mu_0 - \frac{1}{2}\alpha(1+\alpha)\sigma^2 + \bar{\lambda}_t^S - U_t$, and by (E14) $m(S_t)$, are functionals of $\bar{\lambda}_t^S$ and $g_t(\cdot)$ by (b)–(c); the diffusive premium $\alpha\sigma^2$ (E26) is constant.

The displays in (a)–(d) involve Ψ_t only through S_t ; the coordinates $(\mathcal{R}, \tau, \varepsilon)$ of Definition 4 appear in none of them. Therefore $\pi_{\phi}\Psi_t = \pi_{\phi}\Psi'_t$ forces equality of every quantity in (a)–(d), the stated equivalence. That the instantaneous prices admit no *finite-dimensional* reduction—though they factor through the coarser aggregator $g_t(\cdot)$ —is Proposition 3: the map $S \mapsto g_t(\cdot)$ in (a) is into an infinite-dimensional image (by the spanning condition, the family $\{(f_{\theta}/f)^{1/\alpha}\}$ is linearly independent), so no finite collection of *linear* moments of S_t is sufficient (nonlinear summaries are not excluded by this argument, and the intensity channel and the finite

menu do reduce); the claim is the size-disagreement one. For the forward-looking objects, $v(\Psi_t) = \mathbb{E}_t[\int_t^\infty e^{-\delta(s-t)}(C_s/C_t)^{1-\alpha}(\Xi_s/\Xi_t)^\alpha ds]$ (E13) is a conditional expectation of a functional of the future path $\{S_s\}_{s \geq t}$, whose conditional law is governed by the record-resolved generator of Proposition 3, Markov in Ψ_t and not autonomous in S_t (Proposition 13); so v is a conditional expectation of a functional of the future path whose law is fixed by Ψ_t and not by S_t alone; v is therefore a functional of the full record-resolved state, as claimed, and where two such states with a common marginal produce different forward laws the values can differ—the constructive instance is the deadline channel of Theorem 1(iii), established independently of this proposition’s forward clause. The jump ratio $\psi_t = v(\Psi_t^J)/v(\Psi_{t-})$ and the premium level (E27) inherit this dependence. $\square$

B. Dynamics around a disaster

Proof of Proposition 6. (i) Calm erosion. Between disasters the calm derivative is the sum of the selection drift (21) and the calm switch flow (22) (Proposition 3(i)–(ii)). The selection part gives, for bounded h ,

$$\frac{d}{dt}\langle h, S_t \rangle = \frac{1}{\alpha} \int_{\Phi} (\bar{\lambda}_t^S - \lambda) h(\phi) S_t(d\phi) \quad (\text{E33})$$

$$= \frac{1}{\alpha} \left[\bar{\lambda}_t^S \int_{\Phi} h dS_t - \int_{\Phi} \lambda h dS_t \right] \quad (\text{E34})$$

$$= \frac{1}{\alpha} [\mathbb{E}_{S_t}[\lambda] \mathbb{E}_{S_t}[h] - \mathbb{E}_{S_t}[\lambda h]] \quad (\text{E35})$$

$$= -\frac{1}{\alpha} (\mathbb{E}_{S_t}[\lambda h] - \mathbb{E}_{S_t}[\lambda] \mathbb{E}_{S_t}[h]) \quad (\text{E36})$$

$$= -\frac{1}{\alpha} \text{Cov}_{S_t}(\lambda, h), \quad (\text{E37})$$

where (E33) is (21); (E34) distributes $(\bar{\lambda}_t^S - \lambda)h = \bar{\lambda}_t^S h - \lambda h$ and splits the integral; (E35) writes $\int h dS_t = \mathbb{E}_{S_t}[h]$, $\int \lambda h dS_t = \mathbb{E}_{S_t}[\lambda h]$ and substitutes $\bar{\lambda}_t^S = \mathbb{E}_{S_t}[\lambda]$; (E36) factors -1 ; and (E37) is the definition of covariance; the calm switch flow (22) adds $\int_{\Phi} [h(\phi_t^*) - h(\phi)] \Phi_t^{\text{sw}}(d\phi)$. Setting $h(\phi) = \lambda$, the selection part is $-\frac{1}{\alpha} \text{Var}_{S_t}(\lambda)$ and the switch part is $\int_{\Phi} (\lambda(\phi_t^*) - \lambda) \Phi_t^{\text{sw}}(d\phi)$,

which is ≤ 0 on the disaster-free-since-adoption part of the switch set—there the scheduled relief of Proposition 7 gives $\lambda(\phi_t^*) \leq \lambda$ (and $\Phi_t^{\text{sw}} \geq 0$)—and non-positive on the rest under the monotone-reselection condition, so

$$\frac{d\bar{\lambda}_t^S}{dt} = -\frac{1}{\alpha} \text{Var}_{S_t}(\lambda) + \int_{\Phi} (\lambda(\phi_t^*) - \lambda) \Phi_t^{\text{sw}}(d\phi), \quad (\text{E38})$$

the selection term non-positive ($\text{Var}_{S_t}(\lambda) \geq 0$, $\alpha > 0$) and the switch term non-positive under that condition; this is (26). Equation (E38) is the absolutely continuous (dt-rate) part of the calm derivative; where a positive G -mass synchronizes at a calm deadline, Φ_t^{sw} carries a predictable atom and $\bar{\lambda}_t^S$ a predictable jump, which by Lemma 6 moves no traded price.

(ii) *Disaster reweight.* At a disaster of size x , before triggered switches, the cross-section reweights by $\tilde{S}_t^x(d\phi) = r(\phi, x)^{1/\alpha} g_t(x)^{-1} S_{t-}(d\phi)$ (23), with $g_t(x) = \mathbb{E}_{S_{t-}}[r(\cdot, x)^{1/\alpha}]$ (15). Hence

$$\begin{aligned} \bar{\lambda}_t^{\tilde{S}} &= \int_{\Phi} \lambda \tilde{S}_t^x(d\phi) \\ &= \int_{\Phi} \lambda r(\phi, x)^{1/\alpha} g_t(x)^{-1} S_{t-}(d\phi) \end{aligned} \quad (\text{E39})$$

$$= \frac{\mathbb{E}_{S_{t-}}[\lambda r(\cdot, x)^{1/\alpha}]}{\mathbb{E}_{S_{t-}}[r(\cdot, x)^{1/\alpha}]}, \quad (\text{E40})$$

(E39) substituting the reweight and (E40) pulling out the constant $g_t(x)^{-1}$ and writing numerator and denominator ($= g_t(x)$) as expectations under S_{t-} . Subtract $\bar{\lambda}_{t-}^S = \mathbb{E}_{S_{t-}}[\lambda]$:

$$\begin{aligned} \bar{\lambda}_t^{\tilde{S}} - \bar{\lambda}_{t-}^S &= \frac{\mathbb{E}_{S_{t-}}[\lambda r(\cdot, x)^{1/\alpha}]}{\mathbb{E}_{S_{t-}}[r(\cdot, x)^{1/\alpha}]} - \mathbb{E}_{S_{t-}}[\lambda] \\ &= \frac{\mathbb{E}_{S_{t-}}[\lambda r(\cdot, x)^{1/\alpha}] - \mathbb{E}_{S_{t-}}[\lambda] \mathbb{E}_{S_{t-}}[r(\cdot, x)^{1/\alpha}]}{\mathbb{E}_{S_{t-}}[r(\cdot, x)^{1/\alpha}]} \end{aligned} \quad (\text{E41})$$

$$= \frac{\text{Cov}_{S_{t-}}(\lambda, r(\cdot, x)^{1/\alpha})}{\mathbb{E}_{S_{t-}}[r(\cdot, x)^{1/\alpha}]}, \quad (\text{E42})$$

with $\bar{\lambda}_t^{\tilde{S}}$ the post-reweight mean; the final $\bar{\lambda}_t^S$ adds the triggered switchers' relocation term.

(E41) over the common denominator and (E42) the covariance definition; this is (27).

To sign the numerator, expand the covariance with an independent copy. Let (λ, ϕ) and (λ', ϕ') be two independent draws from S_{t-} and write $k := r(\phi, x)^{1/\alpha}$, $k' := r(\phi', x)^{1/\alpha}$; all expectations below are under S_{t-} . Then

$$\mathbb{E}[(\lambda - \lambda')(k - k')] = \mathbb{E}[\lambda k] - \mathbb{E}[\lambda k'] - \mathbb{E}[\lambda' k] + \mathbb{E}[\lambda' k'] \quad (\text{E43})$$

$$= \mathbb{E}[\lambda k] - \mathbb{E}[\lambda] \mathbb{E}[k'] - \mathbb{E}[\lambda'] \mathbb{E}[k] + \mathbb{E}[\lambda' k'] \quad (\text{E44})$$

$$= 2 \mathbb{E}[\lambda k] - 2 \mathbb{E}[\lambda] \mathbb{E}[k] \quad (\text{E45})$$

$$= 2 \text{Cov}_{S_{t-}}(\lambda, k), \quad (\text{E46})$$

(E43) expanding the product inside the expectation, (E44) factoring the cross terms $\mathbb{E}[\lambda k'] = \mathbb{E}[\lambda] \mathbb{E}[k']$ and $\mathbb{E}[\lambda' k] = \mathbb{E}[\lambda'] \mathbb{E}[k]$ by independence of the two draws, (E45) using that the draws are identically distributed ($\mathbb{E}[\lambda' k'] = \mathbb{E}[\lambda k]$, $\mathbb{E}[\lambda'] = \mathbb{E}[\lambda]$, $\mathbb{E}[k'] = \mathbb{E}[k]$), and (E46) the covariance definition. Dividing (E46) by 2 and writing $k = r(\cdot, x)^{1/\alpha}$,

$$\text{Cov}_{S_{t-}}(\lambda, r(\cdot, x)^{1/\alpha}) = \frac{1}{2} \mathbb{E}[(\lambda - \lambda')(r(\phi, x)^{1/\alpha} - r(\phi', x)^{1/\alpha})]. \quad (\text{E47})$$

In the intensity channel $r(\phi, x) = \lambda/\lambda^*$, so $r(\phi, x)^{1/\alpha} = (\lambda/\lambda^*)^{1/\alpha}$ is non-decreasing in λ ; the two factors in (E47) then carry the common sign of $\lambda - \lambda'$, their product is ≥ 0 pointwise, and the expectation is ≥ 0 , so $\text{Cov}_{S_{t-}}(\lambda, r(\cdot, x)^{1/\alpha}) \geq 0$, strict iff $\text{Var}_{S_{t-}}(\lambda) > 0$. Thus the reweight raises $\bar{\lambda}^S$. For the relocation mode: by the proof of Lemma 4 the intensity log-posterior is $\ell_t^\Lambda(\lambda) = \log \rho_0^\Lambda(\lambda) - \lambda t + N_t \log \lambda$, and at a disaster the count N_t rises by one at fixed t . We show the intensity mode $\lambda_t^m = \min \arg \max_\lambda \ell_t^\Lambda$ is non-decreasing in N_t directly, by revealed preference. Write $\ell^{(N)}(\lambda) := \log \rho_0^\Lambda(\lambda) - \lambda t + N \log \lambda$ and let λ_N, λ_{N+1} be the corresponding min-maximisers. Optimality at each count gives

$$\ell^{(N)}(\lambda_N) \geq \ell^{(N)}(\lambda_{N+1}), \quad \ell^{(N+1)}(\lambda_{N+1}) \geq \ell^{(N+1)}(\lambda_N).$$

Adding the two inequalities and regrouping,

$$[\ell^{(N+1)}(\lambda_{N+1}) - \ell^{(N)}(\lambda_{N+1})] - [\ell^{(N+1)}(\lambda_N) - \ell^{(N)}(\lambda_N)] \geq 0. \quad (\text{E48})$$

Since $\ell^{(N+1)}(\lambda) - \ell^{(N)}(\lambda) = \log \lambda$ for every λ , the left-hand side of (E48) equals $\log \lambda_{N+1} - \log \lambda_N$; hence $\log \lambda_{N+1} \geq \log \lambda_N$, and since $\log$ is strictly increasing, $\lambda_{N+1} \geq \lambda_N$. Thus the intensity mode of ϕ_t^* is non-decreasing in N_t : a disaster raises N_t by one, so the common destination ϕ_t^* weakly rises. This is a statement about the destination across counts, not about the switchers' starting point: the triggered revisers relocate *from* their held intensities *to* $\lambda(\phi_t^*)$, so the relocation adds to the reweight (E42) exactly when $\lambda(\phi_t^*)$ exceeds the switchers' mean held intensity under their post-reweight shares (E47)—the complacent-switcher condition, which is a condition on the post-reweight triggered-switcher mean; a complacent newborn law Π_ϕ makes it plausible in the stationary regime but does not imply it, and we do not derive it from primitives. The reweight (E47) raises $\bar{\lambda}^S$ unconditionally, so the fast-in does not rely on this amplifier.

(iii) *Fear factor.* By the bridge relation (19), $d \log U_t = d \log \kappa_t$, and $U_t = \lambda^* \mathbb{E}_f[e^{\alpha J} g_t(J)^\alpha]$ (17). In calm the held models are fixed and $g_t(x) = \langle r(\cdot, x)^{1/\alpha}, S_t \rangle$, so (E37) with $h = r(\cdot, x)^{1/\alpha}$ gives

$$\frac{dg_t(x)}{dt} = -\frac{1}{\alpha} \text{Cov}_{S_t}(\lambda, r(\cdot, x)^{1/\alpha}). \quad (\text{E49})$$

This is the selection component—the derivative at held models fixed; at active deadline dates the calm switch flow adds the relocation term the theorem's c_{sw} collects, and (i)'s displayed drift is the atomless-flow component.

Differentiate $U_t = \lambda^* \int_{\mathcal{J}} e^{\alpha x} g_t(x)^\alpha f(x) dx$ in t . The t -derivative of the integrand is $\alpha e^{\alpha x} g_t(x)^{\alpha-1} dg_t(x)/dt$, bounded uniformly on the compact $\mathcal{J}$: $g_t(x) = \int_0^1 s_t^i r_i(x)^{1/\alpha} di \in [\underline{r}^{1/\alpha}, \bar{r}^{1/\alpha}]$ with $0 < \underline{r} = \min_{\mathcal{J} \times \Phi} r \leq \bar{r} = \max_{\mathcal{J} \times \Phi} r < \infty$ (continuity and positivity of the densities on the compact $\mathcal{J} \times \Phi$, Assumption 4), and dg_t/dt is bounded by (E49); so dominated convergence

justifies differentiation under the integral, and

$$\begin{aligned}\frac{dU_t}{dt} &= \lambda^* \int_{\mathcal{J}} e^{\alpha x} \alpha g_t(x)^{\alpha-1} \frac{dg_t(x)}{dt} f(x) dx \\ &= \lambda^* \int_{\mathcal{J}} e^{\alpha x} \alpha g_t(x)^{\alpha-1} \left(-\frac{1}{\alpha} \text{Cov}_{S_t}(\lambda, r(\cdot, x)^{1/\alpha}) \right) f(x) dx\end{aligned}\tag{E50}$$

$$= -\lambda^* \int_{\mathcal{J}} e^{\alpha x} g_t(x)^{\alpha-1} \text{Cov}_{S_t}(\lambda, r(\cdot, x)^{1/\alpha}) f(x) dx,\tag{E51}$$

(E50) substituting (E49) and (E51) cancelling $\alpha \cdot \frac{1}{\alpha}$. Divide by $U_t = \lambda^* \mathbb{E}_f[e^{\alpha J} g_t^\alpha]$ (chain rule $d \log U_t / dt = (dU_t / dt) / U_t$):

$$\frac{d \log U_t}{dt} = - \frac{\mathbb{E}_f[e^{\alpha J} g_t(J)^{\alpha-1} \text{Cov}_{S_t}(\lambda, r(\cdot, J)^{1/\alpha})]}{\mathbb{E}_f[e^{\alpha J} g_t(J)^\alpha]},$$

which is (28). In the intensity channel $\text{Cov}_{S_t}(\lambda, r(\cdot, x)^{1/\alpha}) \geq 0$ for every x (E47), and $e^{\alpha J}, g_t^{\alpha-1}, g_t^\alpha > 0$, so both expectations are ≥ 0 and the ratio is ≥ 0 ; hence $d \log U_t / dt \leq 0$, strict iff $\text{Var}_{S_t}(\lambda) > 0$.

At a disaster of realised size x_0 , the reweight $S_t = r(\cdot, x_0)^{1/\alpha} g_{t-}(x_0)^{-1} S_{t-}$ moves g to

$$\begin{aligned}g_t(x) &= \mathbb{E}_{S_t}[r(\cdot, x)^{1/\alpha}] \\ &= \frac{\mathbb{E}_{S_{t-}}[r(\cdot, x)^{1/\alpha} r(\cdot, x_0)^{1/\alpha}]}{\mathbb{E}_{S_{t-}}[r(\cdot, x_0)^{1/\alpha}]},\end{aligned}\tag{E52}$$

(E52) substituting the reweight into $\mathbb{E}_{S_t}$. In the intensity channel this equals $\mathbb{E}_{S_t}[(\lambda/\lambda^*)^{1/\alpha}]$, which rose: apply the association argument of (E47) with $Z := (\lambda/\lambda^*)^{1/\alpha}$ in place of λ —both factors are then non-decreasing in λ , so the product inside (E47) is pointwise ≥ 0 and $\text{Cov}_{S_{t-}}(Z, Z) = \text{Var}_{S_{t-}}(Z) \geq 0$ —so the post-reweight mean satisfies $\mathbb{E}_{S_{t-}}[Z^2] / \mathbb{E}_{S_{t-}}[Z] \geq \mathbb{E}_{S_{t-}}[Z]$, strict iff $\text{Var}_{S_{t-}}(Z) > 0$. The reweighted aggregator therefore rises for every x , and since the integrand $e^{\alpha x} g^\alpha$ of (17) is increasing in g , the reweight-only fear factor $U(\tilde{S}_t^{x_0})$ exceeds U_{t-} —the unconditional statement. The final post-disaster U_t adds the triggered relocation, which in the intensity channel changes the aggregator by $\hat{s}_t^{\text{sw}}(Z(\phi_t^*) - \hat{\mathbb{E}}_t^{\text{sw}}[Z])$, with $\hat{s}_t^{\text{sw}}$ and $\hat{\mathbb{E}}_t^{\text{sw}}$ the post-reweight triggered-switcher mass and mean; the exact sign condition for the reloca-

tion is therefore $Z(\phi_t^*) \geq \widehat{\mathbb{E}}_t^{\text{sw}}[Z]$, the likelihood-ratio form of the complacent-switcher condition. For $\alpha \geq 1$ the condition of (ii) implies it:

$$Z(\phi_t^*) = \left(\frac{\lambda(\phi_t^*)}{\lambda^*} \right)^{1/\alpha} \geq \left(\frac{\widehat{\mathbb{E}}_t^{\text{sw}}[\lambda]}{\lambda^*} \right)^{1/\alpha} \geq \widehat{\mathbb{E}}_t^{\text{sw}}[(\lambda/\lambda^*)^{1/\alpha}],$$

the first inequality by the condition and monotonicity of $z \mapsto z^{1/\alpha}$, the second by Jensen, $z \mapsto z^{1/\alpha}$ concave for $\alpha \geq 1$. Hence the relocation weakly raises the aggregator and with it U_t , so $U_t \geq U(\tilde{S}_t^{x_0}) > U_{t-}$: the jump holds under the complacent-switcher condition for $\alpha \geq 1$ (under its likelihood-ratio form for all α), sufficiency not equivalence, and the reweight component holds unconditionally. The disaster premium loads on U_t (Proposition 14) and inherits the direction; its level also involves ψ_t , hence the valuation operator v (Proposition 13).

(iv) *Staggered recovery.* The drifts of (i),(iii) hold on any disaster- and switch-free interval. By Proposition 7 a calm spell ends at the deadline $\Delta^*(\phi, \varepsilon)$ (B4), where the reviser re-selects the mode prevailing at her deadline; by Lemma 4 the mode is non-increasing through a disaster-free spell, so her held intensity weakly falls—a downward step in $\bar{\lambda}^S$ and, in the intensity channel, in U_t . By Proposition 8 the deadlines are weakly ordered by ε , so the steps are staggered, occurring at distinct times where the deadline map is non-constant; and by Proposition 7 Δ^* is finite and increases as $\varepsilon \rightarrow 0$ or the held intensity $\lambda(\phi) \rightarrow 0$ (the dependence on λ^* is indirect, through which held models survive). The jump of (ii) is instantaneous (fast in); the drift of (i),(iii) and the staggered steps act over the deadline horizon (slow out). $\square$

Proof of Theorem 1. (i) By Proposition 2 and Proposition 15, $F_t(0) = \mathbb{E}[U_t \mid \mathcal{F}_t] = U_t = U^{\text{RA}} \kappa_t$ is a functional of S_t .

(ii) *The slope is the generator applied to U .* The fear factor $U_t = \lambda^* \mathbb{E}_f[e^{\alpha J} g_t^\alpha]$ is a bounded functional of the cross-section ($g_t \in [\underline{r}^{1/\alpha}, \bar{r}^{1/\alpha}]$ on the compact $\mathcal{J} \times \Phi$), hence a bounded functional $U \circ \pi_\phi$ of the record-resolved state Ψ_t , which evolves as the piecewise-deterministic Markov process of Proposition 3—deterministic calm flow, disaster jumps at rate

λ^* . By the extended-generator (Dynkin) identity for piecewise-deterministic Markov processes (Davis, 1984; Jacobsen, 2006), with $U \circ \pi_\phi$ in the domain (bounded, absolutely continuous along the calm flow),

$$\partial_s^+ F_t(0) = \frac{d}{ds} \mathbb{E}[U_{t+s} \mid \mathcal{F}_t] \Big|_{s=0+} = (\mathcal{A}_\Psi U)(\Psi_t) = \frac{dU_t}{dt} \Big|_{\text{calm}} + \lambda^* \int_{\mathcal{J}} (U(\Psi_t^x) - U(\Psi_{t-})) f(x) dx, \quad (\text{E53})$$

the generator splitting into the calm drift and the disaster compensator. The identity is evaluated at a predictable t (a.s. every fixed date: disasters are totally inaccessible), where $\Psi_{t-} = \Psi_t$; at a jump date the right slope is generated from the post-jump state.

Calm drift. Differentiate $U_t = \lambda^* \int_{\mathcal{J}} e^{\alpha x} g_t(x)^\alpha f(x) dx$; the integrand's t -derivative $\alpha e^{\alpha x} g_t(x)^{\alpha-1} dg_t(x)$ is bounded uniformly on the compact $\mathcal{J}$ (the aggregator $g_t(x)$ lies between the menu's extreme size ratios, uniformly in t), so dominated convergence justifies differentiating under the integral,

$$\frac{dU_t}{dt} \Big|_{\text{calm}} = \alpha \lambda^* \int_{\mathcal{J}} e^{\alpha x} g_t(x)^{\alpha-1} \frac{dg_t(x)}{dt} \Big|_{\text{calm}} f(x) dx. \quad (\text{E54})$$

The calm motion of $g_t(x) = \langle r(\cdot, x)^{1/\alpha}, S_t \rangle$ has two parts—the selection drift (21) and the calm switch flow (22) of Proposition 3, at $h = r(\cdot, x)^{1/\alpha}$:

$$\frac{dg_t(x)}{dt} \Big|_{\text{sel}} = -\frac{1}{\alpha} \text{Cov}_{S_t}(\lambda, r(\cdot, x)^{1/\alpha}), \quad \frac{dg_t(x)}{dt} \Big|_{\text{sw}} = \int_{\Phi} (r(\phi_t^*, x)^{1/\alpha} - r(\phi, x)^{1/\alpha}) \Phi_t^{\text{sw}}(d\phi), \quad (\text{E55})$$

the first by (E49), the second because a switcher relocating $\phi \rightarrow \phi_t^*$ changes her contribution to g from $r(\phi, x)^{1/\alpha}$ to $r(\phi_t^*, x)^{1/\alpha}$, at the wealth-weighted deadline rate Φ_t^{sw} . Substituting each

part of (E55) into (E54),

$$\begin{aligned}
\left. \frac{dU_t}{dt} \right|_{\text{sel}} &= \alpha \lambda^* \int_{\mathcal{J}} e^{\alpha x} g_t^{\alpha-1} \left(-\frac{1}{\alpha} \text{Cov}_{S_t}(\lambda, r(\cdot, x)^{1/\alpha}) \right) f dx = -\lambda^* \mathbb{E}_f[e^{\alpha J} g_t^{\alpha-1} \text{Cov}_{S_t}(\lambda, r(\cdot, J)^{1/\alpha})] \\
&= -U_t b_{\text{sel}}(S_t), \\
\left. \frac{dU_t}{dt} \right|_{\text{sw}} &= \alpha \lambda^* \int_{\mathcal{J}} e^{\alpha x} g_t^{\alpha-1} \left(\int_{\Phi} (r(\phi_t^*, x)^{1/\alpha} - r(\phi, x)^{1/\alpha}) \Phi_t^{\text{sw}}(d\phi) \right) f dx \\
&= U_t c_{\text{sw}}(\Phi_t^{\text{sw}}),
\end{aligned} \tag{E56}$$

with $b_{\text{sel}}(S_t) = \mathbb{E}_f[e^{\alpha J} g_t^{\alpha-1} \text{Cov}_{S_t}(\lambda, r(\cdot, J)^{1/\alpha})] / \mathbb{E}_f[e^{\alpha J} g_t^{\alpha}] \geq 0$ and c_{sw} the ratio of the statement. *The factor α cancels in the selection term* ($\alpha \cdot \frac{1}{\alpha} = 1$, the share drift (21) carrying the $\frac{1}{\alpha}$), recovering the calm log-drift (28); *it does not cancel in the switch term*, where the deadline flux (22) carries no $\frac{1}{\alpha}$ —so c_{sw} retains the α , the asymmetry between the two calm channels. Under monotone re-selection the switch term is signed, $c_{\text{sw}} \leq 0$: on the *disaster-free* switch set scheduled relief gives $r(\phi_t^*, \cdot) \leq r(\phi, \cdot)$ (Proposition 7); switchers whose records carry non-rejecting disasters need not satisfy it—exactly the condition’s content, and the theorem leaves c_{sw} unsigned.

Disaster compensator. The post-disaster state Ψ_t^x applies the reweight (23) and then relocates the triggered investors (D21), so the jump in U splits as

$$\lambda^* \int_{\mathcal{J}} (U(\Psi_t^x) - U(\Psi_{t-})) f dx = \underbrace{\lambda^* \Theta_{\text{rw}}(S_t)}_{\text{reweight}} + \underbrace{\lambda^* \Theta_{\text{sw}}(\Psi_t)}_{\text{relocation}}, \tag{E57}$$

$\Theta_{\text{rw}}(S_t)$ the change in $U = \lambda^* \mathbb{E}_f[e^{\alpha J} g^{\alpha}]$ produced by the size- x reweight of g_{t-} (27)—a functional of S_{t-} , since the reweight $r(\cdot, x)^{1/\alpha} / g_{t-}(x)$ and the reweighted aggregator are functionals of S_{t-} ; and $\Theta_{\text{sw}}(\Psi_t)$ the additional change from relocating to ϕ_t^* those investors whose updated record leaves the typical set, at the record-dependent rate $\lambda^* \pi(\phi, \tau, \mathcal{R}; \varepsilon)$ (D21)—a functional of Ψ_t , not of S_t , signed under the complacent-switcher condition (for $\alpha \geq 1$, or its likelihood-ratio form)— $\lambda(\phi_t^*)$ above the switchers’ post-reweight mean—which the destination’s count-monotonicity

(Proposition 6(ii)) supports but does not imply; the theorem leaves Θ_{sw} unsigned.

Collecting. Substituting (E56)–(E57) into (E53),

$$\partial_s^+ F_t(0) = \underbrace{-U_t b_{\text{sel}}(S_t) + \lambda^* \Theta_{\text{rw}}(S_t)}_{b_S(S_t)} + U_t c_{\text{sw}}(\Phi_t^{\text{sw}}) + \lambda^* \Theta_{\text{sw}}(\Psi_t), \quad (\text{E58})$$

the first group a functional of S_t through g_t (Proposition 15); the calm flow Φ_t^{sw} and the trigger propensity π are functionals of Ψ_t and not of S_t , since the deadline-hitting flow and the triggered set depend on the joint law of held models, ages, and records, which the held-model marginal does not fix. This is the decomposition of (ii).

(iii) For Ψ_t, Ψ'_t with $\pi_\phi \Psi_t = \pi_\phi \Psi'_t = S_t$: the levels coincide by (i), $F_t(0) = F'_t(0) = U_t$, and by (E58) the $b_S(S_t)$ part of the slope coincides, leaving the two staggered-revision channels $U_t c_{\text{sw}} + \lambda^* \Theta_{\text{sw}}$. Conditioning on no disaster over the horizon suppresses Θ_{sw} and leaves the calm channel $U_t c_{\text{sw}}$; under monotone re-selection $c_{\text{sw}} \leq 0$, so the state with the larger imminent deadline flow Φ_t^{sw} carries the weakly steeper decline, $\partial_s^+ F_t^{0'}(0) \leq \partial_s^+ F_t^0(0)$; under that condition and the complacent-switcher condition, as qualified above, the two channels are oppositely signed. The wedge $F_t(s) - F'_t(s)$ is zero at $s = 0$ and first order in s ; its full-horizon law is Markov in Ψ_t , not autonomous in S_t (Proposition 13).

(iv) Under heterogeneous Bayesian learning each posterior is a sufficient statistic for its own future, so the wealth-weighted cross-section is Markov in itself; there is no first-passage deadline and no rejection trigger, hence neither Φ_t^{sw} nor Θ_{sw} ; the Bayesian generator carries its own posterior-drift terms in place of the lazy calm structure, so the slope is *some* S_t -measurable functional—the benchmark’s own, which Internet Appendix IA.9 computes—not the lazy formula’s $b_S(S_t)$. The forward curve is then pinned by the current cross-section at every horizon (Internet Appendix IA.9); an affine latent-state model and a representative filtering economy give the same conclusion through their own Markov states. $\square$

Proof of Corollary 1. Couple the two economies as in Theorem 1(iii), on a common driving

path with common public posterior. Write $D := e^{-\int_t^{t+s_1} r_u du}$; to first order in $s_2 - s_1$ the calm-conditional tranche is the window's expected discounted count at the prevailing risk-neutral intensity, $P_t^0 = \mathbb{E}_t^{\mathbb{Q},0}[D U_{t+s_1}](s_2 - s_1) + o(s_2 - s_1)$ (Remark 2; discounting inside the window is second order). Hence

$$P_t^0 - P_t'^0 = (s_2 - s_1) \left\{ \mathbb{E}_t^{\mathbb{Q},0}[D (U_{t+s_1} - U'_{t+s_1})] + \mathbb{E}_t^{\mathbb{Q},0}[(D - D') U'_{t+s_1}] \right\} + o(s_2 - s_1).$$

The fear-path term: conditional on no disaster the wedge $U - U'$ is generated by the calm channel—Theorem 1(iii) under the S_{t-} -measurable tilt of Remark 2, which moves only the S -loadings—so it is measurable in Ψ_t and vanishes under a Markov-in- S_t benchmark, whose calm U -path is an autonomous S -functional shared by the pair. The discount-path term: $D - D' = -D \int_t^{t+s_1} (r_u - r'_u) du (1 + o(1))$, and $r_u - r'_u = r(S_u) - r(S'_u)$ with r a bounded-Lipschitz functional of S in total variation ((E1) is affine in $\bar{\lambda}^S$ and U , both bounded Lipschitz on the compact menu), while along the calm coupling $\|S_u - S'_u\|_{\text{TV}}$ is generated by the differing switch flows; so this term is likewise measurable in Ψ_t , and under the Markov benchmark the shared S -path forces $r_u \equiv r'_u$ and it vanishes. For the unconditional tranche, decompose on the first disaster in $(t, t + s_1]$: its compensator adds the disaster-triggered relocation channel through the post-jump state, entering the difference at first order through Θ_{sw} . $\square$

Appendix F. Demographic turnover and the overlapping-generations economy

Assumption 5 (Demographic turnover: perpetual youth): *We extend Assumption 1 with overlapping generations in the manner of Yaari (1965) and Blanchard (1985), as used for heterogeneous agents by Gârleanu and Panageas (2015):*

- (i) *each living investor faces a constant death hazard $\Omega > 0$ —an independent Poisson clock,*

the same for all and independent of the aggregate path;

- (ii) newborns arrive at mass Ω per unit time (population mass constant), each drawn independently with type $(\varepsilon, \phi_0) \sim \Pi$, a fixed law on $(0, 1) \times \Phi$ with ϕ_0 -marginal Π_ϕ —the surprise threshold $\varepsilon \in (0, 1)$ as in Assumption 2, so the typical-set test of Definition 1 is well defined for every newborn type;
- (iii) (allocation of cohorts) the symmetric allocation of Assumption 3(iv) applies to every cohort, so the effective cohort multiplier is common (no preference- or endowment-dispersion is priced) and each newborn enters with no since-birth record ($M = 1$ at birth). Demographic turnover is then a primitive share-replacement rule: the newborn cohort carries aggregate consumption share Ωdt , distributed over held models by the type law Π_ϕ . If formally decentralized (Appendix F), each entering cohort receives the born-at-par Negishi weight $\tilde{\psi}_i^{-1/\alpha} = \Xi_{b_i}$ —the entrant’s weight equals the living population’s current average—which is exactly what makes its consumption share Ωdt and the replacement law (F1) exact. We take the share-replacement law as the primitive; no result below requires its decentralization, which Appendix F supplies only to exhibit the supporting weights. The budget behind them uses two separate instruments: fair annuities insure mortality among the living—estates fund the survivor credit Ω (Yaari, 1965)—while entry is financed by diluting incumbents’ tree claims at rate Ω , each entrant receiving new shares worth the per-capita claim; the two Ω -flows offset in every survivor’s return, so the Euler equation and the short rate are untouched. Explicitly, a survivor holding the (diluted) tree earns $\frac{C_t}{P_t} dt + \frac{dP_t}{P_t} - \Omega dt + \Omega dt$ —dividend yield and capital gain, less the dilution drag, plus the fair annuity credit—so Ω cancels from her return; on the goods side the dying cohorts’ consumption share is Ωdt (death is wealth-proportional across the living), exactly the entrants’ assigned share, so the resource constraint is undisturbed. Death is actuarially fairly annuitized (Yaari, 1965), so Ω does not enter any survivor’s effective discount; the turnover acts on the share simplex and is aggregate-resource-neutral by construction, $\langle 1, S_t \rangle \equiv 1$.

(iv) (newborn information). A newborn inherits the common second-order posterior ρ_t prevailing at her birth date—the public object of Lemma 4, a deterministic functional of the aggregate record, shared by every living investor—and enters holding her drawn model ϕ_0 with a reset test state: regime age $\tau = 0$ and empty since-adoption record ($n = 0$). Inheriting ρ_t keeps the re-selection destination $\mathbf{m}(\rho_t)$ common across cohorts (Lemma 4, Proposition 8); resetting the held model and test clock is what refreshes the cross-section, since young cohorts hold ϕ_0 and have not yet revised. The birth holding is a boundary condition distinct from the post-rejection adoption rule of Definition 1: at birth the held model is drawn from Π_ϕ , not set to the posterior mode $\mathbf{m}(\rho_t)$, and the MAP rule governs only subsequent rejections. This birth-specific endowment—not unconcentrated newborn posteriors—is what sustains the stationary dispersion. Disagreement is thus carried by the held models, not by posterior dispersion. Economically, ρ_t is a public statistic—a deterministic function of the public record, on the same footing as an equilibrium price—so this clause is the standard common-information convention: every investor, newborn or incumbent, conditions on the same public posterior, while the only datum that resets at birth is the agent’s private adoption clock. The primitives of Assumptions 1–4 do not by themselves fix a newborn’s posterior; this selects the common-information reading, a specification choice within the existing information structure rather than a new economic primitive.

We now specialize to a finite menu $\Phi = \{\phi_1, \dots, \phi_K\}$ (the setting of Corollary 3), an additional restriction for the stationary and quantitative exercise—in general Φ is a compact continuum (Assumption 4), not a finite set.

Antecedent and difference. Perpetual youth is Blanchard’s (1985) continuous-time overlapping-generations device, built on Yaari (1965); Gârleanu and Panageas (2015) use it precisely to keep heterogeneity from washing out and to obtain a nondegenerate stationary equilibrium, with the dead’s wealth redistributed to the living. We inherit that structure and its purpose,

and differ in the dimension of heterogeneity (the surprise threshold ε and the initial held model ϕ_0 , not risk aversion) and the environment (CRRA with disaster jumps and lazy revision, not recursive-preference diffusion). The convention in (iii) is the [Gârleanu and Panageas](#) reduced form, adopted to avoid the annuity bookkeeping of the full Yaari (1965) market; it makes the demographic effect on the cross-section a single replacement term.

Under Assumption 5(iii) the law of motion of the pricing state (Proposition 3) gains exactly one term. Over $[t, t + dt]$ the population mass is held at one by balanced birth and death: a fraction Ωdt dies (proportionally in wealth, so the redistribution to survivors leaves their relative shares—hence the normalized measure—unchanged), while newborn mass Ωdt enters with type law Π_ϕ . The post-turnover normalized cross-section is therefore the mixture of the surviving S_t (mass $1 - \Omega dt$) and the newborn Π_ϕ (mass Ωdt):

$$S_{t+dt}|_{\text{demog}} = (1 - \Omega dt) S_t + \Omega dt \Pi_\phi, \quad \text{hence} \quad dS_t|_{\text{demog}} = S_{t+dt}|_{\text{demog}} - S_t = \Omega (\Pi_\phi - S_t) dt. \quad (\text{F1})$$

Pairing the measure identity (F1) with any bounded h ,

$$\frac{d}{dt} \langle h, S_t \rangle \Big|_{\text{demog}} = \Omega (\langle h, \Pi_\phi \rangle - \langle h, S_t \rangle), \quad (\text{F2})$$

added to the selection, switch, and disaster channels of (21)–(23): a mean-reversion of the cross-section toward the newborn law Π_ϕ at rate Ω .

The replacement is resource-neutral by construction: it operates in consumption-share space, where the inflow $\Omega \Pi_\phi dt$ and outflow $\Omega S_t dt$ of (F1) preserve $\int_{\text{living}} s_t^i di = 1$, so goods continue to clear, $\int_{\text{living}} c_t^i di = C_t$, with no second claim on the Lucas tree. The share-endowment birth rule of Assumption 5(iii) is the [Gârleanu and Panageas](#) (2015) reduced form for a Pareto-weight (human-capital) convention—a newborn receives a consumption share, not an additional financial claim funded out of the dead’s annuitized wealth—which is what makes the demographic effect a single share-replacement term rather than a wealth-accounting transfer between the

dead, the survivors, and the newborns. Formally the demographic turnover is a primitive sharing rule on the simplex; if decentralized, it is the Negishi construction of Appendix F with a *state-dependent* cohort weight ζ_t that implements the Ωdt share assignment, priced under fair annuities, so that the death hazard cancels from every survivor's Euler equation (IA.1.1) and enters the equilibrium only through the composition of the living. The share law (F1) is in turn invariant to the death-share bookkeeping—whether the dead's wealth-proportional share is annuitized to survivors and the block then diluted by the newborn inflow, or the dead's share is replaced directly by newborns, both give $(1 - \Omega dt)S_t + \Omega dt \Pi_\phi$ to first order in dt —which is what licenses writing the demographic effect as the single replacement term (F2) without committing to either ledger.

Proposition 16 (The kernel under demographic turnover): *Under Assumptions 1–4 and 5, an overlapping-generations equilibrium (Appendix F) exists, and its state-price density retains the form of Proposition 1,*

$$\xi_t = e^{-\delta t} C_t^{-\alpha} \Xi_t^\alpha, \quad \Xi_t = \int_{\text{living}} \tilde{\psi}_i^{-1/\alpha} (M_t^i)^{1/\alpha} di, \quad (\text{F3})$$

now the $\frac{1}{\alpha}$ -power mean of beliefs over the living population, with born-at-par cohort weights $\tilde{\psi}_i^{-1/\alpha} = \Xi_{b_i}$. Under par entry the inflow and outflow of weight offset exactly, so Ξ_t carries no demographic term: it solves the selection–jump dynamics of Proposition 10 applied to the living cross-section—calm drift $-\frac{1}{\alpha}(\bar{\lambda}_t^S - \lambda^*)$, disaster factor $g_t(x)$ —and (E1) holds verbatim in the turnover economy. The death hazard Ω does not enter the time preference of the kernel—the fair annuity of Assumption 5(iii) offsets mortality in each survivor's Euler equation—and enters the equilibrium only through the composition of the living, i.e. through the demographic channel (F2) of the cross-section. The bridge relation (Proposition 2) and the selection drift (21)—instantaneous relations of the kernel to the cross-section—hold verbatim in the turnover economy; the cross-section dynamics are those of Proposition 3 augmented by (F2), so the calm erosion of Proposition 6(i) acquires a mean-reversion of $\bar{\lambda}_t^S$ toward the newborn mean at rate

Ω , while the disaster jump and staggered recovery are unchanged in form.

Three readings of the proposition. Demographic turnover is a primitive share-replacement law on the simplex (Assumption 5(iii), eq. (F2)): newborns enter carrying an aggregate consumption share Ωdt drawn from Π_ϕ , the dead's wealth-proportional share is released, and $\langle 1, S_t \rangle \equiv 1$ is preserved. The born-at-par weight is what reconciles the two requirements: a *common* underlying weight would give the entering cohort the share $\zeta^{-1/\alpha}/\Xi_b$, fluctuating with the aggregator and growing with the birth date, incompatible with the constant- Ωdt inflow the demographic law requires; entry at par delivers the Ωdt share identically. Only the ratios and the drift of Ξ_t enter prices, so the explicit $e^{-\delta t}$ in (F3) is untouched and the riskless-rate *level* keeps its $+\delta$ term. The *growth* of Ξ_t is stationary under the stationary cross-section; its level is a multiplicative selection functional and need not be, and no priced object reads it.¹⁰ The entrant thus arrives at the living population's current average weight—at par—and the aggregator's composition turns over while its dynamics do not.

Finite lives change *who* sits in the power mean. With fair annuities the mortality risk is fully insured, so each survivor optimises as if infinitely-lived at rate δ (the annuity offsets the death hazard Ω in the Euler equation); clearing on the exogenous Lucas endowment returns the kernel $\xi_t = e^{-\delta t} C_t^{-\alpha} \Xi_t^\alpha$ over the living cross-section, which turnover keeps non-degenerate. Under par entry the aggregate Ξ_t carries no demographic term: the entering weight $\Omega \Xi_t dt$ and the dying weight $\Omega \Xi_t dt$ offset exactly, so Ξ_t evolves by the within-life terms alone—the calm drift $d\Xi_t/\Xi_t = -\frac{1}{\alpha}(\bar{\lambda}_t^S - \lambda^*) dt$ from each cohort's $d(M^i)^{1/\alpha}$, and the disaster factor $g_t(x)$ —Proposition 10 applied to the living cross-section. Hence $\Xi_t > 0$ and finite a.s. (bounded drift, finitely many bounded jump factors on compacts), its *growth* is a stationary functional of the cross-section, and its level is a multiplicative selection functional that no priced object reads: every pricing formula uses Ξ -ratios and the drift alone. The whole apparatus of Sections 3–5.C, including the finite-menu fear factor (D17), therefore transfers to the stationary economy

¹⁰Par entry precludes the overlapping-generations δ -reduction in the rate *level*; the low safe rate in calm rests entirely on the fear-factor mechanism $\bar{\lambda}_t^S - U_t$, not on a demographic accounting offset.

of Proposition 17 without re-derivation; turnover supplies the stationary S_t these results take as input. The level formula (E1) is unchanged: with the *growth* of Ξ_t stationary and its level unread by prices, the explicit $e^{-\delta t}$ survives, so r_t^f keeps its $+\delta$ term, and turnover enters the rate only through the demographic mean-reversion of $\bar{\lambda}_t^S$ and U_t in (F2), which smooths its realized path—the stationary safe rate of the perpetual-youth economy, its fluctuations smoothed—not removed—by the demographic mean-reversion (Gârleanu and Panageas, 2015).

Primitives. Under Assumption 5 an investor born at b with type (ε, ϕ_0) and belief $\mathbb{P}^i$ (change-of-measure M^i) chooses consumption to solve

$$\max \mathbb{E}^i \left[\int_b^\infty e^{-\delta(s-b)} \mathbf{1}\{T_i > s\} u(c_s^i) ds \right] = \max \mathbb{E}^i \left[\int_b^\infty e^{-(\delta+\Omega)(s-b)} u(c_s^i) ds \right], \quad u(c) = \frac{c^{1-\alpha}}{1-\alpha}, \quad (\text{F4})$$

where $T_i \sim \text{Exp}(\Omega)$ is her death time (Assumption 5(i)) and the second equality takes the expectation over T_i , independent of the path. Markets are complete in the aggregate risks (Assumption 3) with state-price density ξ ; under the fair annuity of Assumption 5(iii) a survival-contingent unit of consumption at s costs $\xi_s e^{-\Omega(s-b)}$ (the market price under $\mathbb{P}$ times the survival probability), so the market value of the consumption plan—the budget—is taken under the pricing measure $\mathbb{P}$,

$$\mathbb{E} \left[\int_b^\infty \frac{\xi_s}{\xi_b} e^{-\Omega(s-b)} c_s^i ds \mid \mathcal{F}_b \right] \leq W_b^i, \quad (\text{F5})$$

with birth wealth W_b^i the value the birth endowment of Assumption 5(iii) implies at the birth date b (type-dependent under heterogeneous beliefs, not per-capita), in date- b consumption units and $\mathcal{F}_b$ -measurable (Assumption 5(ii)–(iii)); ξ_b is its date-0 price, and the normalization by ξ_b is absorbed into the cohort multiplier $\tilde{\psi}_i$ in the proof, so the dating affects no equilibrium object. An *overlapping-generations equilibrium* is an SPD ξ and allocations $\{c^i\}$ such that every living investor solves (F4)–(F5) and goods clear, $\int_{\text{living}} c_t^i di = C_t$.

Proofs for this section are in Internet Appendix IA.1.

Appendix G. The stationary cross-section

Proposition 17 (Stationary cross-section under demographic turnover): *Under Assumptions 1–4, the finite menu (Assumption 5), and the demographic turnover of Assumption 5:*

- (i) (Age structure.) *For $\Omega > 0$ the age distribution of the living is stationary with density $\Omega e^{-\Omega a}$ on $a \geq 0$ (at $\Omega = 0$ there is no turnover and no such stationary age law).*
- (ii) (No refresh $\Rightarrow$ concentration.) *If $\Omega = 0$, the second-order posterior is consistent, $\text{MAP}(\rho_t^i) \rightarrow \phi^*$ a.s. (Berk, 1966); every re-selection (11) then returns ϕ^* and the held-model cross-section converges to the single vertex δ_{ϕ^*} .*
- (iii) (Refresh $\Rightarrow$ unique nondegenerate stationary cross-section.) *If $\Omega > 0$, the cohort-resolved cross-section Ψ_t —the wealth measure over (held model $\phi \in \Phi$, since-adoption record $\mathcal{R}$, regime age τ , t cohort age $a \geq 0$ carried alongside for the demographics—is, in the limiting regime where the public posterior has concentrated ($\text{MAP}(\rho_t) = \phi^*$, so the re-selection destination is calendar-invariant), a time-homogeneous measurable functional of the marked disaster point process N on the two-sided stationary line and the stationary demographic-and-type structure (Proposition 9 with Assumption 5). On that two-sided process Ψ_t is a stationary, ergodic process with a unique stationary law Ψ^∞ , to which the economy from any finite prior converges, geometrically at a rate proportional to Ω under the turnover-dominance condition $\Omega > \kappa_\Omega := \lambda_{hi}/\alpha + \lambda^*(\underline{g}^{-1} - 1)$, $\underline{g} := \inf_{\phi, x} r(\phi, x)^{1/\alpha}$ (the two-sided construction, the stationary laws, and every time-average moment below hold without this condition)—in the bounded-Lipschitz (Wasserstein-1) metric on laws over $\mathcal{P}(\Sigma)$, where Σ carries the bounded product metric $d_\Sigma((\phi, \mathcal{R}, \tau, \varepsilon), (\phi', \mathcal{R}', \tau', \varepsilon')) = |\lambda - \lambda'| + |\theta - \theta'| + (\mathbf{1}\{n \neq n'\} + \mathbf{1}\{n = n'\}(\sum_{k \leq n} |x_k - x'_k|) \wedge 1) + |\tau - \tau'| \wedge 1 + |\varepsilon - \varepsilon'|$ —each coordinate bounded (Φ compact, marks in $\mathcal{J}$), so d_Σ is a bounded metric—and not in total variation (Step 5)—as $\text{MAP}(\rho_t) \rightarrow \phi^*$; the pricing state $S_t \in \Delta^{K-1}$ (the ϕ -marginal of Ψ_t) and its instantaneous functionals $(U_t, \bar{\lambda}_t^S)$, together with the price-dividend ratio $v(\Psi_t)$, the jump ratio*

ψ_t , and the disaster premium DRP_t —measurable and integrable functionals of the full Ψ_t —inherit unique stationary laws, with unconditional moments equal to long-run time averages (Birkhoff). Ψ^∞ places mass on at least two models whenever Π_ϕ does; the dependence of Ψ_t on the remote past (and on any initial cross-section) decays geometrically, at a rate proportional to Ω .

Four notes on the statement. In the selection-only reduction with re-selection suppressed every vertex of Δ^{K-1} is absorbing; under the full re-selecting dynamics, by contrast, only δ_{ϕ^*} is invariant, since any other vertex is left at the next rejection—Remark 6. The record coordinates of d_Σ are compared entrywise at equal counts and at distance one otherwise; the coupling bounds of the proof are insensitive to this choice up to constants, and the turnover-dominance constant κ_Ω is a crude sufficient bound. The rate is the demographic one for the head-count component; for the wealth-weighted state the coupling delivers convergence with the rate qualified by the selection and reweight perturbations (a Foster–Lyapunov strengthening in the sense of Costa and Dufour 2008 would sharpen it). The jump ratio reads the full state through $\psi_t(x) = v(\Psi_t^x)/v(\Psi_{t-})$ (Proposition 13), and integrability rather than continuity is what the Birkhoff step needs. A fifth note, on the destination: in this locked regime the re-selection destination is the menu’s Kullback–Leibler projection—the fearful point at the binary menu—so the calm switch flow is non-negative there and the stationary economy’s calm relaxation rests on selection and turnover; the scheduled-relief reading of the calm channel is a finite-history statement (Proposition 6).

The turnover does exactly one thing to the mechanism and one thing to the mathematics. To the mechanism (eq. F2): it adds a constant pull of the cross-section back toward the newborn dispersion Π_ϕ , at rate Ω , which is precisely the force that offsets the selection-and-consistency concentration of Remark 6—young investors, whose *held models* have not yet moved to ϕ^* —they inherit the already-concentrated common posterior but enter with a drawn held model—are continually reborn. To the mathematics: with the menu finite and the inflow of fresh types

constant, the cross-section becomes a time-homogeneous functional of a stationary, ergodic disaster clock, so it is itself stationary and ergodic, and the unconditional moments the calibration needs are well-defined and computable as time averages. The parameter that governs the stationary *dispersion*—and hence, through Proposition 6, the amplitude of the synchronized revision wave—is the newborn joint law Π over birth held models *and* thresholds (its model marginal is Π_ϕ), together with the turnover rate Ω : more dispersed births leave more disagreement, the threshold law sets how long birth beliefs survive before revision (through the rejection and deadline times), and Ω governs how quickly fresh cohorts replace the selected cross-section. This is the sense in which, as anticipated, stationarity rests on the assumed joint law of threshold and intensity at birth.

Two beliefs, two clocks. The refresh and the synchronization live on different objects, and the stationary economy keeps them apart. The common second-order posterior ρ_t (over the menu, the source of the re-selection destination) is a public functional of the record, inherited by every cohort (Assumption 5(iv)); by part (ii) it has concentrated on the fixed pseudo-true model, $\mathbf{m}(\rho_t) \equiv \phi^*$, so in the stationary regime the common destination is constant and carries no remote-past dependence. The per-agent state $(\phi, a, \mathcal{R})$ is reset at birth and its dependence on the past decays geometrically, at a rate proportional to Ω (Step 5 of the proof). Non-degeneracy is therefore a held-model phenomenon: a stationary age-mixture of ϕ_0 -holding young and ϕ^* -absorbed old. The fast-in jump at a disaster is correspondingly carried by the reweighting channel—wealth flowing to the higher-intensity holders (Proposition 6(ii))—and by relocation to the fixed ϕ^* , not by a jumping destination; the destination’s disaster-time movement of Lemma 4 is a transient, finite-record feature. The *direction* of that jump is a restriction, not an identity: relocation raises mean intensity only under the complacent-switcher condition of Proposition 6— $\lambda(\phi^*)$ above the *post-reweight mean of the triggered switchers*, a cross-sectional condition on the stationary state. A complacent newborn law makes it plausible, but Π_ϕ bears on the switcher distribution only through the stationary composition; we state the condition rather than derive it.

Two boundaries, kept sharp. First, stationarity delivers the invariant *distribution* Ψ^∞ ; it does not by itself close the price level. Even under the finite menu the price–dividend ratio $v(\Psi_t)$ solves the Feynman–Kac operator equation (IA.7.1) on the record-resolved state Ψ_t (Corollary IA.1), collapsing to a finite-dimensional-state operator equation in the shares only in the selection-only specialization with no record-dependent switching; with a continuous menu v likewise solves the analogous infinite-dimensional Feynman–Kac valuation problem of Proposition 13 (the displayed operator equation (IA.7.1) is the finite-menu form of Corollary IA.1). So the invariant-distribution statement and the price-level statement are distinct. Second, the turnover modifies the equilibrium kernel only through this cross-section: Proposition 16 showed the kernel form is preserved, with Ω entering the dynamics of Ξ_t (and hence r_t^f) but not the time preference, so the pricing of Sections 3–5.C transfers to the stationary economy: the realized record-resolved state Ψ_t is then *distributed* according to the invariant law Ψ^∞ of Proposition 17, and the state-contingent pricing formulas are evaluated at the realized Ψ_t (with marginal S_t), not at the invariant law itself.

The proof is in Internet Appendix IA.2.

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